F78PA - Probability and Statistics A - Core Notes

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Chapter 1

Probability Models

Probability models offer a mathematical framework for describing and understanding random phenomena.

Examples of random phenomena include:

- the outcome of a die roll
- the lifetime of a computer chip
- the time until the next bus arrives
- the price of a stock one year from now

Probability models will be used in advanced courses that you will study at Heriot-Watt, including Stochastic Processes (F79SP), Survival Models (F79SU), Risk Theory (F70RT) etc.

Key point: A solid grasp of probability theory is essential for success in many advanced courses.

1.1 Formal definition of a probability model

We begin with the formal, mathematical definition of a probability model. This material is quite abstract, but we want to understand the abstract foundations before moving on to practical applications.

A probability model has 3 components, the sample space Ω, the event space $\mathcal{F}$ and the probability measure $\mathbb{P}$.

1. Sample space Ω (or sometimes S)
   - Ω is the set of all possible outcomes of a random experiment.
   - We use the symbol ω to denote an element of Ω, and we call ω a sample point.

   Example 1.1. Roll a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. One possible outcome is $\omega = 4$, meaning that we roll a 4.

   Example 1.2. Measure lifetime of a computer chip. Since (theoretically) the lifetime could be any non-negative number, we take $\Omega = [0, \infty)$.

   Example 1.3. Two tosses of a coin, $\Omega = \{\text{HH,HT,TH,TT}\}$, where H and T represent the Head and Tail of a coin.

2. Event space $\mathcal{F}$

   - $\mathcal{F}$ is a collection of subsets of the sample space Ω, which satisfies the following properties:

(a) $\Omega \in \mathcal{F}$;
(b) if $A \in \mathcal{F}$ (where A is a subset of Ω), then $\Omega \setminus A = A^c \in \mathcal{F}$;
(c) if $A_1, A_2, \cdots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (countable unions are in $\mathcal{F}$).

- Each subset in the collection $\mathcal{F}$ is called an event.

A collection of subsets satisfying properties (a)–(c) above is called a **σ-algebra** or **σ-field**.

Example 1.1 continued. Roll a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. Then 'roll an even number' is represented as $A = \{2, 4, 6\} \subset \Omega$, and we can define our event space $\mathcal{F}$ to include A as one of its elements, so that 'roll an even number' is an event; that is, $A \in \mathcal{F}$. By property (b), the event space $\mathcal{F}$ must then also include the set $A^c = \{1, 3, 5\}$, which is the event 'roll an odd number'.

In the definition of an event space we require that the whole state space should be an event and that we can combine events by asking that A or B is an event if $A, B \in \mathcal{F}$. There are additional ways to combine events. For instance, the empty set is considered to be an event, i.e. nothing happening is an event. Also if $A, B \in \mathcal{F}$ are events then the event that both A and B occur is also an event. Both of these follow from the definition of the event space.

Lemma 1.1. *Let $\mathcal{F}$ be an event space then*

- $\emptyset \in \mathcal{F}$
- *If $A_1, A_2, \cdots \in \mathcal{F}$, then $\bigcap_i A_i \in \mathcal{F}$*

Proof

- By property (a), $\Omega \in \mathcal{F}$. Then by property (b), $\Omega^c \in \mathcal{F}$; that is, $\emptyset \in \mathcal{F}$, as required.
- If $A_1, A_2, \cdots \in \mathcal{F}$, then

$$\begin{aligned}
&\text{By Property (b)} \Rightarrow A_1^c, A_2^c, \cdots \in \mathcal{F} \\
&\text{By Property (c)} \Rightarrow \bigcup_i A_i^c \in \mathcal{F} \\
&\text{By Property (b)} \Rightarrow \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c \quad \text{(de Morgan's Laws from set theory)}
\end{aligned}$$

□

Example 1.1 continued. In the example of rolling a die, 'Roll a multiple of 3' is represented as $B = \{3, 6\}$, and we can define $\mathcal{F}$ to include B; that is, $B \in \mathcal{F}$ and 'roll a multiple of 3' is an event. Then by property (c), $A \cup B = \{2, 3, 4, 6\} \in \mathcal{F}$ is also an event. (Notice that property (c) as originally stated applies to an infinite sequence of events, but we just need to take $A_3 = A_4 = \cdots = \emptyset$ to be able to apply property (c) to the union of just 2 events.) Similarly, $A \cap B = \{6\}$ is an event.

It's natural in the above example to take $\mathcal{F}$ to consist of all subsets of Ω, and then 'an event' simply means any subset of Ω. We will see later that in more complicated examples, this is not possible.

3. Probability measure $\mathbb{P}$

   A probability measure $\mathbb{P}$ is a *function* $\mathbb{P} : \mathcal{F} \to \mathbb{R}$ defined on the event space $\mathcal{F}$ which satisfies the following properties:

   (I) $\forall A \in \mathcal{F},\ \mathbb{P}(A) \geqslant 0$

   (II) $\mathbb{P}(\Omega) = 1$

   (III) For any *mutually exclusive* events $A_1, A_2, \cdots \in \mathcal{F}$ we have $\mathbb{P}(\bigcup A_i) = \sum \mathbb{P}(A_i)$.
   Note: The events $A_1, A_2, \cdots \in \mathcal{F}$ are *mutually exclusive* if for any two of the events A_i and A_j with $i \neq j$, we have $A_i \cap A_j = \emptyset$.

   The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability model (or probability space).

   Interpretation of $\mathbb{P}(A)$

   The quantity $\mathbb{P}(A)$, the probability of event A, represents how likely it is that the outcome of the random experiment lies in A. This likelihood is measured on a scale from 0 to 1.

   Example 1.1 continued. Roll a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. Take the event space to be $\mathcal{F}$ = all subsets of Ω. If the die is fair, then $\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{6\})$. From property (III) and property (II) it then follows that $\mathbb{P}(\{\omega\}) = \frac{1}{6}$ $\forall \omega \in \Omega$. Applying property (III), then

$$\begin{aligned} \mathbb{P}(\text{roll an even number}) &= \mathbb{P}(\{2, 4, 6\}) \\ &= \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} \end{aligned}$$

1.2 Properties of a probability measure $\mathbb{P}$

Using the defining properties of a probability measure $\mathbb{P}$, we can deduce other important properties that are useful for computing probability values.

1. Law of Complements.

   If $A \in \mathcal{F}$, then $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$

2. Inclusion–Exclusion.

   If $A, B \in \mathcal{F}$, then $\mathbb{P}(A \bigcup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \bigcap B)$

3. Monotone Property of Probability Measures

   If $A, B \in \mathcal{F}$ and $A \subseteq B$, then $\mathbb{P}(A) \leqslant \mathbb{P}(B)$

Proofs

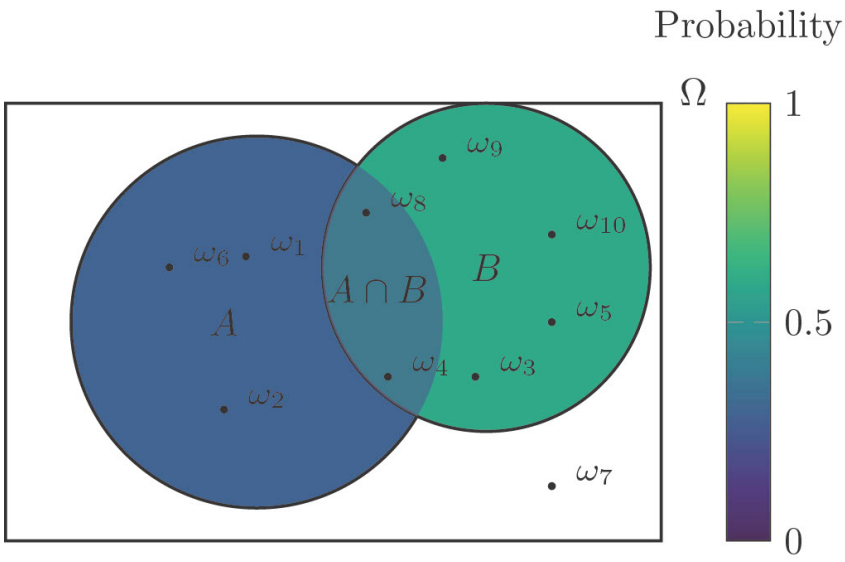


Figure 1.1: Illustration of a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ showing events A, B and their intersection $A \cap B$. Individual outcomes ω_i are represented as points within the sample space Ω. The colour bar encodes probability values of the events A, B and $A \cap B$, with shading corresponding to event likelihoods.

1. For any $A \in \mathcal{F}$, we have $A^c \cap A = \emptyset$, so
$$\begin{aligned} \mathbb{P}(A^c \cup A) &= \mathbb{P}(A^c) + \mathbb{P}(A) && \text{by property (III)} \\ \Rightarrow \quad \mathbb{P}(\Omega) &= \mathbb{P}(A^c) + \mathbb{P}(A) \\ \Rightarrow \quad 1 &= \mathbb{P}(A^c) + \mathbb{P}(A) && \text{by property (II)} \\ \Rightarrow \quad \mathbb{P}(A^c) &= 1 - \mathbb{P}(A) \end{aligned}$$
2. For any $A, B \in \mathcal{F}$, notice that
$$A \cup B = A \cup (B \setminus A)$$
But $A \cap (B \setminus A) = \emptyset$, a disjoint union, so we can now apply property (III):
$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B \setminus A) \\ \Rightarrow \quad \mathbb{P}(B \setminus A) &= \mathbb{P}(A \cup B) - \mathbb{P}(A) \end{aligned}$$
Next, notice that
$$\begin{aligned} B &= (A \cap B) \cup (B \setminus A) && \text{(disjoint union)} \\ \Rightarrow \quad \mathbb{P}(B) &= \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) && \text{property (III)} \\ &= \mathbb{P}(A \cap B) + (\mathbb{P}(A \cup B) - \mathbb{P}(A)) \\ \Rightarrow \quad \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \end{aligned}$$
3. For any $A, B \in \mathcal{F}$ with $A \subseteq B$,
$$\begin{aligned} B &= A \cup (B \setminus A) && \text{(disjoint union)} \\ \Rightarrow \quad \mathbb{P}(B) &= \mathbb{P}(A) + \mathbb{P}(B \setminus A) && \text{property (III)} \\ \Rightarrow \quad \mathbb{P}(B) &\geqslant \mathbb{P}(A) && \text{property (I) applied to } B \setminus A \end{aligned}$$

1.3 Some examples and discussion

Example 1.1 continued. Random experiment: Roll a die.

Probability model:

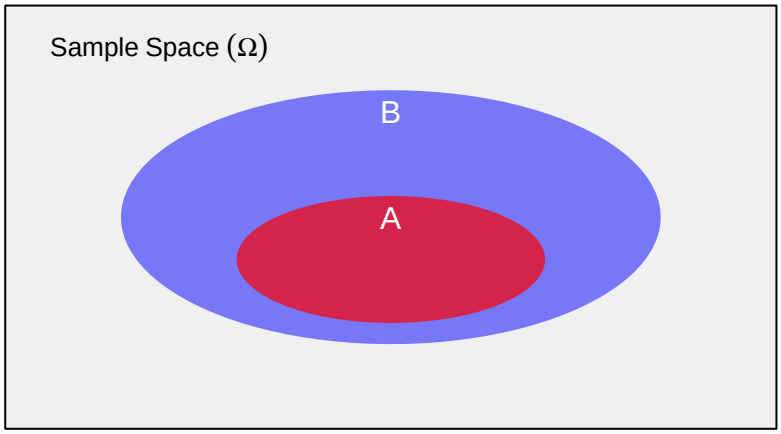


Figure 1.2: Illustration of the monotonicity property of probability measures. The red region represents event A, and the blue region represents event B with $A \subseteq B$. By the monotonicity property, we have $\mathbb{P}(A) \leq \mathbb{P}(B)$.

- Sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Event space $\mathcal{F}$ = all subsets of Ω
- Specify probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$

Since the sample space can be written as the disjoint union $\Omega = \{1\} \cup \{2\} \cup \{3\} \cup \{4\} \cup \{5\} \cup \{6\}$, then

$$\left.\begin{array}{l}\text{Property (III)} \Rightarrow \mathbb{P}(\Omega) = \mathbb{P}(1) + \mathbb{P}(2) + \cdots + \mathbb{P}(6) \\ \text{Property (II)} \ \Rightarrow \mathbb{P}(\Omega) = 1\end{array}\right\} \Rightarrow \begin{array}{l}\text{the probability measure must satisfy} \\ \mathbb{P}(1) + \mathbb{P}(2) + \cdots + \mathbb{P}(6) = 1\end{array}$$

For example, in a biased die, we might assign probabilities as follows:

$$\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \mathbb{P}(5) = \frac{1}{10} \quad \text{and} \quad \mathbb{P}(6) = \frac{1}{2} \qquad (\textit{biased} \text{ die})$$

Once we have specified the values of $\mathbb{P}(\omega)$ $\forall \omega \in \Omega$, we can use Property (III) to compute $\mathbb{P}(A)$ for other $A \in \mathcal{F}$.

Suppose $A = \{2, 4, 6\} = \{\text{outcome on the roll is even}\}$, then

$$\mathbb{P}(A) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{7}{10} \text{ by property (III)}$$

If the sample space Ω is finite, then we can take $\mathcal{F}$ to be the power set of Ω i.e. $\mathcal{F}$ = (all subsets of Ω) and we can define the probability measure $\mathbb{P}$ by specifying $\mathbb{P}(\{\omega\})$ $\forall \omega \in \Omega$ s.t. $\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1$.

Remark 1.1. Recall a set Ω is countable if there exists an injective map $\psi : \Omega \to \mathbb{N}$. Examples of countable sets are: any finite set, the natural numbers $\mathbb{N}$, integers $\mathbb{Z}$ and the rational numbers $\mathbb{Q}$. Examples of sets that are not countable are non-trivial intervals $[a, b]$ with $a < b$ and the set of real numbers $\mathbb{R}$.

Note: You are expected to have learned the fundamental *counting principle* such as **combinations** and **permutations** before taking this course.

Example 1.4. Random experiment: Flip a coin until you obtain a 'Head'.

Probability model:

- Sample space $\Omega = \{1, 2, 3, \dots\}$, where sample point m corresponds to 'first Head occurs on the mth flip'
- Event space $\mathcal{F}$ = all subsets of Ω
- Specify probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$

In this case, Ω is a *countable infinite* set. Since any subset A of a countable set Ω is also countable, then for any $A \in \mathcal{F}$, by Property (III), we have $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\})$.

For example, we could set $\mathbb{P}(m) = \left(\frac{1}{2}\right)^m$ for $m = 1, 2, \dots$
(Remark: This corresponds to the *assumption* that the coin is 'fair' — see later.)

Check: $\mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \cdots = \sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = 1$ (using the formula for summing geometric series).

Let $A = \{2, 4, 6, \dots\} = \{\text{first Head occurs on an even flip}\}$. Then by Property (III) and the summation formula for geometric series, we have

$$\begin{aligned}
\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) &= \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) + \cdots \text{ by property (III)} \\
&= \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \\
&= \frac{1}{4}\left(1 + \frac{1}{4} + \left(\frac{1}{4}\right)^2 + \dots\right) \\
&= \frac{1}{4}\left(\frac{1}{1 - \frac{1}{4}}\right) \text{ using formula for sum of geometric series} \\
&= \frac{1}{3}
\end{aligned}$$

Then can compute, for example,

$$\begin{aligned}
\mathbb{P}(\text{first Head occurs on an odd flip}) &= \mathbb{P}(A^c) \\
&= 1 - \mathbb{P}(A) \text{ by Law of Complements} \\
&= 1 - \frac{1}{3} = \frac{2}{3}
\end{aligned}$$

When the sample space is countable we can still take $\mathcal{F}$ = all subsets of Ω, and we can still define the probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$ s.t. $\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1$.

When we move beyond countable sample spaces we need to be much more careful!

Example 1.5. Suppose we want to model the time spent waiting for a bus to arrive. Let ω be the time we have to wait for a bus from the time we arrive at the bus stop. What would be our sample space?

A natural choice would be $\Omega = [0, \infty)$, note this is an uncountable set! How can we construct an event space for this example?

If we attempt to construct $\mathcal{F}$ as in the countable setting we have issues. Imagine that you went to the bus stop now, there is a small probability that the next bus will arrive in 10 minutes to the nearest minute. The probability that the next bus will arrive in 10 minutes to the nearest second is extremely small, The probability that the next bus will arrive at exactly 10 minutes is zero. Indeed, for any $\omega > 0$ the probability that the bus will arrive at exactly time ω is zero. Therefore we cannot build $\mathcal{F}$ from just singleton sets. However, if we also construct the probability on intervals then it is possible, as we will see in the following example.

Example 1.6. Random experiment: Choose a number 'at random' from the interval [0, 1] in such a way that any number is 'equally likely' to be chosen. For example consider what happens when you type `runif()` in R.

Probability model:

- Sample space $\Omega = [0, 1]$ — an uncountable set.

- Event space. If we try taking $\mathcal{F}$ to be 'all subsets of Ω', then it turns out to be impossible to define a probability measure $\mathbb{P}$ on $\mathcal{F}$ that satisfies properties (I)–(III). When Ω is uncountable there are 'too many' subsets of Ω; there is no probability measure $\mathbb{P}$ on all subsets of Ω which satisfies Properties (I)–(III).

  Instead, we define the event space $\mathcal{F}$ and the probability measure $\mathbb{P}$ using some 'basic sets'.

  - Take the 'basic sets' of [0, 1] to be closed intervals of the form [a, b]. (NB - This is not the only possible choice for 'basic sets'.)
  - Let event space $\mathcal{F}$ = all subsets of [0, 1] that can be generated from the 'basic sets' using the set operations of countable unions, countable intersections, and complement.
  - The name of this event space is the Borel σ-algebra on $[0, 1]$.

- Specify probability measure $\mathbb{P}$ by specifying the value of $\mathbb{P}(B)$ for all basic sets B. Then compute $\mathbb{P}(A)$ for any $A \in \mathcal{F}$ using the properties (I)–(III) of probability measures and the definition of $\mathbb{P}$ on basic sets.

  For this example, specify probability measure $\mathbb{P}$ on the *basic sets* to reflect our assumption that any point in the interval $[0, 1]$ is *equally likely.*

  e.g.

$$\mathbb{P}\left(\left[\frac{1}{2}, \frac{3}{4}\right]\right) = \mathbb{P}\left(\text{chosen number is between } \frac{1}{2} \text{ and } \frac{3}{4}\right)$$
$$= \text{length of the interval } \left[\frac{1}{2}, \frac{3}{4}\right] = \frac{1}{4}$$

  More generally, we take $\mathbb{P}([a, b]) = b - a$ for any a, b with $0 \leqslant a \leqslant b \leqslant 1$.

Examples

1. $[0, \frac{1}{2}) \cup (\frac{3}{4}, 1] = \underbrace{[\frac{1}{2}, \frac{3}{4}]^c}_{\text{complement of a basic set}} \in \mathcal{F}$

   $\mathbb{P}\left([0, \frac{1}{2}) \cup (\frac{3}{4}, 1]\right) = \mathbb{P}\left([\frac{1}{2}, \frac{3}{4}]^c\right) = 1 - \mathbb{P}\left([\frac{1}{2}, \frac{3}{4}]\right) = 1 - \frac{1}{4} = \frac{3}{4}$

   by Law of Complements and definition of $\mathbb{P}$ on the basic set $[\frac{1}{2}, \frac{3}{4}]$.

2. $\underbrace{\left[\frac{1}{4},\frac{1}{2}\right]\cup\left[\frac{2}{3},\frac{3}{4}\right]}_{\text{union of two basic sets}}\in\mathcal{F}$

$\mathbb{P}\left([\frac{1}{4},\frac{1}{2}]\cup[\frac{2}{3},\frac{3}{4}]\right)=\mathbb{P}\left([\frac{1}{4},\frac{1}{2}]\right)+\mathbb{P}\left([\frac{2}{3},\frac{3}{4}]\right)=\frac{1}{4}+\frac{1}{12}=\frac{1}{3}$

by Property (III) of probability measures and the definition of $\mathbb{P}$ on the basic sets $[\frac{1}{4},\frac{1}{2}]$ and $[\frac{2}{3},\frac{3}{4}]$.

3. $(\frac{1}{2},\frac{3}{4})=\left([0,\frac{1}{2}]\cup[\frac{3}{4},1]\right)^c\in\mathcal{F}$

$$\begin{aligned}\mathbb{P}\left(\left(\frac{1}{2},\frac{3}{4}\right)\right) &= 1-\mathbb{P}\left(\left[0,\frac{1}{2}\right]\cup\left[\frac{3}{4},1\right]\right)\\ &= 1-\left(\mathbb{P}\left(\left[0,\frac{1}{2}\right]\right)+\mathbb{P}\left(\left[\frac{3}{4},1\right]\right)\right)\\ &= 1-\left(\frac{1}{2}+\frac{1}{4}\right)=\frac{1}{4}\end{aligned}$$

4. $\{\frac{1}{2}\}=\underbrace{\bigcap_{m=2}^{\infty}[\frac{1}{2}-\frac{1}{m},\ \frac{1}{2}+\frac{1}{m}]}_{\text{countable intersection of basic sets}}\in\mathcal{F}$

To find $\mathbb{P}(\{\frac{1}{2}\})$, we'll use the following important result, valid for any probability measure.

Continuity of probability measure $\mathbb{P}$

Suppose $A_1\subseteq A_2\subseteq\ldots$ is a nested, increasing sequence of events in $\mathcal{F}$ and let $A=\cup_{k=1}^{\infty}A_k$. Then $\mathbb{P}(A)=\lim_{k\to\infty}\mathbb{P}(A_k)$.

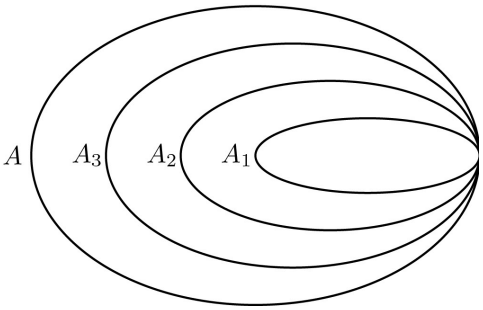


Figure 1.3: An illustration of a series of nested increasing sets A_1, A_2, A_3.

Corollary: Suppose $B_1\supseteq B_2\supseteq\ldots$ is a nested, decreasing sequence of events in $\mathcal{F}$ and let $B=\cap_{k=1}^{\infty}B_k$. Then $\mathbb{P}(B)=\lim_{k\to\infty}\mathbb{P}(B_k)$.

Application of the continuity of $\mathbb{P}$:

Since $\{\frac{1}{2}\}=\bigcap_{m=2}^{\infty}[\frac{1}{2}-\frac{1}{m},\ \frac{1}{2}+\frac{1}{m}]$, intersection of nested decreasing sequence of events, then

$$\begin{aligned}\mathbb{P}\left(\left\{\frac{1}{2}\right\}\right) &= \lim_{m\to\infty}\mathbb{P}\left(\left[\frac{1}{2}-\frac{1}{m},\frac{1}{2}+\frac{1}{m}\right]\right) \qquad \text{(by continuity of } \mathbb{P}\text{)}\\ &= \lim_{m\to\infty}\frac{2}{m}=0.\end{aligned}$$

Remarks:

(i) In general, for *this probability model*, $\mathbb{P}((a,b)) = b-a$ for any $0 \leqslant a < b \leqslant 1$, and $\mathbb{P}(\{c\}) = 0$ for *any* $c \in [0,1]$.

(ii) Although not all subsets of Ω are 'events' here, in practice any subset you might be interested in will be an event. For instance, we've seen that open intervals, closed intervals, half-open intervals, individual point values, are all events. It is possible to construct a subset of $\Omega = [0,1]$ that is *not* an event, but it's quite hard work to do this.

Summary: Building probability models

Discrete case	Non-discrete case
• $\Omega = \{w_1, w_2, \dots\}$ is a *countable set*	• Ω is an *uncountable set*
• $\mathcal{F}$ = all subsets of Ω	• $\mathcal{F}$ is *not* all subsets of Ω; instead, $\mathcal{F}$ contains 'basic sets' and all other sets that can be formed by intersections, unions and complements of basic sets
• Define $\mathbb{P}(\{\omega\})$ for every $\omega \in \Omega$ s.t. $\sum_{\omega\in\Omega} \mathbb{P}(\{\omega\}) = 1$, and then use the properties of $\mathbb{P}$ to determine $\mathbb{P}(A)$ for any $A \in \mathcal{F}$	• Define $\mathbb{P}(B)$ (in a consistent way) for every basic set B and then use the properties of $\mathbb{P}$ to determine $\mathbb{P}(A)$ for any $A \in \mathcal{F}$

1.4 Conditional probability

In this section we introduce the concept of *conditional probability*:

Suppose that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability model, and for some event $E \in \mathcal{F}$, we have $\mathbb{P}(E) > 0$. If we *know* that the event E has occurred but we don't know anything else about the outcome of the random experiment modelled by $(\Omega, \mathcal{F}, \mathbb{P})$. *Given* this information, we can calculate a *revised* probability for every other event $F \in \mathcal{F}$ which we call the *conditional probability* of F *given* the event E.

Conditional probability arises in many applications of statistics, for example if we can not observe directly the variable we are interested in. Consider a medical trial of n patients then what is the probability that a drug is effective given a proportion p patients recovered?

Before giving the definition of conditional probability let us consider the following example to motivate the definition.

Example 1.7. Suppose we flip a fair coin 3 times.
The sample space is $\Omega = \{$HHH, HHT, HTH, THH, TTH, THT, HTT, TTT$\}$.
Define events E, F to be

E = {exactly 2 flips are Heads}

F = {first flip is Heads}

'Fair' means $\mathbb{P}$(Heads) = $\mathbb{P}$(Tails) for each coin flip, and since the coin is fair, each outcome in Ω is equally likely. So we have

$$\mathbb{P}(E) = \mathbb{P}(\{\text{HHT, HTH, THH}\}) = \frac{3}{8}$$

$$\mathbb{P}(F) = \mathbb{P}(\{\text{HHH, HHT, HTH, HTT}\}) = \frac{1}{2}$$

Now suppose you know that the event E has occurred (i.e. exactly 2 flips were Heads). Since we know that E has occurred, we can essentially treat E as if it was the whole sample space. So now think of the sample space as being $E = \{\text{HHT, HTH, THH}\}$, and because the coin is fair, each outcome in E is equally likely. That is, the *conditional probabilities* are

$$\begin{aligned}\mathbb{P}(\{\text{HHT}\} \mid E) &= \frac{1}{3}\\ \mathbb{P}(\{\text{HTH}\} \mid E) &= \frac{1}{3}\\ \mathbb{P}(\{\text{THH}\} \mid E) &= \frac{1}{3}\end{aligned}$$

Also notice that outcomes that are not in E have conditional probability 0, given that E is known to have occurred. That is, $\mathbb{P}(\text{HHH} \mid E) = \mathbb{P}(\text{TTH} \mid E) = \mathbb{P}(\text{THT} \mid E) = \mathbb{P}(\text{HTT} \mid E) = \mathbb{P}(\text{TTT} \mid E) = 0$.

So we have

$$\begin{aligned}\mathbb{P}(F \mid E) &= \mathbb{P}(\{\text{HHH, HHT, HTH, HTT}\} \mid E)\\ &= \mathbb{P}(\{\text{HHH}\} \mid E) + \mathbb{P}(\{\text{HHT}\} \mid E) + \mathbb{P}(\{\text{HTH}\} \mid E) + \mathbb{P}(\{\text{HTT}\} \mid E)\\ &= 0 + \frac{1}{3} + \frac{1}{3} + 0 = \frac{2}{3}\end{aligned}$$

Notice that in this case, $\mathbb{P}(F \mid E) > \mathbb{P}(F)$. That is, knowledge that exactly 2 out of 3 flips are Heads increases the chance that the first flip is Heads.

In general (for a countable sample space), if we know that an event E has occurred, then to compute conditional probabilities:

(i) Outcomes $\omega \in E^c$ have conditional probability $\mathbb{P}(\{\omega\} \mid E) = 0$

(ii) Outcomes $\omega \in E$ have conditional probabilites computed by re-normalising the set of probabilities $\{\mathbb{P}(\{\omega\} : \omega \in E\}$ to add up to 1. That is, we multiply the original probabilities $\mathbb{P}(\{\omega\})$ by the normalising factor $\frac{1}{\mathbb{P}(E)}$ to obtain the conditional probabilites $\mathbb{P}(\{\omega\} \mid E) = \frac{\mathbb{P}(\{\omega\})}{\mathbb{P}(E)}$ for all $\omega \in E$ (exercise: check that these add up to 1).

(iii) For any event F, the conditional probability $\mathbb{P}(F \mid E)$ is computed by adding up the conditional probabilities for all outcomes $\omega \in F$. That is,

$$\begin{aligned}\mathbb{P}(F \mid E) &= \sum_{\omega \in F} \mathbb{P}(\{\omega\} \mid E)\\ &= \sum_{\omega \in E \cap F} \mathbb{P}(\{\omega\} \mid E) + \sum_{\omega \in E^c \cap F} \mathbb{P}(\{\omega\} \mid E)\\ &= \sum_{\omega \in E \cap F} \frac{\mathbb{P}(\{\omega\})}{\mathbb{P}(E)} + 0 \text{ using (i) and (ii) above}\\ &= \frac{\sum_{\omega \in E \cap F} \mathbb{P}(\{\omega\})}{\mathbb{P}(E)}\\ &= \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}\end{aligned}$$

Formally, we *define* conditional probability as follows (for any sample space, countable or uncountable).

Definition
Suppose $E, F \in \mathcal{F}$ are two events, and $\mathbb{P}(E) > 0$. Then the *conditional probability* $\mathbb{P}(F|E)$ of

the event F given that the event E has occurred is defined as

$$\boxed{\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}}$$

Example 1.7 continued. From the definition of conditional probability, we compute

$$\mathbb{P}(F|E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(\{\text{HHT, HTH}\})}{\mathbb{P}(\{\text{HHT, HTH, THH}\})} = \frac{\frac{2}{8}}{\frac{3}{8}} = \frac{2}{3}$$

Note: The answer is the same as we obtained before, but the calculation is quicker.

Properties

Conditional probabilities satisfy the *same* properties as the unconditional probability measure $\mathbb{P}$. That is, for any event $E \in \mathcal{F}$ such that $\mathbb{P}(E) > 0$, we have

(I) $\forall A \in \mathcal{F},\ \mathbb{P}(A \mid E) \geqslant 0$

(II) $\mathbb{P}(\Omega \mid E) = 1$ and $\mathbb{P}(\emptyset \mid E) = 0$

(III) For any *mutually exclusive* events $A_1, A_2, \cdots \in \mathcal{F}$, we have $\mathbb{P}(\bigcup A_i \mid E) = \sum \mathbb{P}(A_i \mid E)$

Remark Here, the event E is one specific fixed event, whereas we consider the conditional probabilities $\mathbb{P}(A \mid E)$ for *all* events $A \in \mathcal{F}$. Sometimes indicate this by writing the *conditional probability measure* as $\mathbb{P}(\,\cdot \mid E)$

Exercise: Use the definition of conditional probability and the properties of the unconditional probability measure $\mathbb{P}$ to show that properties (I)–(III) above hold for the conditional probability measure $\mathbb{P}(\cdot \mid E)$, for any event E such that $\mathbb{P}(E) > 0$.

Since properties (I)-(III) above hold, all the rules of probability that can be deduced from (I)-(III) also hold. For example,

$\mathbb{P}(A \cup B \mid E) = \mathbb{P}(A \mid E) + \mathbb{P}(B \mid E) - \mathbb{P}(A \cap B \mid E)$ (Inclusion-Exclusion)
$\mathbb{P}(A \mid E) = 1 - \mathbb{P}(A^c \mid E)$ (Law of Complements)

and so on.

Rearranging the definition of conditional probability gives the following very useful formula for computing the probability of the intersection of two events.

Chain rule:
For any events $E, F \in \mathcal{F}$ with $\mathbb{P}(E) > 0$,

$$\boxed{\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F \mid E)}$$

This is sometimes referred to as the *Chain Rule* for probabilities.
Note: The Chain rule follows from the definition of conditional probability $\mathbb{P}(F|E) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(E)}$.

Example 1.8. Suppose an urn contains six balls, numbered $1, 2, 3, 4, 5, 6$, and we remove two balls, one at a time, from the urn without replacement, such that each of 30 possible outcomes are equally likely. What is the probability that the sum of the numbers on the balls is *even*?

For $i = 1, 2$, define events

O_i = Odd number on ith draw

E_i = Even number on ith draw $= O_i^c$

Then

$$\begin{aligned}\mathbb{P}(\text{sum is even}) &= \mathbb{P}(\text{both numbers are odd}) + \mathbb{P}(\text{both numbers are even}) \\ &= \mathbb{P}(O_1 \cap O_2) + \mathbb{P}(E_1 \cap E_2) \\ &= \mathbb{P}(O_1)\mathbb{P}(O_2 \mid O_1) + \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1) \qquad \text{(Chain Rule)} \\ &= \frac{3}{6} \cdot \frac{2}{5} + \frac{3}{6} \cdot \frac{2}{5} = \frac{2}{5}\end{aligned}$$

Note: $\mathbb{P}(O_2 \mid O_1) = \frac{2}{5}$ since, given that the first ball chosen is odd and it is removed, there are then 2 odd balls left in the remaining 5 balls. Similarly, $\mathbb{P}(E_2 \mid E_1) = \frac{2}{5}$.

(We could alternatively have solved the above example by noting that there are 6×5 possible outcomes in this experiment and all are equally likely, so each has probability $\frac{1}{30}$. So we could count how many outcomes result in an even sum, and multiply by $\frac{1}{30}$. However, it's a bit fiddly to count all the ways to get an even sum; the chain rule argument is neater.)

The chain rule can be extended to give us a method for computing intersections of more than two events in terms of conditional probabilities:

Extended Chain Rule:
The chain rule can be extended to intersections of three events:

$$\mathbb{P}(E \cap F \cap G) = \mathbb{P}(E)\mathbb{P}(F \mid E)\mathbb{P}(G \mid E \cap F)$$

and more generally:

$$\mathbb{P}(E_1 \cap E_2 \cap \cdots \cap E_n) = \mathbb{P}(E_1)\mathbb{P}(E_2 \mid E_1)\mathbb{P}(E_3 \mid E_1 \cap E_2) \ldots \mathbb{P}(E_n \mid E_1 \cap \cdots \cap E_{n-1})$$

The following theorem is an important application of the Chain Rule.

Law of Total Probability (Partition Theorem)
Suppose events $E_1, E_2, \ldots$ form a *partition* of Ω — that is, events $E_1, E_2, \ldots$ are disjoint with $\Omega = \cup_i E_i$, and $\mathbb{P}(E_i) > 0 \ \forall i$. Then for *any* event $F \in \mathcal{F}$,

$$\boxed{\mathbb{P}(F) = \mathbb{P}(E_1)\mathbb{P}(F \mid E_1) + \mathbb{P}(E_2)\mathbb{P}(F \mid E_2) + \cdots = \sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}$$

Proof
Suppose $E_1, E_2, \ldots$ form a partition of Ω. Then for any $F \in \mathcal{F}$,

$$\begin{aligned}\mathbb{P}(F) &= \mathbb{P}(F \cap \Omega) \\ &= \mathbb{P}\left(F \cap (\cup_i E_i)\right) \\ &= \mathbb{P}\left(\cup_i (F \cap E_i)\right) && \text{(Distributive Property from set theory)} \\ &= \sum_i \mathbb{P}(F \cap E_i) && \left(\begin{smallmatrix}\text{property (III) of probability measures} \\ \text{since the events } F \cap E_1, F \cap E_2, \ldots \text{ are disjoint}\end{smallmatrix}\right) \\ &= \sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i) && \text{(chain rule)}\end{aligned}$$

Example 1.9. The probability that a randomly chosen individual will develop a particular disease is 10^{-4}. Call this event A, so $\mathbb{P}(A) = 10^{-4}$. There is a test for this disease which returns the result Y (positive) or N (negative), and it is known that $\mathbb{P}(Y \mid A) = 0.95$, and that $\mathbb{P}(Y \mid A^c) = 0.1$. An individual is selected at random from the population.

(i) What is the probability that this individual tests positive?

Since the events A and A^c form a *partition*, then by the Law of Total Probability,

$$\mathbb{P}(Y) = \mathbb{P}(Y \mid A)\mathbb{P}(A) + \mathbb{P}(Y \mid A^c)\mathbb{P}(A^c) = 0.95 \cdot 10^{-4} + 0.1 \cdot (1 - 10^{-4}) = 0.100085$$

(ii) Now suppose the individual *has* tested positive. What is the probability that the individual will develop the disease?

Using the definition of conditional probability and then the chain rule,

$$\mathbb{P}(A \mid Y) = \frac{\mathbb{P}(A \cap Y)}{\mathbb{P}(Y)} = \frac{\mathbb{P}(Y \mid A)\mathbb{P}(A)}{\mathbb{P}(Y)} = \frac{0.95 \times 0.0001}{0.100085} = 0.00095$$

This is about 10 times more likely than the unconditional probability $\mathbb{P}(A) = 0.0001$ but is still very small. A person who has tested positive is still unlikely to develop the disease.

The second calculation above illustrates another very useful formula for computing probabilities: for any two events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

Using the Law of Total Probability together with the definition of conditional probability, we obtain the following generalisation.

Bayes Theorem
Suppose events $E_1, E_2, \ldots$ form a partition of Ω with $\mathbb{P}(E_i) > 0 \ \forall i$. Then for *any* event $F \in \mathcal{F}$, for each i,

$$\boxed{\mathbb{P}(E_i \mid F) = \frac{\mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}{\sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}}$$

Proof:

$$\begin{aligned}
\mathbb{P}(E_i \mid F) &= \frac{\mathbb{P}(E_i \cap F)}{\mathbb{P}(F)} && \text{(definition of conditional probability)} \\
&= \frac{\mathbb{P}(E_i)\mathbb{P}(F \mid E_i)}{\sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)} && \text{(chain rule + Law of Total Probability)}
\end{aligned}$$

1.5 Independence

Statistical independence between events is defined as follows.

Definition
Two events E and F are independent with respect to a probability measure $\mathbb{P}$ if and only if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F)$$

Meaning of statistical independence
Suppose E, F are independent events with $\mathbb{P}(E) > 0$. Then we have

$$\mathbb{P}(F \mid E) = \frac{\mathbb{P}(F \cap E)}{\mathbb{P}(E)} \overset{\text{independence}}{=} \frac{\mathbb{P}(F)\mathbb{P}(E)}{\mathbb{P}(E)} = \mathbb{P}(F)$$

That is, if events E and F are independent, then the *conditional probability of F given E* is the *same* as the unconditional probability of F. Similarly, provided $\mathbb{P}(F) > 0$, then $\mathbb{P}(E \mid F) = \mathbb{P}(E)$. So for independent events, knowledge that one event has occurred does not affect the probability that the other has occurred.

Example 1.10. Suppose a die is rolled once, so $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$. Suppose the die is *fair*. Are the events A and B independent?

Solution:
Since the die is fair, $\mathbb{P}(\{i\}) = \frac{1}{6} \quad \forall i \in \Omega$. To check whether the definition of independence is satisfied for events A and B, first notice that $A \cap B = \{2, 4\}$. Then

$$\mathbb{P}(A) = \frac{4}{6} = \frac{2}{3}$$
$$\mathbb{P}(B) = \frac{3}{6} = \frac{1}{2}$$
$$\mathbb{P}(A \cap B) = \mathbb{P}(\{2, 4\}) = \frac{2}{6} = \frac{1}{3}$$

Since $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, the events A and B are independent.

Now suppose we have a *biased* die, and the probabilities for each outcome are:

$$\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \mathbb{P}(4) = \mathbb{P}(5) = \frac{1}{10}, \quad \mathbb{P}(6) = \frac{1}{2}$$

Are the events A and B independent for this biased die?

Check whether the definition of independence is satisfied:

$$\mathbb{P}(A) = \mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(4) = \frac{4}{10}$$
$$\mathbb{P}(B) = \mathbb{P}(2) + \mathbb{P}(4) + \mathbb{P}(6) = \frac{1}{10} + \frac{1}{10} + \frac{1}{2} = \frac{7}{10}$$
$$\mathbb{P}(A \cap B) = \mathbb{P}(2) + \mathbb{P}(4) = \frac{2}{10}$$

In this case, $\mathbb{P}(A \cap B) = \frac{2}{10} \neq \frac{4}{10} \cdot \frac{7}{10} = \mathbb{P}(A)\mathbb{P}(B)$, so the events A and B are not independent.

This example illustrates that we cannot determine whether events are independent without computing *probabilities*, i.e. whether two events are independent depends on the probability measure.

The definition of independence can be generalised as follows.

Definition
The events $E_1, E_2, \ldots$ are independent if and only if for every finite collection of events $E_{i_1}, \ldots, E_{i_k}$,

$$\mathbb{P}(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_k}) = \mathbb{P}(E_{i_1})\mathbb{P}(E_{i_2}) \ldots \mathbb{P}(E_{i_k})$$

Using independence to construct probability models
In practice, when building probability models we often make the *model assumption* (based on physical considerations) that certain outcomes (or events) are independent. We then use this independence assumption to build the model. The next example illustrates this.

Example 1.11. Suppose that we flip a *biased* coin 3 times and that each time we flip the coin, $\mathbb{P}(\text{Heads}) = \frac{1}{3}$ and $\mathbb{P}(\text{Tails}) = \frac{2}{3}$.

Since the outcome on one flip cannot physically affect the outcome on any other flip (that is, flips are *physically independent*), we will *assume* that the flips are statistically independent.

The sample space is $\Omega = \{\text{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}\}$, and we define the probability measure $\mathbb{P}$ using the assumption that the outcomes on different flips are independent, so

$$\mathbb{P}(\text{HHH}) \underset{\text{by indep of flips}}{=} \mathbb{P}(\text{H})\mathbb{P}(\text{H})\mathbb{P}(\text{H}) = \frac{1}{3}\cdot\frac{1}{3}\cdot\frac{1}{3} = \left(\frac{1}{3}\right)^3$$

$$\mathbb{P}(\text{HHT}) \underset{\text{by indep of flips}}{=} \mathbb{P}(\text{H})\mathbb{P}(\text{H})\mathbb{P}(\text{T}) = \left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)$$

and so on.

Warning
A common mistake is to apply the formula $\mathbb{P}(E\cap F) = \mathbb{P}(E)\mathbb{P}(F)$ to events E, F that are not independent — this will give the wrong answer. To avoid this, always start from the more fundamental formula $\mathbb{P}(E\cap F) = \mathbb{P}(E)\mathbb{P}(F \mid E)$, and then think carefully about whether you are justified in replacing $\mathbb{P}(F \mid E)$ with $\mathbb{P}(F)$; this is only justified if you know that events E, F are independent.

Chapter 2

Random Variables: General Theory

In this chapter we introduce **random variables**. In most situations and applications you will be working with random variables rather than with the underlying probability model for a random experiment. It is a peculiarity of the terminology used in probability theory that a random variable is actually a function! Here is the formal definition of a random variable:

Definition 2.1 (Random variable)**.** Suppose $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability model for a random experiment, and suppose that $X : \Omega \to \mathbb{R}$ is a *function* which assigns a numerical value $X(\omega)$ to every outcome $\omega \in \Omega$ such that

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}, \quad \text{for any } x \in \mathbb{R}, \tag{2.1}$$

then we say that X is a random variable.

Remark 2.1.
- When working with random variables we often use the shorthand for any $A \subseteq \mathbb{R}$ to write
  $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.
- We will often consider sets of the form $\{a \leq X \leq b\}$ and $\{X \leq x\}$. Note that these are subsets of Ω and we wish to assign probabilities to these sets, the condition (2.1) states that $\{X \leq x\}$ is an event and hence we can assign a probability to this set.

Example 2.1. Let $\Omega = \{\omega_0, \omega_1\}$ with $\mathcal{F}$ the set of all subsets of Ω, i.e. $\mathcal{F} = \{\emptyset, \{\omega_0\}, \{\omega_1\}, \Omega\}$. Define $X : \Omega \to \mathbb{R}$ by $X(\omega_0) = 0$ and $X(\omega_1) = 1$. This is an example of the simplest non-trivial random variable, it takes the value 1 with probability $p = \mathbb{P}(\omega_1)$ and takes the value 0 with probability $1 - p$. This is known as a Bernoulli Random variable.

This can be used to model a coin flip, let $\omega_0 = T$ be the outcome of seeing "tails" and $\omega_1 = H$ be the outcome of seeing "heads". Then X counts how many heads are seen in one flip of the coin.

Definition 2.2 (Image of a random variable)**.** Suppose $X : \Omega \to \mathbb{R}$ is a random variable. Then the image of X (sometimes called the support of X), denoted by $\mathrm{Im}X$, is defined as

$$\mathrm{Im}(X) = \{x \in \mathbb{R} : X(\omega) = x \text{ for some } w \in \Omega\} = \text{set of all possible values taken by } X$$

Definition 2.3 (Discrete random variables)**.** Suppose that $X : \Omega \to \mathbb{R}$ is a random variable. We say that X is a discrete random variable if and only if $\mathrm{Im}X$ is a countable set.

Example 2.2 (Example 2.1 continued)**.** The image of X is $\mathrm{Im}(X) = \{0, 1\}$. This set is finite and hence countable, therefore X is a discrete random variable.

Example 2.3. Consider repeatedly flipping a coin until it shows "Heads". Let $X : \Omega \to \mathbb{R}$ denote the number of flips until "Heads" appears for the first time.

If we want to construct the probability space we could set

$$\Omega = \{\omega = (\omega_i)_{i=1}^n : n \in \mathbb{N}, \omega_i = T \text{ for } i < n, \omega_n = H\}.$$

that is, the set of all finite sequences where all but the last entry is T and the last entry is H. Then we can define $X(\omega)$ to be the length of the sequence ω, representing the number of flips until the first "Heads". Alternatively, we could set $\Omega = \mathbb{N}$ where $\omega \in \Omega$ represents the number of flips until the first "Heads" appears, in which case $X(\omega) = \omega$. In both cases, X is a discrete random variable since the image of X is $\mathrm{Im}(X) = \mathbb{N}$ which is countable.

Example 2.4. Let X denote the amount of time waiting at a bus stop until the next bus appears. Then the image of X is $[0, \infty)$ which is uncountable, hence X is not a discrete random variable.

2.1 Probability mass function (pmf)

Suppose $X : \Omega \to \mathbb{R}$ is a discrete random variable (i.e. $\mathrm{Im}(X)$ is countable). We define the probability mass function $p_X : \mathrm{Im}X \to [0, 1]$ by

$$\boxed{p_X(x) = \mathbb{P}(X = x) = \mathbb{P}\left(\underbrace{\{\omega : X(\omega) = x\}}_{\text{event in } \Omega}\right)}$$

Example 2.5 (Example 1.7 continued.)**.** $X = \#$heads in 3 flips of a fair coin. $\mathrm{Im}(X) = \{0, 1, 2, 3\}$. Probability mass function is

$p_X(0) = \mathbb{P}(X = 0) = \mathbb{P}\big(\{\mathrm{TTT}\}\big) = 1/8$
$p_X(1) = \mathbb{P}(X = 1) = \mathbb{P}\big(\{\mathrm{HTT}, \mathrm{THT}, \mathrm{TTH}\}\big) = 3/8$
$p_X(2) = \mathbb{P}(X = 2) = \mathbb{P}\big(\{\mathrm{THH}, \mathrm{HTH}, \mathrm{HHT}\}\big) = 3/8$
$p_X(3) = \mathbb{P}(X = 3) = \mathbb{P}\big(\{\mathrm{HHH}\}\big) = 1/8$

Note: In the notation $\mathbb{P}(X = x)$, notice that X and x are very different things.
X is the name of the random variable, and doesn't change so long as we're considering the same random variable.
Whereas x is a particular value that the random variable can take, and varies through all values in $\mathrm{Im}(X)$, so $x = 0, 1, 2, 3$ in Example 2.5 above.

Properties of probability mass functions

Suppose that p_X is the probability mass function of a (discrete) random variable X. Then:

1. $\forall x$, $p_X(x) \geq 0$
2. $\sum_x p_X(x) = 1$

Conversely, any function satisfying the above properties is a valid probability mass function for some discrete random variable.

Definition 2.4 (Cumulative distribution function)**.** Suppose $X : \Omega \to \mathbb{R}$ is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.
Then the cumulative distribution function (cdf) $F_X : \mathbb{R} \to [0, 1]$ is defined by

$$F_X(x) = \mathbb{P}(X \leqslant x) = \mathbb{P}\big(\{\omega : X(\omega) \leqslant x\}\big) \quad \forall x \in \mathbb{R}$$

Remark: *Every* random variable has a cumulative distribution function, whereas only discrete random variables have a probability mass function.

2.2 Computing probabilities using the cdf

Suppose X is a random variable and F_X is the distribution function for X.

1. By *definition* of F_X, we have $\mathbb{P}(X \leqslant b) = F_X(b) \ \ \forall b \in \mathbb{R}$.

2. Suppose $a < b$, then

$$\mathbb{P}(a < X \leqslant b) = \mathbb{P}(X \leqslant b) - \mathbb{P}(X \leqslant a) = F_X(b) - F_X(a)$$

   by definition of F_X.

3. Suppose $b \in \mathbb{R}$. Then the Continuity Property of the probability measure $\mathbb{P}$ implies that

$$\boxed{\mathbb{P}(X < b) = \lim_{x \uparrow b} F_X(x)}$$

4. For any $b \in \mathbb{R}$:

$$\begin{aligned}\mathbb{P}(X = b) &= \mathbb{P}(X \leqslant b) - \mathbb{P}(X < b) \\ &= F_X(b) - \lim_{x \uparrow b} F_X(x) \\ &= \underline{\text{size of the jump}} \text{ in } \ F_X(x) \text{ at } x = b\end{aligned}$$

   Remark: If the df F_X is a continuous function, then $\lim_{x \uparrow b} F_X(x) = F_X(b)$, and it follows that when F_X is a continuous function, we have

   $\forall x \in \mathbb{R} : \ \mathbb{P}(X = x) = 0.$

5. Suppose $a < b$, then

$$\mathbb{P}(a < X < b) = \lim_{x \uparrow b} F_X(x) - F_X(a)$$
$$\mathbb{P}(a \leqslant X \leqslant b) = F_X(b) - \lim_{x \uparrow a} F_X(x)$$

   Remark: When F_X is a continuous function, we have

   $\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a < X \leqslant b) = \mathbb{P}(a < X < b) = \mathbb{P}(a \leqslant X < b) = F_X(b) - F_X(a).$

Example 2.6 (Example 2.5 continued.)**.** Recall Example 2.5: the random variable X denotes the number of Heads in 3 flips of a fair coin. Then X has cumulative distribution function, $F_X(x)$, given by

$$F_X(x) = \mathbb{P}(X \leqslant x) = \begin{cases} 0 & x < 0 \\ p_X(0) & 0 \leqslant x < 1 \\ p_X(0) + p_X(1) & 1 \leqslant x < 2 \\ p_X(0) + p_X(1) + p_X(2) & 2 \leqslant x < 3 \\ 1 & x \geqslant 3 \end{cases} = \begin{cases} 0 & x < 0 \\ 1/8 & 0 \leqslant x < 1 \\ 1/2 & 1 \leqslant x < 2 \\ 7/8 & 2 \leqslant x < 3 \\ 1 & x \geqslant 3 \end{cases}$$

The cumulative distribution function F_X is a step function.

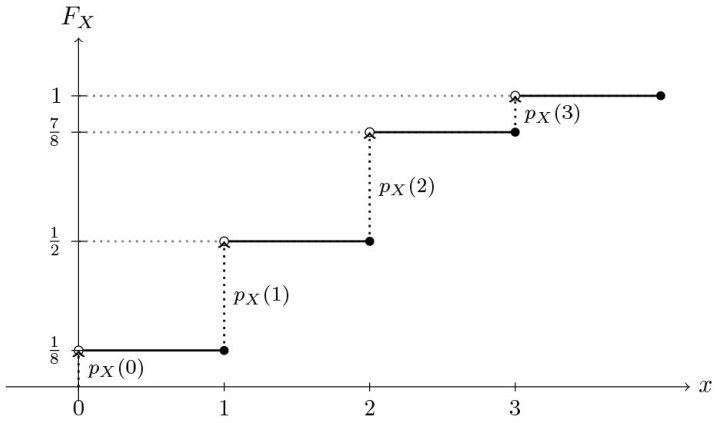


Figure 2.1: Sketch of $F_X(x)$

Given the cumulative distribution function, F_X, we can calculate probabilities:
From Property 4, we obtain: $\mathbb{P}(X = 1) = F_X(1) - \lim_{x\uparrow 1} F_X(x) = 1/2 - 1/8 = 3/8$.
From Property 2, we obtain: $\mathbb{P}(5/4 < X \leqslant 3/2) = F_X(3/2) - F_X(5/4) = 1/2 - 1/2 = 0$.
(This is as expected, since X only takes the values $\{0, 1, 2, 3\}$ and none of these values are contained in the interval $(5/4, 3/2]$, so $\mathbb{P}(5/4 < X \leqslant 3/2) = 0$.)
Remark: If X is a discrete random variable, then F_X is a step function. If F_X has a step at $x = b$, then the size of the step at $x = b$ equals $\mathbb{P}(X = b)$. All of the information about X's distribution is 'stored' in the sizes of the jumps in the cdf F_X. Therefore, it is just as easy to work with the probability mass function p_X (which also describes the distribution of X) as it is to work with F_X.

2.3 Properties of distribution functions

What properties do *all* cumulative distribution functions have?

Suppose that F_X is a cumulative distribution function for a random variable X. Then:

1. $\forall x \in \mathbb{R},\ 0 \leqslant F_X(x) \leqslant 1$

2. If $x < y$ then $F_X(x) \leqslant F_X(y)$, i.e. F_X is a non-decreasing function.

3. $\lim_{x\uparrow\infty} F_X(x) = 1$ and $\lim_{x\downarrow-\infty} F_X(x) = 0$.

4. The cumulative distribution function F_X is right continuous - this means that $\forall b \in \mathbb{R},\ \lim_{x\downarrow b} F_X(x) = F_X(b)$ and the cumulative distribution function F_X has left limits - this means that $\forall b \in \mathbb{R},\ \lim_{x\uparrow b} F_X(x)$ *exists*, but limit might not be equal to $F_X(b)$.

Proof:
1. This follows from the definition of $F_X(x) = \mathbb{P}(X \leqslant x)$ and since $0 \leqslant \mathbb{P}(X \leqslant x) \leqslant 1$.
2. Since $x < y$, we have $\{X \leqslant x\} \subseteq \{X \leqslant y\rangle$. So the monotone property of probability measures $\Rightarrow F_X(x) = \mathbb{P}(X \leqslant x) \leqslant \mathbb{P}(X \leqslant y) = F_X(y)$.
3. This follows from the definition of F_X and the continuity of probability measures.
4. This follows from the definition of F_X and the continuity of probability measures.

Any function that satisfies the above 4 properties is a distribution function for some random variable. So, to decide whether a function is a valid distribution function - sketch it and check whether the above 4 properties are satisfied.

2.4 Independent random variables

The definition of independent random variables is a natural extension of the definition of independent events.

Definition
The random variables $X_1, X_2, \ldots X_n$ are independent if and only if for all subsets $A_1, A_2, \ldots, A_n \subseteq (-\infty, \infty)$, we have

$$\mathbb{P}(X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n) = \mathbb{P}(X_1 \in A_1)\mathbb{P}(X_2 \in A_2)\ldots\mathbb{P}(X_n \in A_n).$$

Remarks

1. The above definition applies to both discrete and non-discrete random variables.
2. In practice, if the random variables $X_1, X_2, \ldots, X_n$ depend on physically independent 'subexperiments' then we make the assumption that the random variables $X_1, X_2, \ldots, X_n$ are (statistically) *independent* r.v.s.
   Example. Suppose that we roll a die twice. Then each roll is a 'physically independent' subexperiment — i.e. the outcome on the first roll doesn't physically influence the outcome on the second roll. So if X_1 is the outcome on the first roll and X_2 is the outcome on the second roll, we will *assume* that the variables X_1 and X_2 are (statistically) independent because the rolls are 'physically independent'.
3. If $X_1, X_2, \ldots, X_n$ are independent random variables then so are the random variables $g_1(X_1), g_2(X_2), \ldots, g_n(X_n)$ for any functions $g_1, g_2, \ldots, g_n$. This is an important result which will be used many times in this course and in other courses.

Example 2.7. Suppose a *fair* die is rolled twice. Let X_1 be the score on the first roll, and X_2 the score on the second roll. Assuming that X_1, X_2 are independent random variables, what is the probability that both scores are less than 4?

Answer: For $i = 1, 2$, each X_i has probability mass function p_{X_i} given by

$$p_{X_i}(x) = \begin{cases} 1/6 & x = 1 \\ 1/6 & x = 2 \\ 1/6 & x = 3 \\ 1/6 & x = 4 \\ 1/6 & x = 5 \\ 1/6 & x = 6 \end{cases}$$

The required probability is

$$\begin{aligned} \mathbb{P}(X_1 < 4, X_2 < 4) &= \mathbb{P}(X_1 \leqslant 3, X_2 \leqslant 3) \\ &= \mathbb{P}(X_1 \leqslant 3)\mathbb{P}(X_2 \leqslant 3) \qquad \text{(by independence of } X_1, X_2) \\ &= (3/6)(3/6) = 1/4 \end{aligned}$$

2.5 Common examples of discrete random variables

In this section we will go through a range of important distributions which will regularly appear in probability and statistics applications. Some of these you may have seen before, here we will be explaining them in the context of this chapter. Note that you will find a summary of these

distributions in the "Formula Sheet For Probability and Statistics Exams" available on Canvas and provided in the exam.

2.5.1 Bernoulli distribution

Bernoulli variables are special, simple random variables which only take two possible values: 0 or 1. We say that the random variable X is a Bernoulli random variable if and only if $\mathrm{Im}X = \{0, 1\}$.

Examples

1. Flip a coin. The sample space is $\Omega = \{\mathrm{H}, \mathrm{T}\}$ and we can define the Bernoulli random variable $X : \Omega \to \mathbb{R}$ by $X(\mathrm{H}) = 1$ and $X(\mathrm{T}) = 0$.
2. Turn on a light and measure the lifetime (in hours) of the light bulb. The sample space is $\Omega = [0, \infty)$. If we are interested in whether the light bulb lasts more than 20 hours, we can define the Bernoulli random variable $Y : \Omega \to \mathbb{R}$ by $Y(\omega) = 1$ if the lifetime $\omega \geq 20$, and $Y(\omega) = 0$ if the lifetime $\omega < 20$. So the random variable Y answers the simple Yes/No question: 'Is the lifetime greater than 20 hours?', where the value 1 corresponds to 'Yes' and the value 0 corresponds to 'No'.

Properties of Bernoulli random variables
Suppose X is a Bernoulli random variable. Then we must have:

$$\mathbb{P}(X = 1) = p \quad \text{for some} \quad 0 \leqslant p \leqslant 1$$
$$\textit{and} \quad \mathbb{P}(X = 0) = 1 - p$$

Definition
We define the **indicator variable** $\mathbb{I}_A$ for the event A by

$$\mathbb{I}_A = \begin{cases} 1 & \text{if event } A \text{ happens} \\ 0 & \text{if event } A \text{ does not happen} \end{cases}$$

Typically, Bernoulli variables arise as **indicator variables** for events of interest.

2.5.2 Binomial distribution

Suppose we have a sequence of n independent, identical, subexperiments (often called 'trials'), each having two possible outcomes, 'Success' or 'Failure', with
$\mathbb{P}(\text{'Success'}) = p$ and $\mathbb{P}(\text{'Failure'}) = 1 - p$.

Let $X =$ total number of successes in the n trials.

Then we say that X is a Binomial random variable with parameters n (number of trials) and p (probability of success in a trial) and we write $X \sim \mathrm{Bin}(n, p)$.

It is clear from the definition of X that $\mathrm{Im}X = \{0, 1, 2, \ldots, n\}$.

The probability mass function for $X \sim \mathrm{Bin}(n, p)$ is given by

$$\boxed{p_X(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \text{ for } k = 0, 1, 2, \ldots n.}$$

Explanation:

$\binom{n}{k}$ = # ways to choose a subset of size k from $\{1, 2, \ldots, n\}$
= # ways to position k successes in a sequence of k successes and $(n-k)$ failures.

Also, each different sequence of k successes and $(n-k)$ failures has probability $p^k(1-p)^{n-k}$ since the trials are independent so we can multiply the probabilities of successes and failures together.

Example 2.8. Flip a *biased* coin 3 times and suppose that $\mathbb{P}(\text{'Heads'}) = 1/3$ and $\mathbb{P}(\text{'Tails'}) = 2/3$. Let X = number of Heads in 3 flips.
Then $X \sim \text{Bin}(3, \frac{1}{3})$, Im$X$=$\{0, 1, 2, 3\}$ and the probability mass function for X is given by:

$$p_X(0) = \mathbb{P}(X=0) = \binom{3}{0}\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^3 = \frac{8}{27}$$
$$p_X(1) = \mathbb{P}(X=1) = \binom{3}{1}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^2 = \frac{12}{27}$$
$$p_X(2) = \mathbb{P}(X=2) = \binom{3}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^1 = \frac{6}{27}$$
$$p_X(3) = \mathbb{P}(X=3) = \binom{3}{3}\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^0 = \frac{1}{27}$$

Representation in terms of indicator variables
If we define the indicator random variables $\mathbb{I}_k$ for $k = 1, 2, \ldots, n$ to be

$$\mathbb{I}_k = \begin{cases} 1 & \text{if trial } k \text{ is a Success} \\ 0 & \text{if trial } k \text{ is a Failure} \end{cases}$$

then we can represent $X \sim \text{Bin}(n, p)$ as $X = \mathbb{I}_1 + \mathbb{I}_2 + \cdots + \mathbb{I}_n$

2.5.3 Geometric distribution

Suppose now that we have an *infinite* sequence of independent, identical trials and each trial has only two possible outcomes, 'Success' or 'Failure', with $\mathbb{P}(\text{'Success'}) = p$ and $\mathbb{P}(\text{'Failure'}) = 1-p$.

Let Y = number of trials until the first success is obtained.

Then we say that Y is a Geometric random variable with parameter p and we write $Y \sim \text{Geo}(p)$. It is clear from the definition of Y that ImY=$\{1, 2, 3, \ldots\}$. The event $\{Y = y\}$ occurs when the first $(y-1)$ trials are all Failures, and then trial y is a Success. Using the independence of the trials, the probability of this is $(1-p)^{y-1}p$.
That is, the probability mass function for $Y \sim \text{Geo}(p)$ is given by

$$\boxed{p_Y(y) = \mathbb{P}(Y=y) = (1-p)^{y-1}p \quad \text{for } y = 1, 2, 3, \ldots}$$

Example 2.9. Flip a *biased* coin repeatedly until the first flip to land 'Heads', and suppose that for each flip, $\mathbb{P}(\text{'Heads'}) = 1/3$ and $\mathbb{P}(\text{'Tails'}) = 2/3$. Find the probability that more than 2 flips are needed.

Answer: Denoting by Y the number of flips needed, then $Y \sim \text{Geo}(1/3)$, with probability mass function

$$p_Y(y) = \mathbb{P}(Y=y) = (2/3)^{y-1}\,1/3 \quad \text{for } y = 1, 2, 3, \ldots$$

We are asked to find $\mathbb{P}(Y > 2)$:

$$\begin{aligned}\mathbb{P}(Y > 2) &= 1 - \mathbb{P}(Y \leq 2) \\ &= 1 - \mathbb{P}(Y = 2) - \mathbb{P}(Y = 1) \\ &= 1 - (2/3)\,(1/3) - 1/3 = 4/9 \approx 0.444\end{aligned}$$

Example 2.10. Products produced by a machine have a 20% defective rate. What is the probability that the first defective product occurs in the first five inspections? Then, find the average number of inspections to obtain the first defective product.

Answer: Let X be the number of inspections to obtain the first defective product, then $X \sim \text{Geo}\,(0.2)$. Then the probability that the first defective product occurs within the first five inspections is

$$\begin{aligned}\mathbb{P}(X \leq 5) &= \mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \mathbb{P}(X = 3) + \mathbb{P}(X = 4) + \mathbb{P}(X = 5) \\ &= 0.20 + (0.80)(0.20) + (0.80)^2(0.20) + (0.80)^3(0.20) + (0.80)^4(0.20) \\ &= 0.67232.\end{aligned}$$

and the expected number of inspections to find the first defective product is

$$\mathbb{E}(X) = \frac{1}{0.20} = 5.$$

2.5.4 Poisson distribution

The Poisson distribution is often used as a model for data describing the number of events of a particular kind that occur in a given time interval, for example:

- the number of claims on a portfolio of policies in a year;
- the number of flooding events that might occur on a certain river over a 5-year period;
- the number of industrial accidents that occur in a factory in a year.

Definition 2.5. Suppose $\lambda > 0$ is a fixed constant. We say that the random variable X has a Poisson distribution with parameter λ, denoted $X \sim \text{Poi}(\lambda)$, if $\text{Im}X = \{0, 1, 2, \dots\}$ and

$$\boxed{p_X(x) = \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} \qquad \text{for } x = 0, 1, 2, \dots}$$

We will show in Section 8.4 that the Poisson distribution arises as the limit of a Binomial distribution.

Example 2.11. The number of flooding events that occur on a certain river over a 5 year period is believe to follow a Poisson distribution with parameter $\lambda = 4$. Find the probability that the number of flooding events over a 5 year period is 2 or less.

Answer: Denoting by X the number of flooding events over a 5 year period, then $X \sim \text{Poi}(4)$, with probability mass function

$$p_X(x) = \mathbb{P}(X = x) = \frac{4^x e^{-4}}{x!} \qquad \text{for } x = 0, 1, 2, \dots$$

We are asked to find $\mathbb{P}(X \leqslant 2)$:

$$\begin{aligned}\mathbb{P}(X \leqslant 2) &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) \\ &= \frac{4^0 e^{-4}}{0!} + \frac{4^1 e^{-4}}{1!} + \frac{4^2 e^{-4}}{2!} \\ &= 13e^{-4} \approx 0.238\end{aligned}$$

Chapter 3

(Absolutely) Continuous Random Variables

3.1 Example

Example 3.1. Let X be a random variable representing a number chosen uniformly at random from the interval $[0,1]$. Observe that $\mathrm{Im}(X) = [0,1]$, which is not a countable set, so X is not a discrete random variable. This means that X *does not have a probability mass function.* However, X *does* have a cumulative distribution function, which we can find as follows.
First of all,

$$\begin{array}{lrcll} \text{for } x < 0, & F_X(x) & = & \mathbb{P}(X \leqslant x) = 0 & \text{(since we can't have } X \leqslant x < 0), \\ \text{for } x \geqslant 1, & F_X(x) & = & \mathbb{P}(X \leqslant x) = 1 & \text{(since we certainly do have } X \leqslant 1 \leqslant x) \end{array}$$

Now recall from Example 1.6 that for a number X chosen uniformly at random from the interval $[0,1]$, we have $\mathbb{P}(X \in [a,b]) = b - a$ for any a, b with $0 \leqslant a \leqslant b \leqslant 1$. So,

$$\text{for } 0 \leqslant x < 1, \qquad F_X(x) \;=\; \mathbb{P}(X \leqslant x) = \mathbb{P}(X \in [0,x]) = x - 0 = x$$

Thus, the cumulative distribution function of X is

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

Notice that the distribution function $F_X(x)$ is defined piece-wise.
A sketch of the cdf F_X is given in Figure 3.1.
Notice that $F_X(x)$ is a continuous function, and is differentiable everywhere except at $x = 0$ and $x = 1$.
Observe that for $x \in (0,1)$

$$\mathbb{P}(X = x) = \lim_{n\to\infty} F_X(x) - F_X(x - \frac{1}{n}) = \lim_{n\to\infty} \frac{1}{n} = 0.$$

This means that working with the uniform distribution is fundamentally different from using discrete random variables.

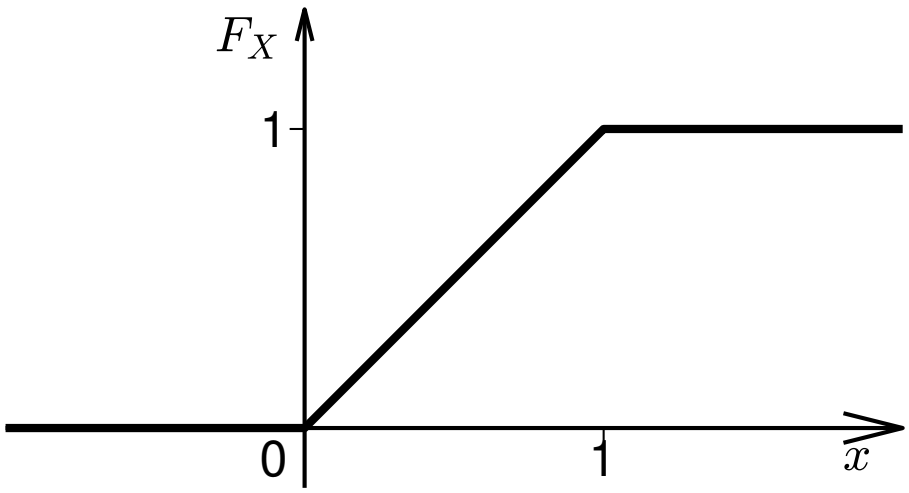


Figure 3.1: Sketch of $F_X(x)$

3.2 Probability density function (pdf)

Although *every* random variable X has a cdf F_X, for discrete random variables it is often easier to work with the probability mass function. Similarly, for (absolutely) continuous random variables it is often easier to work with the probability density function, defined as follows.

Definition: Probability density function
Suppose X is a random variable with distribution function F_X, and suppose that there is a function $f_X : \mathbb{R} \to \mathbb{R}$ s.t. $\forall a < b$,

$$\mathbb{P}(a < X \leqslant b) = F(b) - F(a) = \int_a^b f_X(x)dx \qquad (*)$$

Thenf_X is called the (probability) density function (pdf) of X, and X is said to be an (absolutely) continuous random variable.

Remarks
Suppose that X is a r.v. with probability density function f_X. Then:

1. Letting $a \to -\infty$ in equation $(*)$ above, we obtain

$$\boxed{F(b) = \int_{-\infty}^b f_X(x)dx \qquad \forall b \in \mathbb{R}}$$

   If X has a density function f_X, then we can find the cdf F_X for X by *integrating* f_X

2. Since F_X is obtained by integrating f_X, then conversely, f_X may be obtained by differentiating F_X. That is, the probability density function $f_X(x)$ is given by

$$\boxed{f_X(x) = \frac{dF_X}{dx} \quad \text{for all } x \text{ such that the derivative exists.}}$$

3. Recall that a function $F : [a, b] \to \mathbb{R}$ is absolutely continuous on $[a, b]$ if there exists an integrable function $g : [a, b] \to \mathbb{R}$ such that for all $x \in [a, b]$

   $F(x) = F(a) + \int_a^x g(s)ds.$

Therefore an (absolutely) continuous random variable is a random variable whose cdf is absolutely continuous. There are examples of functions which are continuous but not absolutely continuous, a classic example of this is the Cantor function (see Section 7 of Grimmett, G., & Welsh, D. (1990). Probability: An Introduction for details on the Cantor function).

Example 3.1 continued. Suppose random variable X represents a number picked uniformly at random from the interval $[0, 1]$. The cumulative distribution function of X is

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leqslant x < 1 \\ 1 & x \geqslant 1 \end{cases}$$

To obtain the probability density function $f_X(x)$, differentiate:

$$f_X(x) = \frac{dF_X}{dx} = \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 1 \\ 0 & x < 1 \end{cases}$$

Here is a sketch of the probability density function f_X:

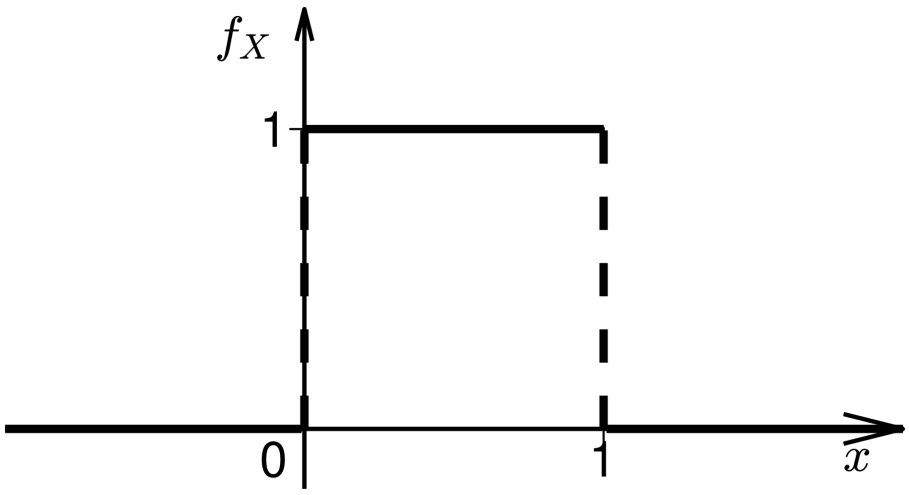


Figure 3.2: Sketch of $f_X(x)$

Notice that for $x \in \mathrm{Im}X$, the value of $f_X(x)$ is constant, corresponding to probability being uniformly distributed across the interval $[0, 1]$.

Remarks
Suppose that X is a random variable with probability density function f_X. Then:

1. From the definition of differentiation, for all x such that the derivative exists,

$$f_X(x) = \frac{dF_X}{dx} = \lim_{\delta x \downarrow 0} \frac{\mathbb{P}(x \leqslant X < x + \delta x)}{\delta x}$$

We can see from this that $f_X(x) \geq 0$, and that the value of $f_X(x)$ indicates, in some sense, how likely the random variable X is to be close to the value x, but that $f_X(x)$ is not a probability. To obtain probabilities, we need to integrate $f_X(x)$.

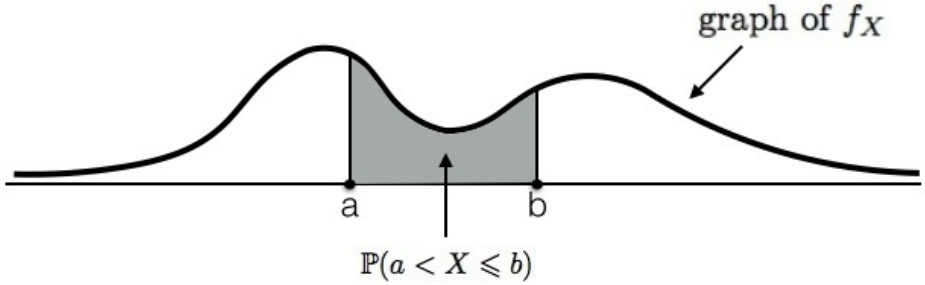
2. $\forall b \in \mathbb{R}$ we have, using continuity of probability measures,

$$\mathbb{P}(X = b) = \lim_{a \uparrow b} \mathbb{P}(a < X \leqslant b) = \lim_{a \uparrow b} \int_a^b f_X(x)\,\mathrm{d}x = 0$$

Since $\mathbb{P}(X = b) = 0$ for all $b \in \mathbb{R}$, this means that the cumulative distribution function F_X does not have jumps. That is, if a random variable X has a density function, then the cumulative distribution function F_X is a *continuous* function. This is why we say that X is an (absolutely) *continuous* random variable when it has a probability density f_X.

3. For any $a < b$,

$$\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a \leqslant X < b) = \mathbb{P}(a < X < b) = \mathbb{P}(a < X \leqslant b)$$
$$= \int_a^b f_X(x)\,\mathrm{d}x = (\text{area under } f_X \text{ between } a \text{ and } b)$$



4. Since $F(b) = \int_{-\infty}^{b} f_X(x)dx$ and $\lim_{b\to\infty} F(b) = 1$ we get:

$$1 = \lim_{b\to\infty} F(b) = \lim_{b\to\infty} \int_{-\infty}^{b} f_X(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = (\text{total area under graph of } f_X)$$

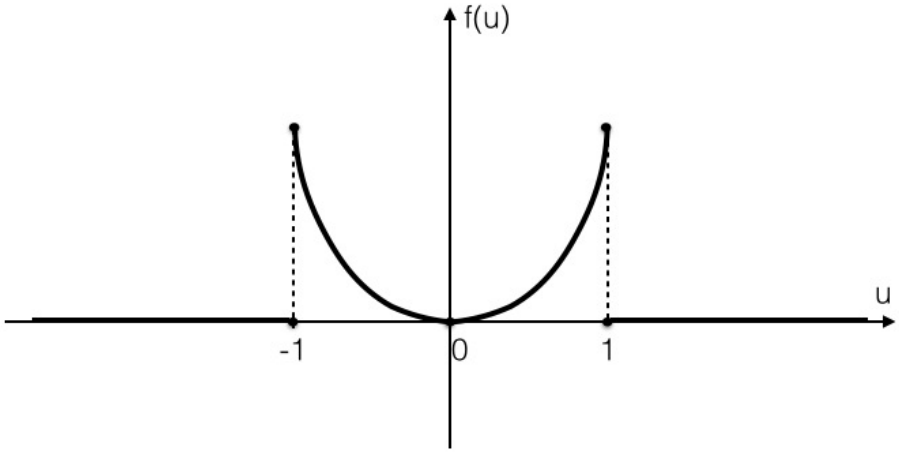
Properties of probability density functions
If X is a r.v. with probability density function f_X, then:

- $f_X(x) \geqslant 0 \quad \forall x \in \mathbb{R}$
- $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$

Conversely, any function satisfying the above properties is a valid probability density function for some (absolutely) continuous random variable.
(Compare with properties of probability mass functions.)

Example 3.2.

Suppose $f(u) = \begin{cases} \frac{3u^2}{2} & -1 \leqslant u \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$



Is f a density function? If so, what is the corresponding distribution function F?

Answer:

- $f(u) \geqslant 0 \ \ \forall u$, so f is non-negative.
- $\int_{-\infty}^{\infty} f(u)\,\mathrm{d}u = \int_{-1}^{1} \frac{3u^2}{2}\,\mathrm{d}u = \left[\frac{u^3}{2}\right]_{-1}^{1} = 1$ (area under the graph)

therefore f *is* a density function.
If X is a random variable with probability density function f, then $\mathrm{Im}X = [-1, 1]$ because $f(u) > 0$ on the interval $[-1, 1]$, and the cumulative distribution function for X is given by $F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u$ for any $x \in \mathbb{R}$.
To determine the cumulative distribution function F_X, we have to consider 3 cases:

$$x < -1: \quad F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \int_{-\infty}^{x} 0 \ \,\mathrm{d}u = 0$$

$$-1 \leqslant x < 1: \quad F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \int_{-1}^{x} \frac{3u^2}{2}\,\mathrm{d}u = \frac{x^3+1}{2}$$

$$1 \leqslant x: \quad F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \int_{-1}^{1} \frac{3u^2}{2}\,\mathrm{d}u = 1$$

So we have

$$F_X(x) = \begin{cases} 0 & x < -1 \\ \frac{x^3+1}{2} & -1 \leqslant x < 1 \\ 1 & 1 \leqslant x \end{cases}$$

Remarks:

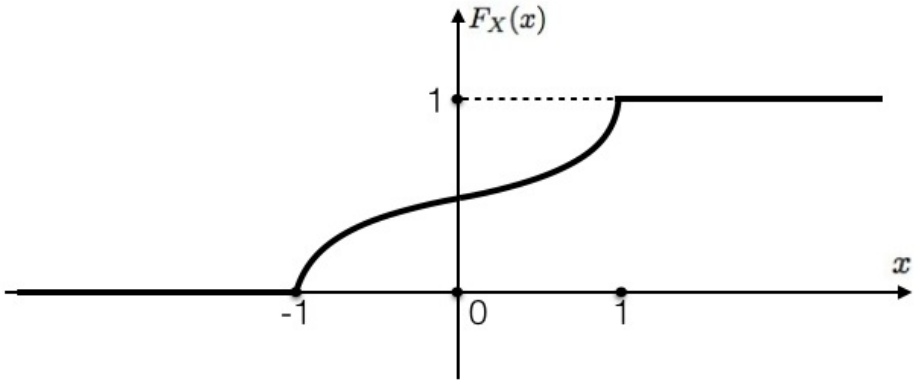


Figure 3.3: Sketch of $F_X(x)$

1. By sketching the graph of the cumulative distribution function F_X we see that it is continuous although the density $f(u)$ is not a continuous function. In general, a probability density function does not have to be continuous. Not all cumulative distribution functions correspond to a probability density function (e.g. discrete random variables don't have a probability density function), but any cumulative distribution function that corresponds to a probability density function is a continuous function.
2. The graph of a density f_X for a random variable X is often easier to interpret than the graph of its cumulative distribution function F_X.

Example 3.2 continued. Suppose X is a random variable with probability density function

$$f_X(u) = \begin{cases} \frac{3u^2}{2} & -1 \leqslant u \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

then we can *see* from the graph of f_X that (for example)

$$\mathbb{P}(|X| \geqslant 1/2) > \mathbb{P}(|X| < 1/2)$$

by looking at the *area* under the graph of f_X.

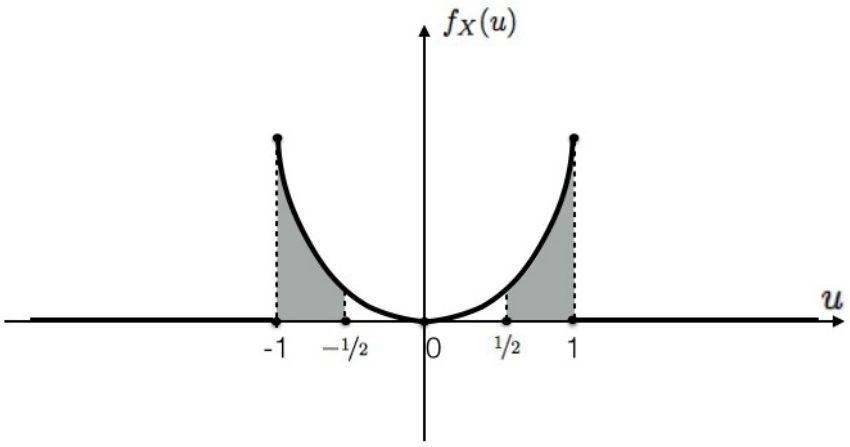


Figure 3.4: Area of shaded region $= \mathbb{P}(|X| \geqslant \frac{1}{2})$

Key point: It can be useful to sketch the probability density function f_X for the r.v. X in order to get an impression about which values X is more or less likely to take.

Question: Suppose X is a random variable with distribution function F_X. How can we tell whether X has a probability density function f_X, and if it does have a probability density function f_X, how can we find f_X?

Answer: If F_X is continuous and differentiable 'almost everywhere' then X has a probability density function which can be found by differentiating F_X

Remark 3.1 (Beyond the course material). For our purposes if F_Y is differentiable at all but a countable set of points then it is differentiable 'almost everywhere'. Although the meaning of 'almost everywhere' is more general than this and goes beyond what we need in this course. A function has a given property 'almost everywhere' is the set of points S where the property fails is sufficiently small. By sufficiently small we mean that if we choose any interval $[a, b]$ with $a < b$ and draw a point U uniformly at random on that interval then the probability $U \in S$ is zero.

Example 3.3. Suppose Y is a random variable with cumulative distribution function F_Y given by

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ \frac{y^2}{2} & 0 \leqslant y \leqslant 1 \\ 1 - \frac{(2-y)^2}{2} & 1 < y \leqslant 2 \\ 1 & 2 < y \end{cases}$$

Is F_Y *continuous*?
To check this, we note that there are 4 'pieces' in the definition of F_Y.
Each piece is continuous, so it is enough to check that the pieces join up (i.e. take the same values) at their boundary points $y = 0$, $y = 1$, and $y = 2$.

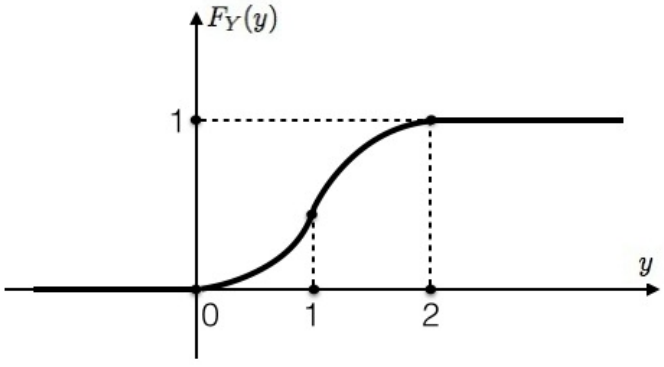


Figure 3.5: Sketch of $F_Y(y)$

From the sketch, we see that F_Y *is* continuous.

Is F_Y differentiable 'almost everywhere'?

Again, since there are 4 pieces in the definition of F_Y and each piece is differentiable, F_Y is differentiable 'almost everywhere' (we say 'almost everywhere' because sometimes there can be

a problem with differentiating at the boundary points).

Therefore Y has a density function $f_Y(y)$ which is given by:

$$f_Y(y) = \begin{cases} 0 & y < 0 \\ y & 0 < y < 1 \\ 2 - y & 1 < y < 2 \\ 0 & 2 < y \end{cases}$$

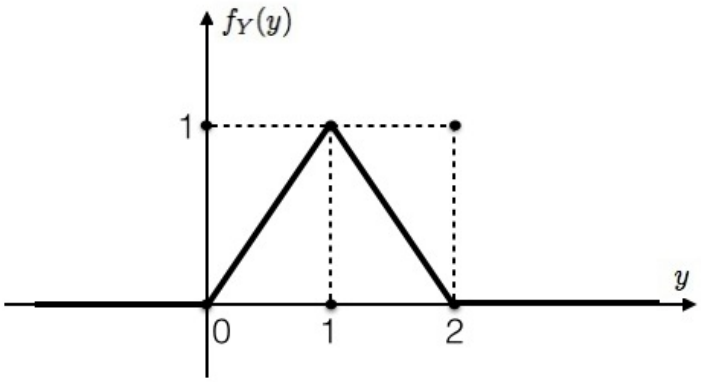


Figure 3.6: Sketch of $f_Y(y)$

3.3 Common examples of continuous random variables

3.3.1 Uniform distribution

Uniform random variables are used to model picking a point at random from the interval $[a, b]$ such that each point in the interval is 'equally likely' to be chosen. If X is a uniform random variable on the interval $[a, b]$, the value taken by X represents the randomly chosen point in the interval $[a, b]$ and $\mathrm{Im}X = [a, b]$.

If X is uniform on $[a, b]$, we write $X \sim \mathrm{U}(a, b)$.

Since X is equally likely to take any value between a and b, we have that for $a \leqslant c \leqslant d \leqslant b$:

$$\mathbb{P}(c \leqslant X \leqslant d) = \frac{\text{length of } [c, d]}{\text{length of } [a, b]} = \frac{d - c}{b - a}$$

(i.e. probability is *proportional* to length of interval).

We can determine the cumulative distribution function F_X and the density f_X for X as follows:

First notice that:

1. For $x < a$, $F_X(x) = \mathbb{P}(X \leqslant x) = 0$ (since X cannot be smaller than a).
2. For $x \geqslant b$, $F_X(x) = \mathbb{P}(X \leqslant x) = 1$ (since for any $x \geqslant b$, X is definitely no greater than x).

Then for $a \leqslant x < b$,

$$F_X(x) = \mathbb{P}(X \leqslant x) = \mathbb{P}(a \leqslant X \leqslant x) \quad = \quad \frac{x - a}{b - a}.$$

So for $X \sim \mathrm{U}(a,b)$, the cumulative distribution function is

$$\boxed{F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leqslant x < b \\ 1 & x \geqslant b \end{cases}}$$

To find the probability density function for $X \sim \mathrm{U}(a,b)$, differentiate the cumulative distribution function:

$$\boxed{f_X(x) = \frac{dF_X}{dx} = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}}$$

Notice that the density f_X is constant on (a,b).

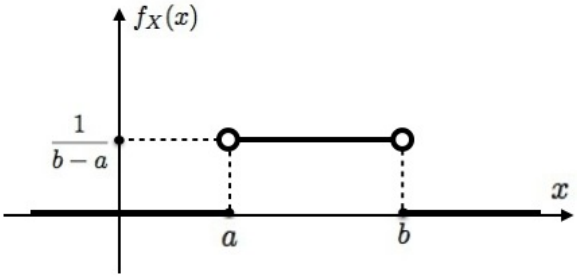


Figure 3.7: Sketch of $f_X(x)$

Properties of Uniform random variables

For $X \sim \mathrm{U}(a,b)$:

$$\mathrm{Im}X = [a,b], \qquad F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leqslant x < b, \\ 1, & b \leqslant x, \end{cases} \qquad f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

Suppose $X \sim \mathrm{U}(a,b)$ and $Y = cX + d$ where $c > 0$. Then $Y \sim \mathrm{U}(ca+d, cb+d)$.
That is, a linear function of a uniform variable is *also* uniform.

Proof: With $X \sim \mathrm{U}(a,b)$ and $Y = cX + d$ where $c > 0$, since $\mathrm{Im}X = [a,b]$ then $\mathrm{Im}Y = [ca+d, cb+d]$. Hence the cumulative distribution function of Y is given by $F_Y(y) = 0$ for $y < ca + d$, and $F_Y(y) = 1$ for $y \geqslant cb + d$. For $ca + d \leqslant y < cb + d$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leqslant y) \\ &= \mathbb{P}(cX + d \leqslant y) \\ &= \mathbb{P}\left(X \leqslant \frac{y-d}{c}\right) \\ &= F_X\left(\frac{y-d}{c}\right) \\ &= \frac{\frac{y-d}{c} - a}{b-a} \\ &= \frac{y-d-ca}{cb-ca} \\ &= \frac{y-(ca+d)}{(cb+d)-(ca+d)} \end{aligned}$$

So Y has cumulative distribution function

$$F_Y(y) = \begin{cases} 0 & y < ca + d \\ \frac{y-(ca+d)}{(cb+d)-(ca+d)} & ca + d \leqslant y < cb + d \\ 1 & y \geqslant cb + d \end{cases}$$

which is the cumulative distribution function of the $\mathrm{U}(ca + d, cb + d)$ distribution.

3.3.2 Exponential distribution

Exponential random variables arise in many areas of statistics and actuarial science; in particular, they can be used to model random waiting times. We say that X has an exponential distribution with parameter $\lambda > 0$, and write $X \sim \mathrm{Exp}(\lambda)$, if and only if its density is given by

$$\boxed{f_X(x) = \begin{cases} 0 & x < 0 \\ \lambda e^{-\lambda x} & x \geqslant 0 \end{cases}}$$

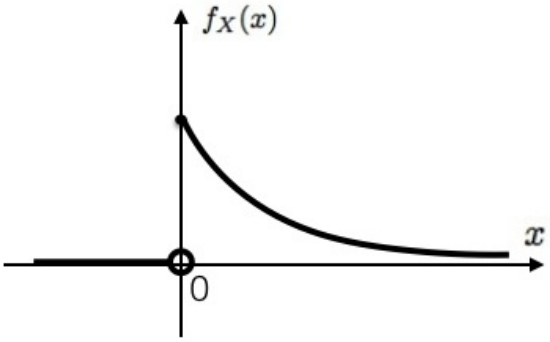


Figure 3.8: Sketch of the density $f_X(x)$

We note that $\mathrm{Im}(X) = [0, \infty)$ and that the cumulative distribution function F_X for $X \sim \mathrm{Exp}(\lambda)$ is given by

$$\boxed{F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geqslant 0 \end{cases}}$$

An Important Property of Exponential Random Variables
Suppose $X \sim \mathrm{Exp}(\lambda)$ and $c > 0$. Let $Y = cX$, then $Y \sim \mathrm{Exp}(\lambda/c)$. That is, Y is *also* an exponential random variable.

Proof: Since $X \sim \mathrm{Exp}(\lambda)$ we have $\mathrm{Im}X = [0, \infty)$, so with $c > 0$ and $Y = cX$, then $\mathrm{Im}Y = [0, \infty)$. So for $y < 0$ we have $F_Y(y) = 0$. To find $F_Y(y)$ for $y \geqslant 0$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leqslant y) \\ &= \mathbb{P}(cX \leqslant y) \\ &= \mathbb{P}(X \leqslant y/c) \\ &= F_X(y/c) \\ &= 1 - e^{-\lambda(y/c)} \end{aligned}$$

So the distribution function of Y is

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 1 - e^{-(\lambda/c)y} & y \geq 0 \end{cases}$$

This is the distribution function of the Exp(λ/c) distribution, so $Y \sim \text{Exp}(\lambda/c)$

Remark 3.2. Exponential random variables arise as models for the length of time (starting from now) until a 'completely random' event occurs. For example, exponential random variables are often used to model:

- the time until an earthquake occurs
- the time until next telephone call is received
- the time until a particle is emitted from a radioactive substance

Example 3.4. The lifetime X (in thousands of hours) of a particular electrical component may be modelled as an exponential random variable with mean 2000 hours. Calculate the probability that the component's lifetime is less than 1000 hours.

Answer: In units of thousands of hours, we are told that $\mathbb{E}(X) = 2$. For $X \sim \text{Exp}(\lambda)$ we know that $\mathbb{E}(X) = 1/\lambda$, so in this example we have $\lambda = 1/2$, and X has distribution function

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-(1/2)x} & x \geq 0 \end{cases}$$

The required probability is

$$\begin{aligned} \mathbb{P}(X < 1) &= \mathbb{P}(X \leqslant 1) \qquad \text{since } X \text{ is a } \textit{continuous} \text{ random variable} \\ &= F_X(1) \\ &= 1 - e^{-1/2} \approx 0.393 \end{aligned}$$

3.3.3 The Gamma distribution

Exponential random variables (see Section 3.3.2) are a special case of a more general class of random variables called gamma random variables. A random variable X has a gamma distribution with parameters $\alpha > 0$ and $\lambda > 0$, written $X \sim \text{Gamma}(\alpha, \lambda)$, if and only if its density is given by

$$\boxed{f_X(x) = \begin{cases} 0 & x < 0 \\ \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & x \geqslant 0 \end{cases}}$$

where the Gamma function $\Gamma(\alpha)$ is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} \mathrm{d}x \text{ for } \alpha > 0$$

Exercise: Show that the function $f_X(x)$ given above is a valid probability density function.

The Gamma function is related to the factorial function as follows. Suppose that $\alpha > 1$. Then applying integration by parts in the definition of the Gamma function we can show that

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$$

Now, since $\Gamma(1) = 1$ (check this!), it is straightforward to deduce (by induction) that if α is a positive integer then

$$\Gamma(\alpha) = (\alpha - 1)!$$

The Gamma function can be considered to be an extension of the factorial function to non-integer values.

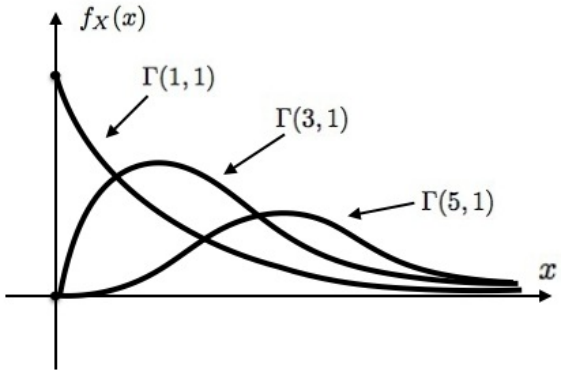
The relationship $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$ is very useful in many situations as it enables to do many important calculations without explicitly evaluating $\Gamma(\alpha)$. In particular, it follows that, for $\alpha > 0, k \in \mathbb{Z}, k > 0$,

$$\frac{\Gamma(\alpha + k)}{\Gamma(\alpha)} = \alpha(\alpha + 1) \times \cdots \times \ldots (\alpha + k - 1)$$

Remarks

1. Gamma random variables provide a family of models for positive quantities such as survival times, etc

2. The parameter α is called the shape parameter and the parameter λ is called the reciprocal scale parameter or the rate parameter.



3. The exponential distribution is a special case of the gamma distribution. In particular, for any $\lambda > 0$, if $X \sim \text{Exp}(\lambda)$, then X has a Gamma $(1, \lambda)$ distribution.

Chapter 4

Expectations of Random Variables

4.1 Expected value

In this section, we introduce the concept of the *expected value* (or *expectation*) of a random variable X, denoted $\mathbb{E}(X)$. Informally, this represents the 'average value' of the random variable X. You may have seen this defined for discrete random variables before.

Definition 4.1. For a discrete random variable X, the *expected value* of X is

$$\boxed{\mathbb{E}(X) = \sum_{x \in \mathrm{Im}X} x \underbrace{p_X(x)}_{\text{pmf of } X} = \sum_{x \in \mathrm{Im}X} x\mathbb{P}(X = x)}$$

For a continuous random variable X, the *expected value* of X is

$$\boxed{\mathbb{E}(X) = \int_{-\infty}^{\infty} x \underbrace{f_X(x)}_{\text{pdf of } X} \mathrm{d}x = \int_{\mathrm{Im}X} x f_X(x) \mathrm{d}x}$$

Before we explore the mathematical definition let us consider different interpretations of what the expected value should represent.

Interpretations

Long run interpretation:
Suppose we run a very large number of simulations of X. We record the outcomes as $x_1, x_2, x_3, \ldots, x_n$. The proportion of times we see the value x we call $\hat{p}(x)$ and this should be approximately $p_X(x)$ given a large enough set of data. Therefore the average outcome, $\overline{x}$, is

$$\overline{x} = \sum_{i=1}^{n} x_i = \sum_{x \in \mathrm{Im}(X)} x\hat{p}(x) \approx \sum_{x \in \mathrm{Im}(X)} x p_X(x).$$

This will be justified using the law of large numbers in Chapter 8 at the end of the course.

Physical interpretation:
Suppose we have a rod of negligible weight with masses of size w_i placed at points x_i, for $i = 1, ..., n$. Then the centre of gravity of the rod is located at

$$\frac{\sum_{i=1}^{n} x_i w_i}{\sum_{i=1}^{n} w_i} = \sum_{i=1}^{n} x_i p_i.$$

Here $p_i = w_i / \sum w_i$ is the proportion of the mass located at x_i. Now if X is a random variable which takes the value x_i with probability p_i, then we can interpret the expected value of X as the centre of gravity of the rod.

Fair Price interpretation:
Suppose you pay a fixed amount £b to place a bet and receive a return of £X which is random. If you place n independent bets then your return is

$$S = X_1 + \ldots + X_n \approx \sum_{x \in \mathrm{Im}(X)} nx p_X(x) = n\mathbb{E}(X). \tag{4.1}$$

Then your average net return per bet is £$(\mathbb{E}(X) - b)$. So $\mathbb{E}(X)$ is the price of fair bet for a return of £X. This price makes wins and loses cancel out over the long run.

Remark 4.1.
- Compare the formula of expectation with the sample mean of frequency data, $\bar{x} = \sum_i \left(\frac{f_i}{n}\right) x_i$, where f_i is the frequency of observation x_i. The relative frequency $\frac{f_i}{n}$ for observation x_i is replaced by the probability $p_X(x)$ for $x \in \mathrm{Im}X$.
- $\mathbb{E}(X)$ is sometimes referred to as the *population mean* or *theoretical mean* in order to distinguish it from the *sample mean*.
- Observe the similarities in the definition of expectation for discrete and continuous random variables, with the main differences being the sum or integral and using the probability mass function or probability density function. Many of the comments we make about expectations will apply to both discrete and continuous random variables.

Example 2.5 continued. $X = \#$ heads in 3 flips of a fair coin. $\mathrm{Im}X = \{0, 1, 2, 3\}$. Probability mass function is

$$p_X(x) = \begin{cases} 1/8 & x = 0 \\ 3/8 & x = 1 \\ 3/8 & x = 2 \\ 1/8 & x = 3 \end{cases}$$

Expected value is

$$\mathbb{E}(X) = \sum_{x \in \mathrm{Im}X} x\, p_X(x) = 0 \times 1/8 + 1 \times 3/8 + 2 \times 3/8 + 3 \times 1/8 = 3/2$$

That is, when a fair coin is flipped 3 times, the expected number of Heads is 3/2 (unsurprisingly).

Example 4.1. Now suppose you play the following game: You pay £1 to select a number ω uniformly at random from $[0, 1]$. If the outcome $\omega > 0.5$, then you win £2; if the outcome $\omega \leq 0.5$, then you win nothing. Let $Z(\omega)$ be your net gain in this game.
The random variable Z can only take two values, i.e. $\mathrm{Im}(Z) = \{-1, 1\}$, so Z is a discrete random variable.
Probability mass function is

$$p_Z(z) = \begin{cases} 1/2 & z = -1 \\ 1/2 & z = 1 \end{cases}$$

Expected value is

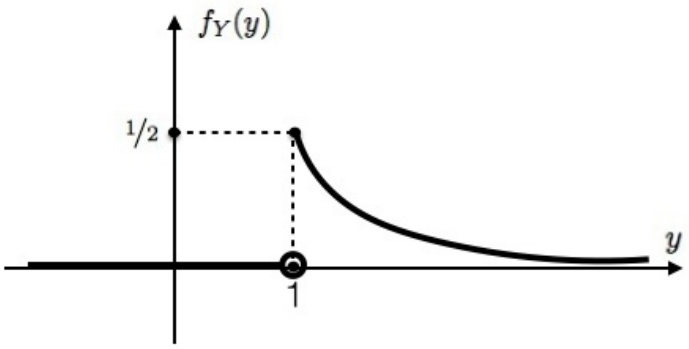
$$\mathbb{E}(X) = \sum_{x \in \mathrm{Im}X} x\, p_X(x) = -1 \times 1/2 + 1 \times 1/2 = 0$$

The expected net gain is zero, so we can say that the game is *fair*.

Example 4.2. Suppose Y is a random variable with density function given by

$$f_Y(y) = \begin{cases} \frac{2}{y^3} & \text{for } y \geqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

The image of Y is $\mathrm{Im}Y = [1, \infty)$.



Then

$$\begin{aligned}\mathbb{E}(Y) &= \int_1^\infty y\, f_Y(y)\mathrm{d}y \\ &= \int_1^\infty y\frac{2}{y^3}\,\mathrm{d}y = \int_1^\infty \frac{2}{y^2}\,\mathrm{d}y = \left[\frac{-2}{y}\right]_1^\infty = \left(0 - \left(\frac{-2}{1}\right)\right) = 2.\end{aligned}$$

Warning: The expectation need not always exist as is shown in the following example.

Example 4.3 (Cauchy Distribution)**.** Let X be a continuous random variable with probability density function

$$f_X(x) = \frac{1}{\pi(1+x^2)}.$$

One can show that this is a valid probability density function however if the expectation exists it would be given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)}\,\mathrm{d}x.$$

Note that $\frac{x}{\pi(1+x^2)} \approx \frac{1}{x\pi}$ for x large and positive, and hence this function is not integrable.

Expected value of $g(X)$

Suppose X is a discrete random variable with probability mass function p_X, and that $Y = g(X)$ for some function $g : \mathbb{R} \to \mathbb{R}$. Then Y is itself a random variable, with its own probability mass function p_Y. If we want to compute the expected value of Y, then one way is to first find the probability mass function $p_Y(y)$, then use the definition

$$\mathbb{E}(Y) \;=\; \sum_{y\in \mathrm{Im}Y} y\, p_Y(y)$$

But there is a more direct way, without needing to explicitly find p_Y. If X is a discrete random variable with probability mass function p_X, and $Y = g(X)$ for some function $g : \mathbb{R} \to \mathbb{R}$, then

$$\boxed{\mathbb{E}(Y) = \mathbb{E}(g(X)) = \sum_{x\in \mathrm{Im}X} g(x)\, p_X(x).}$$

Similarly, if X is a continuous random variable with probability density function f_X, and $Y = g(X)$ for some function $g : \mathbb{R} \to \mathbb{R}$, then

$$\boxed{\mathbb{E}(Y) = \mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)\, f_X(x)\,\mathrm{d}x.}$$

Example 4.4. Fix some event A and let

$$g(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Then

$$\mathbb{E}(g(X)) = \mathbb{P}(X \in A).$$

4.2 Properties of expected value

Proposition 4.1. *Let X be a random variable with finite expectation, and let $a, b \in \mathbb{R}$ be constant. Then the following properties hold:*

1. *The expected value of a constant: $\mathbb{E}(a) = a$.*
2. *Expectation is linear: $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$.*
3. *If X is non-negative (i.e. $\mathbb{P}(X \geq 0) = 1$), then $\mathbb{E}(X) \geq 0$.*

Proof We will establish these properties for continuous random variables, the proof for discrete random variables is analogous and in fact these properties hold for any random variables with finite expectation.
Let X be a continuous random variable with probability density function f_X be such that $\mathbb{E}[X] < \infty$, and let $a, b \in \mathbb{R}$ be constants.

1.

$$\mathbb{E}(a) = a \times \int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = a \times 1 = a.$$

2.

$$\begin{aligned} \mathbb{E}(aX + b) &= \int_{-\infty}^{\infty} (ax + b) f_X(x)\mathrm{d}x \\ &= a\int_{-\infty}^{\infty} x f_X(x)\mathrm{d}x + b\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x \\ &= a\mathbb{E}(X) + b \times 1 = a\mathbb{E}(X) + b \end{aligned}$$

3. Since X is non-negative, we have $f_X(x) = 0$ for all $x < 0$ therefore

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\mathrm{d}x = \int_{0}^{\infty} x f_X(x)\mathrm{d}x.$$

From the definition of a probability density function we have $f_X(x) \geq 0$ for all $x \in \mathbb{R}$. Hence $x f_X(x) \geq 0$ for all $x \geq 0$ and using that the integral of a non-negative function is non-negative we have

$$\mathbb{E}[X] = \int_{0}^{\infty} x f_X(x)\mathrm{d}x \geq 0.$$

□

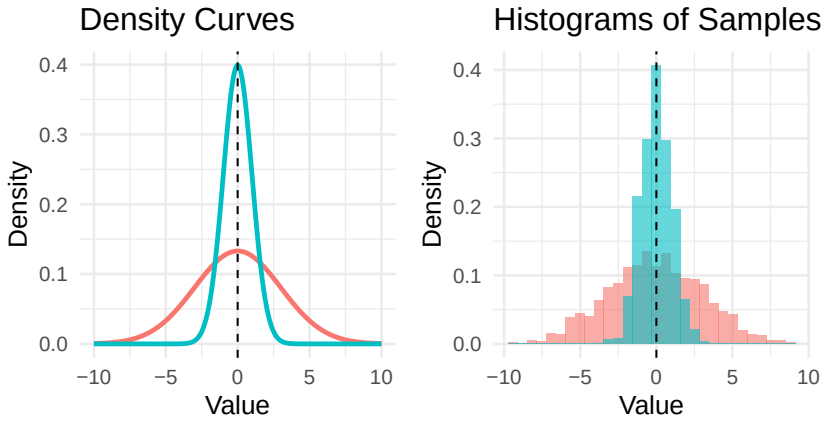


Figure 4.1: The left panel shows the theoretical density curves for two normal distributions with the same mean (indicated by the dashed black line) but different variances: the blue curve corresponds to the distribution with low variance ($\sigma^2 = 1$), and the orange curve to the distribution with high variance ($\sigma^2 = 9$). The right panel displays histograms of samples drawn from these distributions, illustrating how increased variance leads to greater spread in observed values.

4.3 Variance of a random variable

Once we understand expectation as a measure of the "average" or "central" value of a random variable, it's natural to ask how much the values of that variable tend to deviate from this average. This is where variance comes in. Variance quantifies the spread or dispersion of a random variable's values around its expectation. In other words, it tells us how "spread out" the outcomes are, giving insight into the variability or uncertainty inherent in the random process.

Definition 4.2. The variance of a random variable X is the expected value of its *squared deviation* from its expected value $\mathbb{E}(X)$:

$$\boxed{\mathbb{V}\text{ar}(X) = \mathbb{E}\left((X - \mathbb{E}(X))^2\right) = \mathbb{E}\left((X-\mu)^2\right) \quad \text{where } \mu = \mathbb{E}(X)}$$

Notation: We sometimes write σ_X^2 for $\mathbb{V}\text{ar}(X)$

Remark 4.2. To understand the form of the variance formula, recall that variance is meant to measure how far values of a random variable deviate from the mean, i.e. the spread of $X - \mathbb{E}[X]$. Simply averaging the deviations from the mean would always give zero: Since positive and negative deviations cancel out, that is

$$\mathbb{E}\left[X - \mathbb{E}[X]\right] = 0.$$

To avoid this, we square the deviations, ensuring all contributions are non-negative and that larger deviations are penalized more heavily. The variance is then defined as the expected value of these squared deviations. This captures the average squared distance from the mean, providing a natural measure of spread in the distribution.

While variance gives us a measure of spread, it does so in squared units. For example, if X is measured in metres, then $\mathbb{V}\text{ar}(X)$ is in square metres.

This can make interpretation less intuitive. To bring the measure of spread back to the original units of the data, we take the square root of the variance.

This gives us the standard deviation:

$$\boxed{\sigma_X = \sqrt{\mathbb{V}\text{ar}(X)}}$$

The standard deviation is often more interpretable in practice, as it tells us, on average, how far values tend to deviate from the mean — in the same units as the data itself.

Properties of $\mathbb{V}\text{ar}(X)$

Proposition 4.2. *Let X be a random variable with finite second moment, i.e. $\mathbb{E}[X^2] < \infty$, and let $a, b \in \mathbb{R}$ be constant. Then the following properties hold:*

1. *The variance is non-negative:* $\mathbb{V}\text{ar}(X) \geq 0$
2. *The variance of constants is zero:* $\mathbb{V}\text{ar}(a) = 0$
3. *Variance is shift independent but scales quadratically:*

$$\mathbb{V}\text{ar}(aX + b) = a^2\mathbb{V}\text{ar}(X)$$

Special cases of Property 3*:*

$\mathbb{V}\text{ar}(aX) = a^2\mathbb{V}\text{ar}(X)$
$\mathbb{V}\text{ar}(X + b) = \mathbb{V}\text{ar}(X)$

4. *A useful formula for computations:*

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Proof

1. Since $(X - \mu)^2 \geq 0$, by Property 3 of expected values, we have

$$\mathbb{V}\text{ar}(X) = \mathbb{E}((X - \mu)^2) \geq 0$$

2. Since $\mathbb{E}(a) = a$ all the deviations are zero, i.e. $a - \mathbb{E}[a] = 0$.
Hence the squared deviations are zero and using that $\mathbb{E}[0] = 0$ we have $\mathbb{V}\text{ar}(a) = \mathbb{E}((a - \mathbb{E}(a))^2)) = \mathbb{E}(0) = 0$.

3.

$$\begin{aligned}
\mathbb{V}\text{ar}(aX + b) &= \mathbb{E}((aX + b - \mathbb{E}(aX + b))^2) \\
&= \mathbb{E}((aX + b - a\mathbb{E}(X) - b)^2) \qquad \text{(by the linearity of expected value)} \\
&= \mathbb{E}(a^2(X - \mathbb{E}(X))^2) \qquad \text{(by algebra)} \\
&= a^2\mathbb{E}((X - \mathbb{E}(X))^2) \qquad \text{(by the linearity of expected value)} \\
&= a^2\mathbb{V}\text{ar}(X) \qquad \text{(by the definition of } \mathbb{V}\text{ar}(X)
\end{aligned}$$

4. Let $\mu = \mathbb{E}[X]$ then

$$\begin{aligned}
\mathbb{V}\text{ar}(X) = \mathbb{E}((X - \mu)^2) &= \mathbb{E}(X^2 - 2\mu X + \mu^2) \\
&= \mathbb{E}(X^2) - 2\mu\mathbb{E}(X) + \mu^2 \\
\mathbb{V}\text{ar}(X) &= \mathbb{E}(X^2) - \mu^2 \quad (\text{since } \mathbb{E}(X) = \mu)
\end{aligned}$$

□

In practise it is often easier to compute the variance using Property 4 in Proposition 4.2 rather than calculating the variance directly.

Example 2.5 continued. $X = \#$ heads in 3 flips of a fair coin. $\text{Im}X = \{0, 1, 2, 3\}$. Probability mass function is

$$p_X(x) = \begin{cases} 1/8 & x = 0 \\ 3/8 & x = 1 \\ 3/8 & x = 2 \\ 1/8 & x = 3 \end{cases}$$

Find the variance for X, $\mathbb{V}\text{ar}(X)$.

Solution:
From the earlier calculations, we found $\mathbb{E}(X) = \frac{3}{2}$.

$$\mathbb{E}(X^2) = \sum_{x \in \text{Im}(X)} x^2 p_X(x) = 0^2(1/8) + 1^2(3/8) + 2^2(3/8) + 3^2(1/8) = 3.$$

Therefore

$$\begin{aligned} \mathbb{V}\text{ar}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= 3 - \left(\frac{3}{2}\right)^2 \\ &= \frac{3}{4}. \end{aligned}$$

4.4 Expectation and Variance of important examples

4.4.1 Bernoulli Random Variable

Recall $X \sim \text{Ber}(p)$ if and only if X is a discrete random variable with $\text{Im}(X) = \{0, 1\}$ and $p = \mathbb{P}(X = 1)$. Then it follows that

$$\mathbb{E}(X) = 0 \times \mathbb{P}(X = 0) + 1 \times \mathbb{P}(X = 1) = p$$

Also,

$$\mathbb{E}(X^2) = 0^2 \times \mathbb{P}(X = 0) + 1^2 \times \mathbb{P}(X = 1) = p,$$

and so

$$\mathbb{V}\text{ar}(X) = \mathbb{E}(X^2) - \big(\mathbb{E}(X)\big)^2 = p - p^2 = p(1 - p)$$

4.4.2 Binomial Random Variable

Recall $X \sim \text{Bin}(n, p)$ if and only if X is a discrete random variable with $\text{Im}(X) = \{0, 1, \ldots, n\}$ and

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k \in \{0, 1, \ldots, n\}.$$

Given the probability mass function $p_X(x)$, we can find the expected value of $X \sim \text{Bin}(n,p)$:

$$\begin{aligned}
\mathbb{E}(X) = \sum_{k=0}^{n} k p_X(k) &= \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=0}^{n} k \left(\frac{n!}{k!\,(n-k)!} \right) p^k (1-p)^{n-k} && \text{by definition of } \binom{n}{k} \\
&= \sum_{k=1}^{n} \frac{n!}{(k-1)!\,(n-k)!}\, p^k (1-p)^{n-k} && \text{since } k! = k \times (k-1)! \\
&= np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!\,(n-k)!}\, p^{k-1} (1-p)^{n-k} && \text{since } n! = n \times (n-1)! \\
&= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{(n-1)-j} && \text{changing the variable } j = k-1
\end{aligned}$$

Notice that the summation in the final line above is the sum of all terms in the probability mass function of the $\text{Bin}(n-1,p)$ distribution, so must sum to 1, and so we obtain

$$\mathbb{E}(X) = np$$

That is, if we carry out n trials, each with success probability p, the expected number of successes is np.

We will show later that for $X \sim \text{Bin}(n,p)$,

$$\mathbb{V}\text{ar}(X) = np(1-p).$$

4.4.3 Geometric Random Variable

Recall $X \sim \text{Geo}\,(p)$ if and only if X is a discrete random variable with $\text{Im}(X) = \{1,2,3,\ldots\}$ and

$p_X(k) = p(1-p)^{k-1}, k \in \{1,2,3,\ldots\}$.

Now find the expectation of $X \sim \text{Geo}\,(p)$:

$$\mathbb{E}(X) \quad = \quad \sum_{x \in \text{Im}X} x p_Y(x) = \sum_{x=1}^{\infty} x(1-p)^{x-1} p = p \sum_{x=1}^{\infty} x(1-p)^{x-1}$$

To evaluate this infinite sum we use a geometric series. Recall that the sum of a geometric series is given by

$$\sum_{x=0}^{\infty} a^x = \frac{1}{1-a}, \qquad \text{for any } a \text{ such that } |a| < 1.$$

This is not quite of the form we need, however if we differentiate both sides with respect to a we have

$$\sum_{x=1}^{\infty} x a^{x-1} = \frac{1}{(1-a)^2}, \qquad \text{for any } a \text{ such that } |a| < 1.$$

Note here the term corresponding to $x = 0$ was constant therefore once we differentiate it vanishes. Then taking $a = 1-p$, then provided $p < 1$ we obtain

$$\mathbb{E}(X) \quad = \quad p \left(\frac{1}{(1-(1-p))^2} \right) = \frac{p}{p^2} = \frac{1}{p}$$

That is, for $X \sim \text{Geo}(p)$ with $0 \leqslant p < 1$,

$$\mathbb{E}(X) = \frac{1}{p}$$

(Exercise: find $\mathbb{E}(X)$ when $p = 1$. Hint: before doing any calculations, think about the experiment in this case.)

Exercise: Show that for $X \sim \text{Geo}(p)$,

$$\mathbb{V}\text{ar}(X) = \frac{1-p}{p^2}$$

Hint: Note that for $-1 < x < 1$,

$$\frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\sum_{k=0} x^k = \sum_{k=1}^{\infty} kx^{k-1},$$
$$\frac{2}{(1-x)^3} = \frac{d^2}{dx^2}\left(\frac{1}{1-x}\right) = \frac{d^2}{dx^2}\sum_{k=0}^{\infty} = \sum_{k=2}^{\infty} k(k-1)x^{k-2}.$$

4.4.4 Poisson Random Variable

Recall that $X \sim \text{Poi}(\lambda)$ if and only if X is a discrete random variable, with $\text{Im}(X) = \{0, 1, 2, 3, ...\}$ and

$$p_X(x) = \frac{\mathrm{e}^{-\lambda}\lambda^x}{x!}, x \in \{0, 1, 2, 3, \ldots\}.$$

To find the expected value:

$$\begin{aligned}
\mathbb{E}(X) = \sum_{x \in \text{Im}(X)} x p_X(x) &= \sum_{x=0}^{\infty} x\left(\frac{\lambda^x}{x!}\,\mathrm{e}^{-\lambda}\right) \\
&= \mathrm{e}^{-\lambda}\sum_{x=1}^{\infty}\frac{\lambda^x}{(x-1)!} \\
&= \lambda\mathrm{e}^{-\lambda}\sum_{y=0}^{\infty}\frac{\lambda^y}{y!} \text{ where } y = x-1 \\
&= \lambda\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda} \\
\mathbb{E}(X) &= \lambda
\end{aligned}$$

Recall that in deriving the Poisson probability mass function, we originally set λ to be 'the expected number of events'. We have just checked, from the probability mass function and the formula for expected value, that indeed $\mathbb{E}(X) = \lambda$.

It can be shown similarly that the variance of $X \sim \text{Poi}(\lambda)$ is

$$\mathbb{V}\text{ar}(X) = \lambda$$

(Exercise: Show that $\mathbb{V}\text{ar}(X) = \lambda$. Hint: start by finding $\mathbb{E}(X(X-1))$ for $X \sim \text{Poi}(\lambda)$.)

The Poisson distribution has the distinctive property that the mean and variance are the same.

4.4.5 Uniform Random Variable

Recall that $X \sim \mathrm{U}(a,b)$ if and only if X is a continuous random variable with

$$f_X(x) = \frac{1}{b-a}\mathbb{I}_{[a,b]}(x).$$

Find the expected value:

$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{\infty} x f_X(x)\,dx \\ &= \int_a^b x\left(\frac{1}{b-a}\right)dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{b-a}\left(\frac{b^2-a^2}{2}\right) = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}\end{aligned}$$

Notice that this expected value is unsurprising.

Can also show (*exercise*) that $\mathbb{V}\mathrm{ar}(X) = \frac{(b-a)^2}{12}$

4.4.6 Exponential Random Variables

Recall $X \sim \mathrm{Exp}(\lambda)$ if and only if X is a continuous random variable with pdf

$$f_X(x) = \begin{cases}\lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0.\end{cases}$$

$$\begin{aligned}\mathbb{E}(X) = \int_0^\infty x f_X(x)\mathrm{d}x &= \int_0^\infty \lambda x e^{-\lambda x}\mathrm{d}x \\ &= \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\mathrm{d}x \qquad \text{(integrate by parts)} \\ &= \left[-xe^{-\lambda x}\right]_0^\infty - \left[\frac{e^{-\lambda x}}{\lambda}\right]_0^\infty \\ &= \left(-(0)e^{-\lambda(0)} - 0\right) - \left(\frac{e^{-\lambda(0)}}{\lambda} - 0\right) \\ &= \frac{1}{\lambda}\end{aligned}$$

(Note: You can use the exponential series to show $\lim_{M\to\infty} Me^{-\lambda M} = \lim_{M\to\infty} \frac{M}{e^{\lambda M}} = 0$, indeed,

$$0 \leq \frac{M}{e^{\lambda M}} = \frac{M}{1+\lambda M + \frac{1}{2}\lambda^2 M^2 + \ldots} \leq \frac{M}{1+\lambda M + \frac{1}{2}\lambda^2 M^2} \to 0$$

as $M \to \infty$.)

Exercise: Show that $\mathbb{V}\mathrm{ar}(X) = \frac{1}{\lambda^2}$

4.4.7 Gamma Random Variable

Recall $X \sim \mathrm{Gamma}\,(\alpha,\lambda)$ if and only if X is a continuous random variable with probability density function

$$f_X(x) = \begin{cases}\frac{\lambda^\alpha}{\Gamma(\alpha)} e^{-\lambda x} x^{\alpha-1}, & x > 0, \\ 0, & x \leq 0.\end{cases} \tag{4.2}$$

We can use this to find $\mathbb{E}(X)$ and $\mathbb{V}\text{ar}(X)$, as follows.

$$\begin{aligned}\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx &= \int_0^{\infty} x\left(\frac{\lambda^\alpha}{\Gamma(\alpha)}\,x^{\alpha-1}e^{-\lambda x}\right)dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^{\infty} x^\alpha e^{-\lambda x}\,dx\end{aligned}$$

Making the substitution $y = \lambda x$ in the integral,

$$\begin{aligned}\mathbb{E}(X) &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^{\infty}\left(\frac{y}{\lambda}\right)^\alpha e^{-y}\,\frac{dx}{\lambda} \\ &= \frac{1}{\lambda\Gamma(\alpha)}\int_0^{\infty} y^\alpha e^{-y}\,dy \\ &= \frac{1}{\lambda\Gamma(\alpha)}\Gamma(\alpha+1) \qquad \text{(from definition of Gamma function)} \\ &= \frac{\alpha}{\lambda} \qquad \text{(since } \Gamma(\alpha+1) = \alpha\Gamma(\alpha)\text{)}\end{aligned}$$

Exercise: Show that $\mathbb{V}\text{ar}(X) = \frac{\alpha}{\lambda^2}$

Chapter 5

Transformations of Random Variables and the Normal (Gaussian) Distribution

5.1 Transformations of random variables

Example 5.1. Suppose a 2kg robot is programmed to move along a straight track from point A to point B, the speed, X, of the robot when it reaches point B is recorded in meters per second. Due to variations in motor performance and friction we model the final speed as $X \sim \mathrm{U}(0.5, 1.5)$. When considering risks involved with crashes we work with the kinetic energy

$$Y = \frac{1}{2}mX^2$$

where m is the mass of the robot in kg. Calculate:

1. The expected final speed.
2. The expected final kinetic energy.
3. The probability that the final kinetic energy is less than 2.
4. Describe the distribution of the final kinetic energy.

Solution:

1. The expected final speed is $\mathbb{E}[X]$ and is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_{0.5}^{1.5} x dx = \left[\frac{1}{2}x^2\right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{2} = 1.$$

2. The expected final kinetic energy, $\mathbb{E}[Y]$, is given by

$$\mathbb{E}[Y] = \mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_{0.5}^{1.5} x^2 dx = \left[\frac{1}{3}x^3\right]_{0.5}^{1.5} = \frac{1.5^3 - 0.5^3}{2} = \frac{3.25}{3} \approx 1.0833.$$

   Warning: Note that $\mathbb{E}[g(X)]$ is <u>not</u> the same as $g(\mathbb{E}[X])$.

3. We can find probabilities of Y by first finding the cumulative distribution function, F_Y, of Y.

   $$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(X \leq \sqrt{y}) = F_X(\sqrt{y}).$$

   Recall

$$F_X(x) = \begin{cases} 0, & x < 0.5, \\ x - 0.5, & 0.5 \leq x < 1.5, \\ 1, & x \geq 1.5. \end{cases}$$

Therefore

$$F_Y(y) = \begin{cases} 0, & \sqrt{y} < 0.5, \\ \sqrt{y} - 0.5, & 0.5 \leq \sqrt{y} < 1.5, \\ 1, & \sqrt{y} \geq 1.5. \end{cases}$$

In particular, $F_Y(y) = \sqrt{y} - 0.5$ for $0.25 \leq y \leq 2.25$, hence $\mathbb{P}(Y \leq 2) = \sqrt{2} - 0.5 \approx 0.9142$.

4. From the cumulative distribution function of Y we see that Y is a continuous random variable, since F_Y is continuous and piecewise smooth, with probability density function

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2} y^{-\frac{1}{2}}, \quad 0.25 \leq y \leq 2.25$$

and $f_Y(y) = 0$ otherwise.

In many applications of probability theory we are interested in a function (or transformation) of a random variable X. Specifically, suppose X is a random variable and suppose $g : \mathbb{R} \to \mathbb{R}$ is a function. Then $Y = g(X)$ is *also* a random variable and we can ask:

- What is the expected value of $g(X)$?
- What is the probability $g(X) \leq y$ for some $y \in \mathbb{R}$? That is, what is the cumulative distribution function, F_Y, of Y?
- Can we determine the probability density function (if it exists)?

Typically it is easier to calculate the expected value $\mathbb{E}[g(X)]$ directly without first calculating the cumulative distribution function or probability density function so it is important to consider which of these tasks you need to do!

If we know the cumulative distribution function F_X (or probability density f_X) for X, how can we find the cumulative distribution function (or probability density) for the random variable $Y = g(X)$?

Example 5.2. Suppose X is a random variable with cumulative distribution function $F_X(x)$ and probability density function $f_X(x)$ given by:

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & x \geqslant 0 \\ 0 & x < 0 \end{cases} \qquad f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geqslant 0 \\ 0 & x < 0 \end{cases}$$

where λ is a fixed, positive constant.
X is an example of an exponential random variable with parameter $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$. Notice that $\text{Im}X = [0, \infty)$.

Suppose $Y = g(X) = X^3$ (i.e. $g(x) = x^3$). Let's determine the cumulative distribution function F_Y for Y:

Step 1: The first step is to determine $\text{Im}Y$ (i.e. the set of values that $Y = X^3$ can take). Since we know that $\text{Im}X = [0, \infty)$ (i.e. $0 \leqslant X < \infty$), we have $0 \leqslant X^3 < \infty$, that is, $\text{Im}Y = [0, \infty)$ too.

Step 2: The next step is to determine the cumulative distribution function F_Y for all $y \in \mathbb{R}$.

Since $\text{Im}Y = [0, \infty)$, we know that for any $y < 0$, $F_Y(y) = \mathbb{P}(Y \leq y) = 0$.

Next, suppose that $y \geq 0$. To compute $F_Y(y)$ in this case:

$$\begin{aligned}
\text{For } y \geqslant 0,\ F_Y(y) &= \mathbb{P}(Y \leqslant y) && \text{(definition of } F_Y\text{)} \\
&= \mathbb{P}(X^3 \leqslant y) && \text{(since } Y = X^3\text{)} \\
&= \mathbb{P}(X \leqslant y^{1/3}) \\
&= F_X(y^{1/3}) && \text{(express in terms of } F_X\text{)} \\
&= 1 - e^{-\lambda y^{1/3}} && \text{(by substituting } y^{1/3} \text{ into } F_X\text{)}
\end{aligned}$$

So we have:

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 1 - e^{-\lambda y^{1/3}} & y \geqslant 0 \end{cases}$$

You can check that F_Y is continuous and differentiable (almost everywhere). Therefore Y also has a density f_Y which we can find by *differentiating* F_Y. So the probability density function for Y is given by:

$$f_Y(y) = F_Y^{'}(y) = \begin{cases} 0 & y < 0 \\ \frac{\lambda \exp\left(-\lambda y^{1/3}\right)}{3y^{2/3}} & y \geqslant 0 \end{cases}$$

Short cut: If you do not need to know an explicit formula for the cumulative distribution function F_Y of Y, you can find the probability density f_Y for Y without working out the cumulative distribution function F_Y. Since (from the algebra above) $F_Y(y) = F_X(y^{1/3})$ for $y \geqslant 0$, then for $y \geqslant 0$ we have

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}\left(F_X(y^{1/3})\right) \overset{chain\ rule}{=} F_X'(y^{1/3})\frac{\mathrm{d}}{\mathrm{dy}}(y^{1/3}) = f_X(y^{1/3}) \cdot \frac{1}{3y^{2/3}} = \frac{\lambda e^{-\lambda y^{1/3}}}{3y^{2/3}}$$

and $f_Y(y) = 0$ for $y < 0$ since $\mathrm{Im}Y = [0, \infty)$.

Warning: note that $Y = X^3$ but $f_Y(y) \neq \big(f_X(y)\big)^3$

Example 5.3. Suppose W is a random variable with density function given by

$$f_W(w) = \begin{cases} \frac{3w^2}{2} & -1 \leqslant w \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

So, $\mathrm{Im}W = [-1, 1]$ since $f_W(w) > 0$ when $-1 \leq w \leq 1$ (and the density is equal to 0 elsewhere).

Let $Z = W^2$. Find the probability density $f_Z(z)$ for Z.

Step 1. As in the previous example, the first step is to determine $\mathrm{Im}Z$. Since $-1 \leq W \leq 1$ (because $\mathrm{Im}W = [-1, 1]$), we must have $0 \leq Z \leq 1$ (since $Z = W^2$). It follows that $\mathrm{Im}Z$ = [0,1].

Step 2. Since the probability density f_Z is the derivative of the distribution function F_Z, the next step is to determine the cdf F_Z.

Since $\mathrm{Im}Z$ = [0, 1], we must have

$$\begin{aligned}
F_Z(z) &= 0 && \text{for } z < 0 \\
F_Z(z) &= \mathbb{P}(Z \leqslant z) = 1 && \text{for } z \geqslant 1
\end{aligned}$$

For $0 \leqslant z < 1$, we find an expression for F_Z in terms of F_W:

$$\begin{aligned} F_Z(z) \overset{\text{definition of df}}{=} \mathbb{P}(Z \leqslant z) &= \mathbb{P}(W^2 \leqslant z) && \text{(since } Z = W^2\text{)} \\ &= \mathbb{P}(-\sqrt{z} \leqslant W \leqslant \sqrt{z}) && \text{(solve the inequality } W^2 \leqslant Z \text{ for } W\text{)} \\ &= \mathbb{P}(-\sqrt{z} < W \leqslant \sqrt{z}) && \text{(since } W \text{ has a density and is a cts random variable)} \\ &= F_W(\sqrt{z}) - F_W(-\sqrt{z}) \end{aligned}$$

At this stage we have enough information to work out f_Z even though we have not worked out an explicit formula for either F_W or F_Z, as follows.

Step 3. The last step is to differentiate F_Z to obtain f_Z.

From Step 2 above, it is clear that if either $z < 0$ or $z > 1$, $F'(z) = f_z(z) = 0$.

For $0 < z < 1$, we differentiate both sides of the equation

$$F_Z(z) = F_W(\sqrt{z}) - F_W(-\sqrt{z})$$

to obtain

$$\begin{aligned} f_Z(z) = F_Z'(z) &= \frac{\mathrm{d}}{\mathrm{dz}}\big(F_W(\sqrt{z}) - F_W(-\sqrt{z})\big) \\ &= F_W'(\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} + F_W'(-\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} && \text{(chain rule)} \\ &= f_W(\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} + f_W(-\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} && \text{(since } F_W' = f_W\text{)} \\ &= \frac{3}{2}\sqrt{z} && \text{(using the given function } f_W\text{)} \end{aligned}$$

So we have:

$$f_Z(z) = \begin{cases} \frac{3}{2}\sqrt{z} & 0 \leqslant z \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

Linear transformations

Proposition 5.1. *Suppose that X is a random variable with pdf $f_X(x)$, and that $Y = aX + b$ for some $a, b \in \mathbb{R}$ with $a \neq 0$. Then Y has probability density function $f_Y(y)$ given by*

$$f_Y(y) = \begin{cases} \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) & \frac{y-b}{a} \in \mathrm{Im}X \\ 0 & \textit{otherwise} \end{cases}$$

Proof Since $Y = aX + b$, then $X = \frac{Y-b}{a}$, so that $\mathrm{Im}Y = \{y : \frac{y-b}{a} \in \mathrm{Im}X\}$.
Consider the case $a > 0$. For y such that $\frac{y-b}{a} \in \mathrm{Im}X$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leqslant y) \\ &= \mathbb{P}\left(aX + b \leqslant y\right) \\ &= \mathbb{P}\left(X \leqslant \frac{y-b}{a}\right) && \text{(since } a > 0\text{)} \\ &= F_X\left(\frac{y-b}{a}\right) \end{aligned}$$

$$\Rightarrow \qquad f_Y(y) = \frac{dF_Y}{dy} = \frac{d}{dy}F_X\left(\frac{y-b}{a}\right) = f_X\left(\frac{y-b}{a}\right) \times \frac{d}{dy}\left(\frac{y-b}{a}\right) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

as required. □

Exercise: Prove the above result for the case $a < 0$.

5.2 The Normal (Gaussian) distribution

Normal random variables arise in many ways in probability and statistics. We start with the definition of the *standard* normal random variable.

Definition
A random variable Z has the standard normal distribution if and only if its density φ is given by

$$\boxed{\varphi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2} \qquad \text{for } -\infty < z < \infty}$$

$\big(\text{so } \mathrm{Im}Z = (-\infty, \infty)\ \big)$

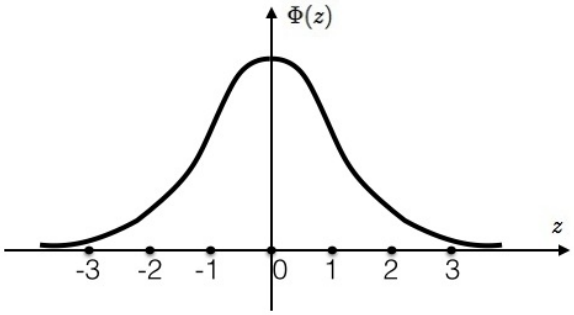


Figure 5.1: The graph of $\varphi(z)$

We will show that $\mathbb{E}(Z) = 0$ and $\mathbb{V}\mathrm{ar}(Z) = 1$. If Z has the standard normal distribution, we write $Z \sim \mathrm{N}(0, 1)$.

Remark 5.1. 1. The density φ is symmetric about $z = 0$, so $\mathbb{P}(Z \leqslant 0) = \mathbb{P}(Z \geqslant 0) = 1/2$

2. Since the density φ is symmetric about 0 it follows that $\mathbb{E}[Z] = 0$, indeed

$$\mathbb{E}[Z] = \int_{-\infty}^{\infty} z f_Z(z)\,\mathrm{d}z = \int_{-\infty}^{\infty} z\varphi(z)\,\mathrm{d}z = \int_{-\infty}^{0} \frac{ze^{-z^2/2}}{\sqrt{2\pi}}\,\mathrm{d}z + \int_{0}^{\infty} \frac{ze^{-z^2/2}}{\sqrt{2\pi}}\,\mathrm{d}z.$$

Now making the substitution $z \mapsto -z$ in the first integral

$$\mathbb{E}[Z] = \int_{0}^{\infty} \frac{-ze^{-z^2/2}}{\sqrt{2\pi}} dz + \int_{0}^{\infty} \frac{ze^{-z^2/2}}{\sqrt{2\pi}} dz = 0.$$

3. In general, it is not possible to express

$$\mathbb{P}(a \leqslant Z \leqslant b) = \int_a^b \frac{e^{-z^2/2}}{\sqrt{2\pi}}\mathrm{d}z$$

in terms of 'standard functions'.

Instead, we define a function $\Phi(x)$ to be

$$\Phi(x) = \int_{-\infty}^{x} \frac{e^{-z^2/2}}{\sqrt{2\pi}}\mathrm{d}z = \mathbb{P}(Z \leqslant x) \text{ for } Z \sim \mathrm{N}(0, 1)$$

4. **Using statistical tables**:
   Most calculators don't have a $\Phi(x)$ button, so instead we have to use statistical tables (or a computer package). Statistical tables give the values of the cumulative distribution function $\Phi(x)$, and then since $\mathbb{P}(a \leqslant Z \leqslant b) = \Phi(b) - \Phi(a)$, we can use the tables to compute probabilities for $Z \sim \mathrm{N}(0,1)$.

   Examples
   The Cambridge Statistical Tables give the values of $\Phi(x)$ for $x \geq 0$. So using the Cambridge Statistical Tables we can compute:

   $\mathbb{P}(Z \leqslant 1.2) = \Phi(1.2) = 0.8849$
   $\mathbb{P}(0.5 \leqslant Z \leqslant 1.2) = \Phi(1.2) - \Phi(0.5) = 0.8849 - 0.6915 = 0.1934$
   $\mathbb{P}(Z \geqslant 0.75) = 1 - \mathbb{P}(Z \leqslant 0.75) = 1 - \Phi(0.75) = 0.2266$

   Important: Using the symmetry of the distribution,

$$\Phi(-z) = \mathbb{P}(Z \leqslant -z) = \mathbb{P}(Z \geqslant z) = 1 - \mathbb{P}(Z \leqslant z) = 1 - \Phi(z)$$

   So:

$$\begin{aligned}\mathbb{P}(Z \leqslant -0.5) &= \Phi(-0.5) = 1 - \Phi(0.5) = 1 - 0.6915 = 0.3085\\ \mathbb{P}(-0.5 \leqslant Z \leqslant 0.75) &= \Phi(0.75) - \Phi(-0.5) = \Phi(0.75) - \big(1 - \Phi(0.5)\big)\\ &= \Phi(0.75) + \Phi(0.5) - 1 = 0.7734 + 0.6915 - 1 = 0.4649\end{aligned}$$

Proposition 5.2. *Let $Z \sim \mathrm{N}(0,1)$ be a standard Gaussian random variable then*

$$\mathbb{E}[Z] = 0$$
$$\mathbb{V}\mathrm{ar}[Z] = 1.$$

Proof As shown in Remark 5.1 the symmetry of the density gives that $\mathbb{E}[Z] = 0$ so it remains to show that $\mathbb{V}\mathrm{ar}[Z] = 1$.
Note that since $\mathbb{E}[Z] = 0$ we have $\mathbb{V}\mathrm{ar}[Z] = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = \mathbb{E}[Z^2]$ so it is sufficient to show that $\mathbb{E}[Z^2] = 1$.
We will show that $\mathbb{E}[Z^2] = 1$ by use of integration by parts. Using the expression for φ we can write

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \varphi(z)\,\mathrm{d}z = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z$$

Recall that the formula for integration by parts is

$$\int_a^b u(z)\,v'(z)\,dz = [u(z)v(z)]_a^b - \int_a^b v(z)\,u'(z)\,\mathrm{d}z.$$

We will apply this with $a = -\infty, b = \infty, u = z$ and $v = -e^{-\frac{z^2}{2}}$, noting that

$$v'(z) = \frac{d}{dz}\left(-e^{-\frac{z^2}{2}}\right) = ze^{-\frac{z^2}{2}}.$$

Then integration by parts gives

$$\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z = \frac{1}{\sqrt{2\pi}}\left[-ze^{-\frac{z^2}{2}}\right]_{-\infty}^{\infty} - \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} -e^{-\frac{z^2}{2}}\,\mathrm{d}z.$$

Observe that the $ze^{-z^2/2} \to 0$ as $z \to -\infty$ or $z \to \infty$ therefore the boundary terms vanish and we have

$$\mathbb{E}[Z^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} \, \mathrm{d}z = \int_{-\infty}^{\infty} \varphi(z) \, \mathrm{d}z = 1.$$

The final equality holds since φ is a probability density function and hence integrates to 1. □

5.2.1 Linear transformation of $Z \sim \mathrm{N}(0,1)$

Let $Z \sim \mathrm{N}(0,1)$ and $X = \mu + \sigma Z$ for some $\mu \in \mathbb{R}$ and $\sigma \neq 0$. Then X has pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Indeed, using Proposition 5.1 about linear transformations we have that if φ is the probability density function of Z then

$$f_X(x) = \frac{1}{|\sigma|}\varphi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{|\sigma|}\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

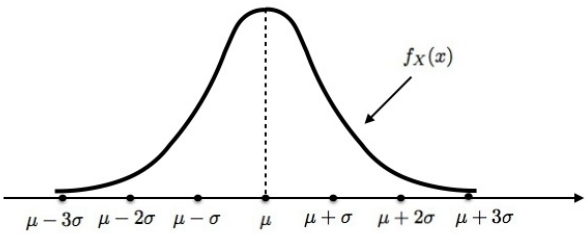


Figure 5.2: Sketch of $f_X(x)$

Observe the graph of f_X is bell-shaped and symmetric about $x = \mu$. We say that the function

$$\boxed{f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad -\infty < x < \infty}$$

is a normal density with *mean* μ and *variance* σ^2, and we write $X \sim \mathrm{N}(\mu, \sigma^2)$ to denote that X has a normal distribution with *mean* μ and *variance* σ^2 and density given by the function f_X above.

Proposition 5.3. *Let $X \sim \mathrm{N}(\mu, \sigma^2)$ be a Gaussian random variable then*

$$\mathbb{E}[Z] = \mu$$
$$\mathbb{V}\mathrm{ar}[Z] = \sigma^2.$$

Proof Let $X \sim \mathrm{N}(0,1)$ and write $X = \mu + \sigma Z$ for some $Z\mathrm{N}(0,1)$, by linearity of the expectation

$$\mathbb{E}[X] = \mathbb{E}[\mu + \sigma Z] = \mu + \sigma\mathbb{E}[Z] = \mu$$

since $\mathbb{E}[Z] = 0$. Moreover

$$\begin{aligned}\mathbb{V}\mathrm{ar}(X) &= \mathbb{V}\mathrm{ar}(\mu + \sigma Z) \\ &= \mathbb{V}\mathrm{ar}(\sigma Z) \\ &= \sigma^2 \mathbb{V}\mathrm{ar}(Z) \\ &= \sigma^2.\end{aligned}$$

Here we used the properties of the variance and that $\mathbb{V}\mathrm{ar}(Z) = 1$. □

Key fact: If $X \sim \mathrm{N}\left(\mu, \sigma^2\right)$, then $\frac{X-\mu}{\sigma} \sim \mathrm{N}\left(0,1\right)$. (This is called standardisation.)

Remark: We can use the Key Fact and the Cambridge Statistical Tables to calculate probabilities for X.

Often measurements of *physical quantities* follow a normal distribution.

Example 5.4. Based on medical data, researchers have concluded that the heights of 25-year old men follow a normal distribution with $\mu = 70$ inches and $\sigma^2 = 6.25$ in^2. If a 25-year old man is selected at random, what is the probability that he will be at least 6 feet tall?

Let X = height of randomly selected man. We need to compute

$$\begin{aligned}
\mathbb{P}(X \geqslant 72) &= \mathbb{P}\left(\frac{X-70}{\sqrt{6.25}} \geqslant \frac{72-70}{\sqrt{6.25}}\right) \\
&= \mathbb{P}(Z \geqslant 0.8) \qquad \left(\text{since if } X \sim \mathrm{N}\left(70, 6.25\right), \text{ then } Z = \frac{X-70}{\sqrt{6.25}} \sim \mathrm{N}\left(0,1\right)\right) \\
&= 1 - \Phi(0.8) = 1 - 0.7881 = 0.2119.
\end{aligned}$$

Chapter 6

Joint distributions, marginal distributions, and conditional distributions

We have seen in Section 2.4 that if random variables X_1 and X_2 are *independent*, then we can use *independence* of X_1 and X_2 to compute

$$\mathbb{P}(X_1 \in A_1,\ X_2 \in A_2) = \mathbb{P}(X_1 \in A_1)\mathbb{P}(X_2 \in A_2)$$

If X_1 and X_2 are not independent, how can we compute $\mathbb{P}(X_1 \in A_1,\ X_2 \in A_2)$?
In many practical situations, random variables are not independent. To compute joint probabilities in such cases, we need to understand their joint distribution.

Example 6.1. Suppose we select a student at random and let

X = student's height in cm

Y = student's weight in kilograms

Intuitively, it is clear that X and Y are not independent (because if you know the value of X, the student's height, then you are better able to predict the student's weight Y than you would be without knowing the value of X).

In order to compute joint probabilities such as

$$\mathbb{P}(68 < X \leqslant 72,\ 65 < Y \leqslant 75)$$

for the variables in the example above, we need to know the joint distribution of X and Y.

6.1 Joint probability mass function for discrete random variables

For a discrete random variable X, the simplest way to express the distribution is through the probability mass function $p_X(x) = \mathbb{P}(X = x)$ for $x \in \mathrm{Im}X$. To deal with two (or more) discrete random variables at once, we can use the joint probability mass function.

Definition. Suppose X and Y are discrete random variables with $\mathrm{Im}X = \{x_1, x_2, \dots\}$ and $\mathrm{Im}Y = \{y_1, y_2, \dots\}$. Then the **joint probability mass function** $p_{X,Y}(x, y)$ is defined to be

$$\boxed{p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y) \qquad \forall x \in \mathrm{Im}X,\ y \in \mathrm{Im}Y}$$

Properties of joint probability mass functions

Suppose that $p_{X,Y}$ is the joint probability mass function of (discrete) random variables X, Y. Then:

1. $p_{X,Y}(x,y) \geqslant 0 \ \forall x, y$
2. $\sum_x \sum_y p_{X,Y}(x,y) = 1$

Conversely, any function satisfying the above properties is a valid joint probability mass function for some pair of discrete random variables.

Example 6.2. Suppose a fair coin is flipped 4 times. Define

X = number of Heads in first 3 flips

Y = number of Heads in last 2 flips

It seems intuitively clear that X and Y will not be independent here. To check this, we first find the joint probability mass function.

We have $\text{Im}X = \{0,1,2,3\}$ and $\text{Im}Y = \{0,1,2\}$.

The sample space Ω consists of the 16 possible outcomes of flipping a coin 4 times. The coin is fair, and it is reasonable to assume that flips are independent of one another, so the 16 possible outcomes are all equally likely. So for each (x,y) with $x \in \text{Im}X$, $y \in \text{Im}Y$ we can find the probability $\mathbb{P}(X = x, Y = y)$ by counting the number of outcomes that result in x Heads in the first 3 flips and y Heads in the last 2 flips.

For instance, $p_{X,Y}(1,1) = \mathbb{P}(X = 1, Y = 1) = \mathbb{P}(\{HTTH,\ THTH,\ TTHT\})$

We can present the joint probability mass function in a table:

$p_{X,Y}(x,y)$	$x=0$	$x=1$	$x=2$	$x=3$	
$y=0$	$1/16$	$2/16$	$1/16$	0	$p_Y(0) = 1/4$
$y=1$	$1/16$	$3/16$	$3/16$	$1/16$	$p_Y(1) = 1/2$
$y=2$	0	$1/16$	$2/16$	$1/16$	$p_Y(2) = 1/4$
	$p_X(0) = 1/8$	$p_X(1) = 3/8$	$p_X(2) = 3/8$	$p_X(3) = 1/8$	

Exercise: By enumerating possible outcomes (HHHH, HHHT, etc.), check that the values in the above joint probability mass function table are correct.

Exercies: Considering X and Y separately, check that $X \sim \text{Bin}(3, 1/2)$ and $Y \sim \text{Bin}(2, 1/2)$. Note that this tells us nothing about the relationship *between* X and Y.

The entries in the table above give the values of the joint probability mass function $p_{X,Y}$. By summing up the rows and columns, we can use this table to compute the *marginal* probability mass functions p_X and p_Y for the random variables X and Y, respectively. The two marginal probability mass functions p_X and p_Y are presented in the margins of the table above.

Marginal probability mass functions

Given the joint probability mass function $p_{X,Y}$ for discrete random variables X and Y, we can find the (*marginal*) probability mass functions p_X and p_Y for X and Y as follows:

Suppose $x \in \text{Im}X$, then

$$p_X(x) \underset{\text{def.}}{=} \mathbb{P}(X = x) = \mathbb{P}\Big(\{X = x\} \cap \big(\cup_{y \in \text{Im}Y} \underbrace{\{Y = y\}}_{\substack{\text{these events} \\ \text{form a partition}}} \big)\Big)$$

$$\begin{aligned} &= \mathbb{P}\big(\cup_{y \in \text{Im}Y} \{X = x,\ Y = y\}\big) && \text{distributive property from set theory} \\ &= \sum_{y \in \text{Im}Y} \mathbb{P}(X = x,\ Y = y) && \text{property (III) of } \mathbb{P} \\ p_X(x) &= \sum_{y \in \text{Im}Y} p_{X,Y}(x, y) \end{aligned}$$

Similarly, for $y \in \text{Im}Y$,

$$p_Y(y) = \mathbb{P}(Y = y) = \sum_{x \in \text{Im}X} p_{X,Y}(x, y).$$

Independent discrete random variables

It follows from the definition of independent random variables in Section 2.4 that discrete random variables X and Y with joint probability mass function $p_{X,Y}$ and marginal probability mass functions p_X and p_Y are *independent* if and only if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \qquad \forall x \in \text{Im}X,\ y \in \text{Im}Y$$

Example 6.2 continued. In Example 6.2 we have $p_{X,Y}(0, 2) = 0$, but the corresponding marginal probabilities are $p_X(0) = 1/8$ and $p_Y(2) = 1/4$, so that $p_{X,Y}(0, 2) \neq p_X(0)p_Y(2)$. The random variables X, Y are not independent.

Question: Suppose we know the joint probability mass function $p_{X,Y}$ for variables X and Y, how can we determine whether X and Y are independent?

Answer: To show that X and Y are independent, we need to check that $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ for *every* pair of values $x \in \text{Im}X,\ y \in \text{Im}Y$.
To show that X and Y are not independent, it is sufficient to find *any* pair of values $x \in \text{Im}X,\ y \in \text{Im}Y$ with $p_{X,Y}(x, y) \neq p_X(x)p_Y(y)$.

Remarks

1. If we know that discrete random variables X and Y are independent, then we can find their joint probability mass function from their marginal probability mass functions using the relationship $p_{X,Y}(x, y) = p_X(x)p_Y(y)$. Can only do this if we know that X, Y are independent.
2. When the variables X and Y depend on *different experiments*, which are *physically independent*, then it is reasonable to assume that the variables X and Y are (statistically) independent and we can compute $p_{X,Y}$, the joint probability mass function for X and Y, by multiplying their marginal probability mass functions.
3. In general, knowing the marginal probability mass functions $p_X(x)$ and $p_Y(y)$ is not sufficient to be able to find the joint probability mass function $p_{X,Y}(x, y)$.

Example 6.3. Suppose X is a random variable with $\mathrm{Im}X=\{1,2\}$ and with probability mass function given by $p_X(1)=p_X(2)=1/2$ and suppose Y is a random variable with $\mathrm{Im}Y=\{1,2,3\}$ and with probability mass function given by $p_Y(1)=p_Y(2)=p_Y(3)=1/3$.

This information gives us the row sums and the column sums for the table that describes the joint probability mass function $p_{X,Y}(x,y)$, but this information doesn't tell us *how* to fill in the table. There are many ways to fill in the table so that the row sums and column sums agree with the values of p_X and p_Y. Here are two possibilities:

	$y=1$	$y=2$	$y=3$	
$x=1$	$1/6$	$1/12$	$1/4$	$1/2$
$x=2$	$1/6$	$1/4$	$1/12$	$1/2$
	$1/3$	$1/3$	$1/3$	

	$y=1$	$y=2$	$y=3$	
$x=1$	$1/12$	$1/6$	$1/4$	$1/2$
$x=2$	$1/4$	$1/6$	$1/12$	$1/2$
	$1/3$	$1/3$	$1/3$	

Both of these tables of values for $p_{X,Y}$ are *consistent* with the given information, and this means that we can't work out $p_{X,Y}(x,y)$ just from knowledge of p_X and p_Y.

Functions of discrete random variables X and Y

Suppose X and Y are discrete random variables with joint probability mass function $p_{X,Y}(x,y)$ and suppose $g:\mathbb{R}^2\to\mathbb{R}$ is a function. Let $Z=g(X,Y)$, then Z is *also* a discrete random variable.

To determine the probability mass function $p_Z(z)$ for Z, we first need to determine $\mathrm{Im}Z$. Then we can use the joint probability mass function $p_{X,Y}(x,y)$ for X and Y to determine

$$p_Z(z)=\mathbb{P}(Z=z)=\mathbb{P}(g(X,Y)=z)=\sum_{(x,y):g(x,y)=z} p_{X,Y}(x,y) \quad \forall z\in \mathrm{Im}Z.$$

Example 6.2 continued. Suppose $Z=XY$. By considering the *possible* values that X and Y can jointly take, we obtain

$$\mathrm{Im}Z=\{0,1,2,3,4,6\}.$$

Looking at the table for the joint probability mass function $p_{X,Y}$ in example 6.2, we find that the probability mass function $p_Z(z)$ is given by

$$\begin{aligned}
p_Z(6)&=\mathbb{P}(Z=6)=\mathbb{P}(XY=6)=\mathbb{P}((X,Y)=(3,2))=\frac{1}{16}\\
p_Z(4)&=\mathbb{P}(Z=4)=\mathbb{P}(XY=4)=\mathbb{P}((X,Y)=(2,2))=\frac{2}{16}\\
p_Z(3)&=\mathbb{P}(Z=3)=\mathbb{P}(XY=3)=\mathbb{P}((X,Y)=(3,1))=\frac{1}{16}\\
p_Z(2)&=\mathbb{P}(Z=2)=\mathbb{P}(XY=2)=\mathbb{P}((X,Y)\in\{(1,2),(2,1)\})=\frac{1}{16}+\frac{3}{16}=\frac{4}{16}\\
p_Z(1)&=\mathbb{P}(Z=1)=\mathbb{P}(XY=1)=\mathbb{P}((X,Y)=(1,1))=\frac{3}{16}\\
p_Z(0)&=\mathbb{P}(Z=0)=\mathbb{P}(XY=0)=\mathbb{P}((X,Y)\in\{(0,0),(0,1),(0,2),(0,3),(1,0),(2,0)\})\\
&=\frac{1}{16}+\frac{2}{16}+\frac{1}{16}+0+\frac{1}{16}+0=\frac{5}{16}
\end{aligned}$$

(Note that $p_Z(5)=0$, since $5\notin\mathrm{Im}Z$.)

Given the probability mass function p_Z for Z, we can compute the expected value $\mathbb{E}(Z)$:

$$\begin{aligned}\mathbb{E}(Z) &= 0 \times p_Z(0) + 1 \times p_Z(1) + 2 \times p_Z(2) + 3 \times p_Z(3) + 4 \times p_Z(4) = 6 \times p_Z(6) \\ &= 0 + \frac{3}{16} + 2 \times \frac{4}{16} + 3 \times \frac{1}{16} + 4 \times \frac{2}{16} + 6 \times \frac{1}{16} \\ &= \frac{28}{16} = \frac{7}{4} = 1.75\end{aligned}$$

Exercise: Compute $\mathbb{V}\mathrm{ar}(Z)$

Also, from the marginal probability mass functions for X and Y we can compute $\mathbb{E}(X) = 3/2$ and $\mathbb{E}(Y) = 1$ (*exercise*: check). Notice that $\mathbb{E}(XY) \neq \mathbb{E}(X)\mathbb{E}(Y)$ in this example.

Expected value of $g(X, Y)$ for discrete X and Y
Just as in the case of a function of a single variable, there is a shortcut for computing $\mathbb{E}\big(g(X, Y)\big)$ which does not require us to first determine the probability mass function for $g(X, Y)$. Specifically:

$$\boxed{\mathbb{E}\left(g(X,Y)\right) = \sum_{\substack{x \in \mathrm{Im}X \\ y \in \mathrm{Im}Y}} g(x,y)\; p_{X,Y}(x,y)}$$

Exercise: Use the above formula to compute $\mathbb{E}(XY)$ in Example 6.2 (you should get the same value as obtained previously).

6.1.1 Conditional probability mass functions

Suppose X and Y are discrete random variables with joint probability mass function $p_{X,Y}$ and suppose y is some specified value such that $p_Y(y) = \mathbb{P}(Y = y) > 0$. Then the **conditional probability mass function** for X *given* $Y = y$ is defined by

$$\boxed{p_{X|Y}(x|y) = \mathbb{P}(X = x | Y = y)}$$

It follows from the definition of conditional probability for *events* that

$$\boxed{p_{X|Y}(x|y) = \mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} = \frac{p_{X,Y}(x,y)}{p_Y(y)}}$$

Remarks

1. If X and Y are *independent* random variables (and $p_Y(y) > 0$), then

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \overset{\text{by indep of } X \text{ and } Y}{=} \frac{p_X(x)p_Y(y)}{p_Y(y)} = p_X(x)$$

   i.e. whenever random variables are *independent*, the conditional distributions and the unconditional (marginal) distributions for the random variables are the *same*.

2. Sometimes the notation $p_{X|Y}(\cdot \mid y)$ is used, to emphasise that we condition upon one specific value y of Y, and then consider the set of conditional probabilities $p_{X|Y}(x \mid y)$ for all values $x \in \mathrm{Im}X$.

3. Conditional probability mass functions satisfy all the same properties as unconditional probability mass functions.

$$\text{E.g.} \qquad \sum_{x \in \mathrm{Im}X} p_{X|Y}(x|y) = 1$$

4. We can use the conditional probability mass function for X given $\{Y = y\}$ to compute the **conditional expectation of X given $\{Y = y\}$**:

$$\boxed{\mathbb{E}(X \mid Y = y) = \sum_{x \in \mathrm{Im}X} x \; p_{X|Y}(x|y) = \sum_{x \in \mathrm{Im}X} x \; \mathbb{P}(X = x \mid Y = y)}$$

We also have:

$$\mathbb{E}(g(X) \mid Y = y) = \sum_{x \in \mathrm{Im}X} g(x) \; p_{X|Y}(x|y)$$

Example 6.2 continued. Using the definition given above, we can determine the **conditional probability mass function** $p_{X|Y}(x|0)$ of X given $\{Y = 0\}$:

$$p_{X|Y}(0|0) = \frac{p_{X,Y}(0,0)}{p_Y(0)} = \frac{1/16}{1/4} = \frac{1}{4}$$

$$p_{X|Y}(1|0) = \frac{p_{X,Y}(1,0)}{p_Y(0)} = \frac{2/16}{1/4} = \frac{1}{2}$$

$$p_{X|Y}(2|0) = \frac{p_{X,Y}(2,0)}{p_Y(0)} = \frac{1/16}{1/4} = \frac{1}{4}$$

$$p_{X|Y}(3|0) = \frac{p_{X,Y}(3,0)}{p_Y(0)} = \frac{0}{1/4} = 0$$

Note:: The conditional probabilities sum to 1

Using the conditional probability mass function $p_{X|Y}(x|0)$, we can compute the *conditional* expected value of X *given* that $Y = 0$:

$$\mathbb{E}(X \mid Y = 0) = 0 \cdot p_{X|Y}(0|0) + 1 \cdot p_{X|Y}(1|0) + 2 \cdot p_{X|Y}(2|0) + 3 \cdot p_{X|Y}(3|0) =$$
$$= 0 + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot 0 = 1$$

For comparison, the *unconditional* expected value of X is $\mathbb{E}(X) = 3/2$.
Question: Why does it make *sense* that $\mathbb{E}(X) > \mathbb{E}(X \mid Y = 0)$ in this example?

Remark: The conditional expected value $\mathbb{E}(X \mid Y = y)$ is a function that depends on the value of y, so $\mathbb{E}(X \mid Y)$ is a *random variable* that depends on the random variable Y. We can compute

the expected value of the random variable $\mathbb{E}(X \mid Y)$ as follows:

$$\begin{aligned}
\mathbb{E}(\mathbb{E}(X \mid Y)) &= \sum_{y \in \mathrm{Im}(Y)} \mathbb{E}(X \mid Y = y) p_Y(y) \\
&= \sum_{y \in \mathrm{Im}(Y)} \left(\sum_{x \in \mathrm{Im}(X)} x \; p_{X|Y}(x|y) \right) p_Y(y) \\
&= \sum_{y \in \mathrm{Im}(Y)} \sum_{x \in \mathrm{Im}(X)} x \; p_{X,Y}(x,y) \\
&= \sum_{x \in \mathrm{Im}(X)} x \sum_{y \in \mathrm{Im}(Y)} p_{X,Y}(x,y) \\
&= \sum_{x \in \mathrm{Im}(X)} x \; p_X(x) \\
&= \mathbb{E}(X)
\end{aligned}$$

Summary: The calculations above show that $\mathbb{E}(\mathbb{E}(X \mid Y)) = \mathbb{E}(X)$. This result is used frequently in the study of stochastic processes.

6.2 Joint distribution function

Only discrete random variables have (joint) probability mass functions. Just as every random variable X has a cumulative distribution function F_X, *every* pair of random variables X and Y has a joint distribution function $F_{X,Y}(x, y)$, defined as follows.

Definition
Suppose X and Y are random variables. Then the joint distribution function $F_{X,Y} : \mathbb{R}^2 \to [0, 1]$ for X and Y is defined by

$$\boxed{F_{X,Y}(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y) \qquad \forall (x, y) \in \mathbb{R}^2}$$

In general, for random variables $X_1, X_2, \ldots, X_n$, we define the joint distribution function $F_{X_1,X_2,\ldots,X_n} : \mathbb{R}^n \to [0, 1]$ for $X_1, X_2, \ldots, X_n$ by

$$F_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n) = \mathbb{P}(X_1 \leqslant x_1, X_2 \leqslant x_2, \ldots, X_n \leqslant x_n) \qquad \forall (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$$

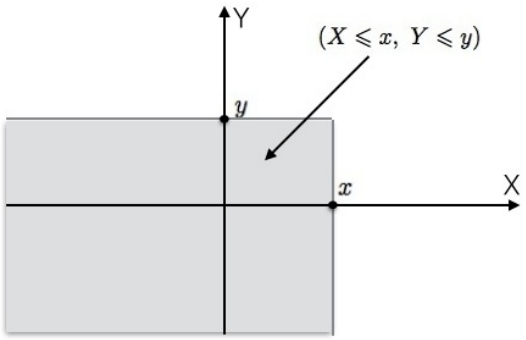
Remarks

1. The definition of joint distribution function $F_{X,Y}(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y)$ is a natural generalisation of the definition of the cumulative distribution function for a single random variable.
2. $F_{X,Y}(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y)$ computes the probability that the ordered pair of variables (X, Y) takes a value in the shaded (infinite) rectangular region:

3. If $x_1 \leqslant x_2$ and $y_1 \leqslant y_2$, then $\{X \leqslant x_1, Y \leqslant y_1\} \subseteq \{X \leqslant x_2, Y \leqslant y_2\}$, so

$$F_{X,Y}(x_1, y_1) \leqslant F_{X,Y}(x_2, y_2)$$

by the monotonicity of $\mathbb{P}$.



4. **Independence of random variables**
   It follows from the definition of independent random variables in Section 2.4 that the random variables $X_1, X_2, ..., X_n$ are independent if and only if for all $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$

$$F_{X_1,X_2,...,X_n}(x_1, x_2, \ldots, x_n) = F_{X_1}(x_1)F_{X_2}(x_2)\cdots F_{X_n}(x_n)$$

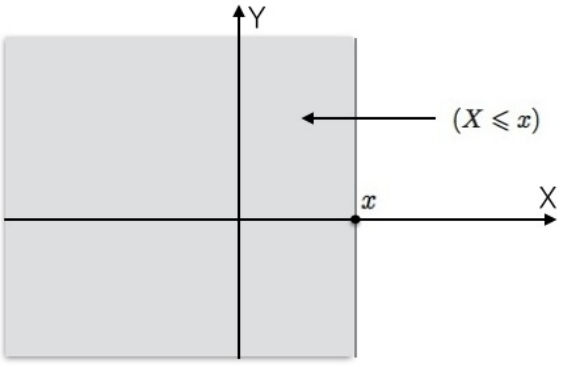
   where $F_{X_1,X_2,...,X_n}$ is the joint distribution function of $X_1, X_2, ..., X_n$ and F_{X_i} is the distribution function of X_i for $i = 1, 2, ..., n$.

Marginal distribution functions

Suppose X and Y have a joint df $F(x, y) = \mathbb{P}(X \leqslant x, Y \leqslant y)$ for $(x, y) \in \mathbb{R}^2$. How can we use $F(x, y)$ to find the (marginal) distribution functions of X and Y individually?

Answer: We have

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leqslant x) \\ &= \lim_{y\uparrow\infty} \mathbb{P}(X \leqslant x, Y \leqslant y) && \text{(continuity of } \mathbb{P}) \\ &= \lim_{y\uparrow\infty} F(x, y) && (x \text{ fixed, } y \to \infty) \end{aligned}$$



Likewise, $F_Y(y) = \lim_{x\uparrow\infty} F(x, y)$.

Example 6.4. Suppose X and Y have joint df given by

$$F(x, y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-x-y} & \text{if } x \geqslant 0 \text{ and } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

To find F_X, the (marginal) distribution function of X, we note:

- if $x < 0$, then $F_X(x) = \lim_{y\to\infty} F(x,y) = 0$
- if $x \geqslant 0$, then

$$F_X(x) = \lim_{y\to\infty} F(x,y) = \lim_{y\to\infty} \left(1 - e^{-x} - e^{-y} + e^{-x-y}\right) = 1 - e^{-x}$$

So

$$F_X(x) = \begin{cases} 1 - e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

This is the distribution function of the Exp(1) distribution, so we conclude that $X \sim \text{Exp}(1)$. Similarly, you can show that $Y \sim \text{Exp}(1)$.

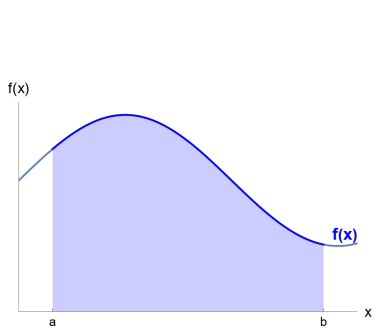
In this example, we can see that $F_{X,Y}(x,y) = F_X(x)F_Y(y)$ for every $(x,y) \in \mathbb{R}^2$, so X and Y are **independent** random variables here.

6.3 Introduction to Double Integration

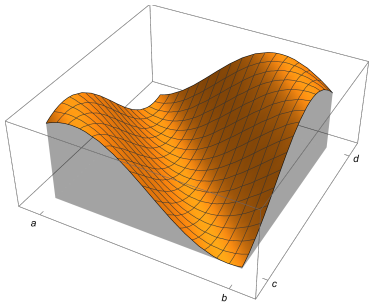
As we saw in Chapter 3, probability density functions often serve as the continuous analogue of probability mass functions. To compute probabilities from a density function, we integrate over an interval. Extending this idea to jointly continuous random variables requires evaluating integrals of functions $f : \mathbb{R}^2 \to \mathbb{R}$, which involves integration over two variables-this is known as double integration.

In Semester 2, the course Multivariable Calculus will introduce double integration. Here, we provide a brief overview to support its use in probability theory. The purpose of this section is not to cover the material in that course but to merely introduce you to key concepts that we can use for joint distributions.

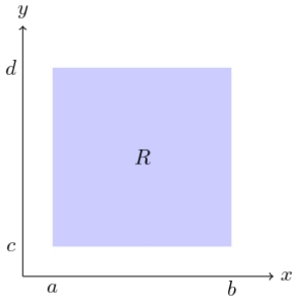
One way to understand integration is as the calculation of the area under a curve. We extend this idea to functions of two variables by computing the volume under a surface. For example, the integral of the indicator function $\mathbb{I}_{[a,b]}$ gives the length of the interval, $b-a$, while the double integral of the indicator function $\mathbb{I}_{[a,b]\times[c,d]}$ yields the area of the rectangle, $(b-a)(d-c)$.



(a) $\int_a^b f(x)\,\mathrm{d}x$



(b) $\int_a^b \int_c^d f(x,y)\,\mathrm{d}x\,\mathrm{d}y$



(c) $R = [a,b] \times [c,d]$

Double integrals such as

$$\iint_R f(x,y)\,\mathrm{d}x\;\mathrm{d}y$$

consist of three components: i) the integrand f; ii) the region of integration R; iii) the order of integration (e.g. $\mathrm{d}x\,\mathrm{d}y$). We can calculate double integrals by reducing them to repeated integrals. We discuss how to do this in a few cases.

Rectangular Domains

The simplest situation to consider is that of rectangular domains, i.e. when the domain of integration is a rectangular region $R = \{(x,y) | a \le x \le b, c \le y \le d\}$. In this case we can write the double integral as the repeated integral

$$\iint_R f(x,y)\,\mathrm{d}x\;\mathrm{d}y = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\;\mathrm{d}y = \int_c^d \left(\int_a^b f(x,y)\,\mathrm{d}x \right) \mathrm{d}y.$$

If we inspect the inner integral we note that y is fixed and the integral is only happening with respect to the variable x. As such we can just treat y as a constant when calculating the inner integral and evaluate

$$\int_a^b f(x,y)\,\mathrm{d}x$$

as a typical one-dimensional integral. Once we have evaluated this integral we have a function of only one variable y (which also depends on a and b) and we need to integrate this from c to d which is again a one dimensional integral. In this way we have reduced the two dimensional integration into 2 one-dimensional integrals.

Example 6.5. Consider

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y.$$

The first step is to plot the area of integration; this is shown in Figure 6.2. After plotting the

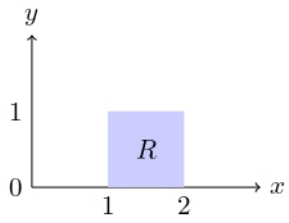


Figure 6.2: The region $R = [1, 2] \times [0, 1]$

area of integration our next step is to evaluate the inner integral, that is, we calculate

$$\int_1^2 xy \, \mathrm{d}x = \left[\frac{1}{2}x^2 y\right]_{x=1}^{x=2} = \frac{1}{2}4y - \frac{1}{2}y = \frac{3}{2}y.$$

Now to calculate the double integral we have

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \frac{3}{2}\int_0^1 y \, \mathrm{d}y = \frac{3}{2}\left[\frac{1}{2}y^2\right]_{y=0}^{y=1} = \frac{3}{4}.$$

Remark 6.1 (A comment on notation). Sometimes we write the integral as

$$\iint_R f(x, y) \, \mathrm{d}x \, \mathrm{d}y \text{ or } \int_c^d \int_{a(y)}^{b(y)} f(x, y) \, \mathrm{d}x \, \mathrm{d}y.$$

Here R is the two dimensional region $\{(x, y)|a(y) \leq x \leq b(y), c \leq y \leq d\}$. In the case of rectangular domains a and b are constants. In Example 6.5 the region of integration is shown in Figure 6.2. Under general assumptions (which will hold for all the examples in this course) you can swap the order of integration and do the y integral first followed by the x integral. Always keep track of the limits when you swap the order! For more details see Multivariable Calculus, Fubini's Theorem.

Example 6.6 (Example 6.5 continued)**.** We have already seen that

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \frac{3}{4}.$$

Let us verify for this example that we can swap the order of integration, i.e.

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \int_1^2 \int_0^1 xy \, \mathrm{d}y \, \mathrm{d}x.$$

We first calculate the inner integral

$$\int_0^1 xy \, \mathrm{d}y = \left[\frac{1}{2}xy^2\right]_{y=0}^{y=1} = \frac{1}{2}x.$$

To calculate the double integral we have

$$\int_0^1 \int_1^2 xy \, \mathrm{d}x \, \mathrm{d}y = \int_1^2 \frac{1}{2}x \, \mathrm{d}x = \left[\frac{1}{4}x^2\right]_{x=1}^{x=2} = \frac{3}{4}.$$

Triangular Domains

The next type of domain we consider are triangular domains, which is when the region of integration R is a triangular shape. For example

$R = \{(x, y) | 0 \leq x \leq y, 0 \leq y \leq 1\}$.

A good first step for any double integration question is to draw the domain. For this example a sketch of R is given in Figure 6.3. A useful approach to sketching R is to first plot the boundary lines, then test a few points to identify the relevant region.

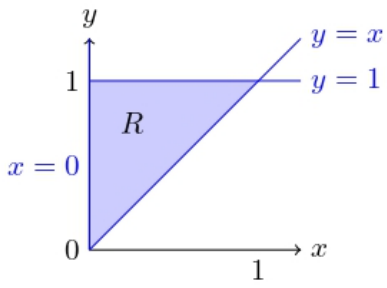


Figure 6.3

As with rectangular domains, we can evaluate double integrals over triangular domains by evaluating repeated integrals, only now we need to be aware that the limits of the integral are also functions of the variable.

Example 6.7. Consider how to calculate

$$\iint_R xy \, \mathrm{d}x \, \mathrm{d}y$$

where $R = \{(x, y) | 0 \leq x \leq y, 0 \leq y \leq 1\}$. Our first step is to draw the region which we have already done in Figure 6.3. In order to write this as a repeated integral we want to write it in the form

$$\int_c^d \int_{a(y)}^{b(y)} f(x, y) \, \mathrm{d}x \, \mathrm{d}y.$$

From the figure we see that y can vary between 0 and 1 then for each fixed y the variable x takes values from 0 to y, so we have $c = 0, d = 1, a(y) = 0$ and $b(y) = y$, that is

$$\iint_R xy \, \mathrm{d}x \, \mathrm{d}y = \int_0^1 \int_0^y xy \, \mathrm{d}x \, \mathrm{d}y.$$

Note that we could have chosen to do the integral in the other order, in which case x is the variable of the outermost integral. In this case x takes values from 0 to 1, then for each fixed x the variable y varies between y to 1. Therefore we have

$$\iint_R xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_0^y xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_x^1 xy \,\mathrm{d}y \,\mathrm{d}x$$

We can use either of these to evaluate the integral

$$\iint_R xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_0^y xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 y \left[\frac{1}{2}x^2\right]_{x=0}^{x=y} \mathrm{d}y = \int_0^1 \frac{1}{2}y^3 \,\mathrm{d}x = \left[\frac{1}{8}y^4\right]_{y=0}^{y=1} = \frac{1}{8}.$$

Note in the examples we have considered so far there has been little reason to change the order of integration. However sometimes it can be very important, consider for example

$$\int_0^1 \int_y^1 e^{-x^2} \,\mathrm{d}x \,\mathrm{d}y.$$

In this case there is no explicit formula for the integral of e^{-x^2} however if we exchange the integrals we have

$$\int_0^1 \int_y^1 e^{-x^2} \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \int_0^x e^{-x^2} \,\mathrm{d}y \,\mathrm{d}x.$$

Note that in order to determine the limits we should draw the region and use that to work out the new limits but as this is the same as the earlier example we refer to Figure 6.3.
Now if we evaluate the inner integral

$$\int_0^1 \int_0^x e^{-x^2} \,\mathrm{d}y \,\mathrm{d}x = \int_0^1 xe^{-x^2} \,\mathrm{d}x.$$

Observe that $-\frac{1}{2}\frac{d}{dx}(e^{-x^2}) = xe^{-x^2}$ so we have

$$\int_0^1 \int_0^x e^{-x^2} \,\mathrm{d}y \,\mathrm{d}x = -\frac{1}{2}\int_0^1 \frac{d}{dx}(e^{-x^2}) \,\mathrm{d}x = -\frac{1}{2}(e^{-1} - 1) \approx 0.316.$$

More complicated regions

In the regions we have considered so far the area R is a rectangle or a triangle however the region can get more complicated. For instance in the following example the boundary of R is not comprised of straight lines.

Example 6.8. Determine

$$\iint_R y \,\mathrm{d}x \,\mathrm{d}y$$

where $R = \{(x, y) : 0 \leq y \leq 1, x^2 + y^2 \leq 1\}$ is a unit semicircle. As always we start by drawing the region, which is done in Figure 6.4.

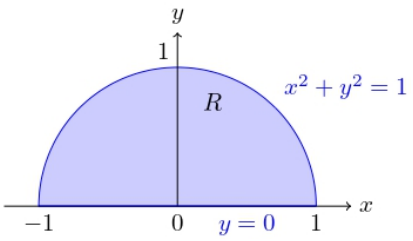


Figure 6.4

Next, we need to choose the order of integration, in this example we will show both starting with using y for the outer integral.

Solution 1: If we use y for the outer integral then we observe that y varies from 0 to 1, for fixed y we have that x runs from $-\sqrt{1-y^2}$ to $\sqrt{1-y^2}$. Therefore

$$\iint_R y\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} y\,\mathrm{d}x\,\mathrm{d}y = 2\int_0^1 y\sqrt{1-y^2}\,\mathrm{d}y.$$

Observe that $\frac{d}{dy}(1-y^2)^{3/2} = -3y(1-y^2)^{1/2}$ so we have

$$\iint_R y\,\mathrm{d}x\,\mathrm{d}y = -\frac{2}{3}\int_0^1 \frac{d}{\mathrm{d}y}(1-y^2)^{3/2}\,\mathrm{d}y = \frac{2}{3}.$$

Solution 2: Alternatively, if we use x for the outermost integral then x varies from -1 to 1, then for fixed x the variable y runs from 0 to $\sqrt{1-x^2}$. In which case

$$\begin{aligned}\iint_R y\,\mathrm{d}x\,\mathrm{d}y &= \int_{-1}^1 \int_0^{\sqrt{1-x^2}} y\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-1}^1 \left[\frac{1}{2}y^2\right]_{y=0}^{y=\sqrt{1-x^2}} \mathrm{d}x \\ &= \frac{1}{2}\int_{-1}^1 1-x^2\,\mathrm{d}x \\ &= \frac{1}{2}\left[x - \frac{1}{3}x^3\right]_{x=-1}^{x=1} = \frac{2}{3}.\end{aligned}$$

Example 6.9. Consider the following

$$\iint_P xy\,\mathrm{d}y\,\mathrm{d}x$$

for $P = \{(x,y)|0 \le x, 0 \le y \le x, y \le 1, x \le 1+y\}$. We start by plotting P, we do this by first plotting all the constraints of P. Then by testing suitable points we judge which region is P, this has been done in Figure 6.5. We see that P consists of the union of the blue-shaded and red cross-hatched regions. We will start by determining the limits of integration using x for the outer integral. Observe that x runs from 0 to 2 however the y limits vary depending on whether $x \le 1$ (blue shaded area) or $x \ge 1$ (red shaded area). We can deal with this issue by splitting the integral and writing

$$\iint_P xy\,\mathrm{d}y\,\mathrm{d}x = \iint_{P_1} xy\,\mathrm{d}y\,\mathrm{d}x + \iint_{P_2} xy\,\mathrm{d}y\,\mathrm{d}x$$

where $P_1 = \{(x,y)|0 \le x \le 1, 0 \le y \le x\}$ is the blue shaded area and $P_2 = \{(x,y)|1 \le x, x-1 \le y \le 1\}$ is the red shaded area. Now we have

$$\iint_P xy \,\mathrm{d}y \,\mathrm{d}x = \int_0^1 \int_0^x xy \,\mathrm{d}y \,\mathrm{d}x + \int_1^2 \int_{x-1}^1 xy \,\mathrm{d}y \,\mathrm{d}x.$$

Alternatively, if we swap the order of integration then we no longer need to split the integration. Indeed if we take y to be the outer integral then y runs from 0 to 1 and for fixed y we have that x runs from y to $1+y$. This gives

$$\begin{aligned}
\iint_P xy \,\mathrm{d}y \,\mathrm{d}x &= \int_0^1 \int_y^{1+y} xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 \left[\frac{1}{2}x^2 y\right]_{x=y}^{x=1+y} \mathrm{d}y \\
&= \frac{1}{2}\int_0^1 (1+y)^2 y - y^3 \,\mathrm{d}y \\
&= \frac{1}{2}\int_0^1 (y + 2y^2) \,\mathrm{d}y \\
&= \frac{1}{2}\left[\frac{1}{2}y^2 + \frac{2}{3}y^3\right]_{y=0}^{y=1} = \frac{7}{12}.
\end{aligned}$$

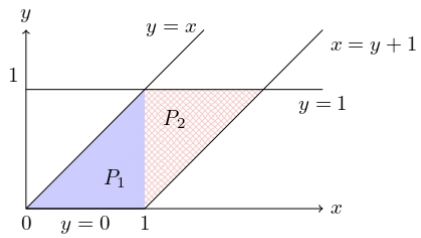


Figure 6.5

Summary of double integration

In each of our examples we have used the following steps to evaluate a double integral:

1. Draw the area of integration;
2. Decide the order of integration to use;
3. Determine the correct limits of the integration (if needed split the integral);
4. Evaluate the double integration using repeated integrals.

Useful checks to avoid errors include:

- Once you have evaluated an integral the variable of integration should no longer appear;
- If $f(x,y) \ge 0$ (for example if f is a probability density) then $\iint_R f(x,y) \,\mathrm{d}x \,\mathrm{d}y \ge 0$.

6.4 Joint probability density function for continuous random variables

Although every pair of random variables (X, Y) has a joint distribution function $F_{X,Y}$, for discrete random variables it is usually more convenient to work with the joint probability mass function. Similarly, for continuous random variables it is often more convenient to use the joint probability density function, defined as follows.

Definition
We say that the random variables X and Y are **jointly continuous** random variables if there is a non-negative function $f_{X,Y}(x, y)$ with

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y = 1$$

such that for any $a \leqslant b$ and $c \leqslant d$,

$$\boxed{\mathbb{P}(a \leqslant X \leqslant b,\ c \leqslant Y \leqslant d) = \int_c^d \int_a^b f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y}$$

and, more generally, for any 'nice' set $A \subseteq \mathbb{R}^2$,

$$\mathbb{P}\left((X,Y) \in A\right) = \iint_A f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

The function $f_{X,Y}$ is called the **joint density function** of X and Y.

Remark: Note that

$$\int_c^d \left(\int_a^b f_{X,Y}(x,y)\ \mathrm{d}x\right)\mathrm{d}y = \int_a^b \left(\int_c^d f_{X,Y}(x,y)\ \mathrm{d}y\right)\mathrm{d}x$$

and more generally,

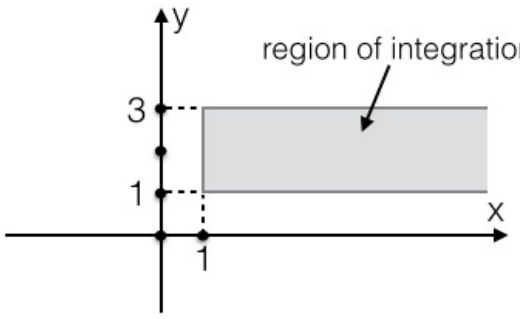
$$\iint_A f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y = \iint_A f_{X,Y}(x,y)\ \mathrm{d}y\ \mathrm{d}x$$

That is, we can do the integrations either way round.

Example 6.10. Suppose that X and Y are jointly continuous with joint density function given by

$$f_{X,Y}(x,y) = \begin{cases} 3e^{-x-3y} & x, y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{P}(1 \leqslant X,\ 1 \leqslant Y \leqslant 3)$
To compute this probability, we need to set up and evaluate a double integral of the joint density over the region in $\mathbb{R}^2$ that corresponds to the inequalities $1 \leq X$ and $1 \leq Y \leq 3$



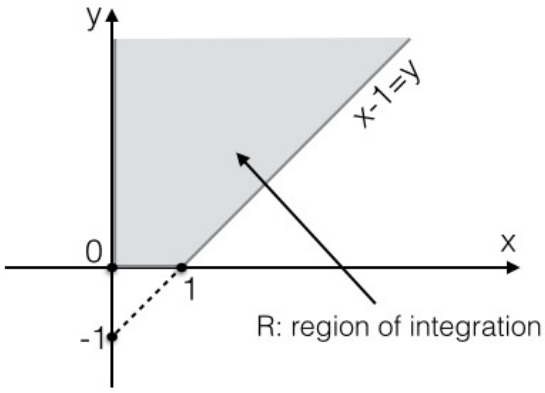
$$\begin{aligned}
\mathbb{P}(1 \leqslant X,\ 1 \leqslant Y \leqslant 3) &= \int_{y=1}^{3} \left(\int_{x=1}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}x \right) \mathrm{d}y \\
&= \int_{y=1}^{3} \left(\int_{x=1}^{\infty} 3e^{-x-3y}\ \mathrm{d}x \right) \mathrm{d}y \\
&= \int_{y=1}^{3} \left(\int_{x=1}^{\infty} e^{-x}\ \mathrm{d}x \right) 3e^{-3y} \mathrm{d}y \\
&= \int_{y=1}^{3} \left[-e^{-x} \right]_{x=1}^{\infty}\ 3e^{-3y} \mathrm{d}y \\
&= \int_{y=1}^{3} \left(e^{-1} - \lim_{x \to \infty} e^{-x} \right)\ 3e^{-3y} \mathrm{d}y \\
&= e^{-1} \int_{y=1}^{3} 3e^{-3y} \mathrm{d}y \\
&= e^{-1} \left[-e^{-3y} \right]_{y=1}^{3} \\
&= e^{-1}(e^{-3} - e^{-9}) \approx 0.0183
\end{aligned}$$

Alternatively (*exercise*), you can compute

$$\mathbb{P}(1 \leqslant X,\ 1 \leqslant Y \leqslant 3) = \int_{x=1}^{\infty} \left(\int_{y=1}^{3} f_{X,Y}(x,y)\ \mathrm{d}y \right) \mathrm{d}x$$

Find $\mathbb{P}(X - 1 \leqslant Y)$
This calculation is trickier because the region of integration for the double integral of the joint density is not a rectangle. This means that we have to think more carefully about the *order* of the integration in the double integral and the *limits* of integration in the double integral.

Step 1:
To work out how to set up the double integral, the first step is sketch the region in the plane where the inequality $X - 1 \leqslant Y$ is satisfied *and* $f_{X,Y}(x,y) > 0$ (we don't need to integrate over regions where $f_{X,Y}(x,y) = 0$).
This (infinite) region looks like this:



Step 2:
The next step is to set up and evaluate the double integral of the joint density over the region

of integration determined in Step 1 above.
Tip: Try to choose the order of integration to make the calculation as easy as possible.
So:

$$\begin{aligned}
\mathbb{P}(X-1 \leqslant Y) = \mathbb{P}\big((X,Y)\in R\big) &= \int_{y=0}^{\infty}\left(\int_{x=0}^{y+1} 3e^{-x}e^{-3y}\,\mathrm{d}x\right)\mathrm{d}y \\
&= \int_{y=0}^{\infty}\left(\int_{x=0}^{y+1} e^{-x}\,\mathrm{d}x\right)3e^{-3y}\,\mathrm{d}y \\
&= \int_{y=0}^{\infty}\left[-e^{-x}\right]_{x=0}^{y+1} 3e^{-3y}\,\mathrm{d}y \\
&= \int_{y=0}^{\infty}\left(1-e^{-(y+1)}\right)3e^{-3y}\,\mathrm{d}y \\
&= \int_{y=0}^{\infty} 3e^{-3y}-3e^{-(4y+1)}\,\mathrm{d}y \\
&= \left[-e^{-3y}+\frac{3}{4}e^{-(4y+1)}\right]_{y=0}^{\infty} \\
&= 1-\frac{3}{4}e^{-1}+\lim_{y\to\infty}\left(-e^{-3y}+\frac{3}{4}e^{-(4y+1)}\right) \\
&= 1-\frac{3}{4}e^{-1}\approx 0.724
\end{aligned}$$

Alternatively (*exercise*), you could integrate with respect to y first and then with respect to x, but to do the calculation in this way, you have to set up two double integrals, because of the shape of the region R (refer to sketch above):

$$\mathbb{P}(X-1\leqslant Y)=\int_{x=0}^{1}\left(\int_{y=0}^{\infty}3e^{-x}e^{-3y}\,\mathrm{d}y\right)\mathrm{d}x\;+\;\int_{x=1}^{\infty}\left(\int_{y=x-1}^{\infty}3e^{-x}e^{-3y}\,\mathrm{d}y\right)\mathrm{d}x=1-\frac{3}{4}e^{-1}$$

Properties of joint probability density functions
If $f:\mathbb{R}^2\to\mathbb{R}$ is the joint probability density function of a pair of random variables X,Y, then:

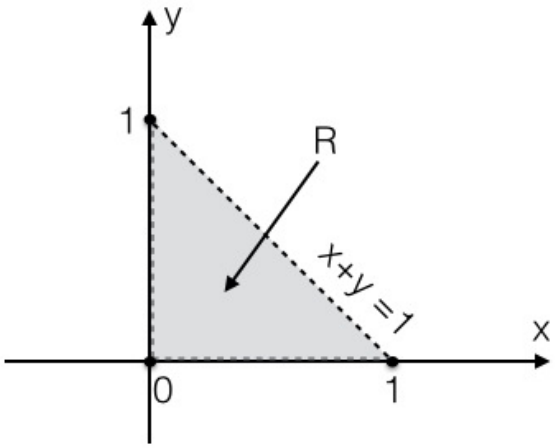
1. $f(x,y)\geq 0$ for all $(x,y)\in\mathbb{R}^2$
2. $\displaystyle\iint_{\mathbb{R}^2} f(x,y)\,\mathrm{d}x\,\mathrm{d}y=1$

Conversely, any function $f:\mathbb{R}^2\to\mathbb{R}$ satisfying the above two properties is a valid joint probability density function.

Example 6.11. Suppose that X and Y are jointly continuous random variables with joint density function given by

$$f_{X,Y}(x,y)=\begin{cases}60xy^2 & x>0,\ y>0,\ x+y<1\\ 0 & \text{otherwise}\end{cases}$$

Remark
The inequalities $x>0,\ y>0$ and $x+y<1$ describe the region of possible values for the ordered pair (X,Y), and this region is *not* a rectangle (due to the constraint $x+y<1$). It is important to understand what this region looks like in order to avoid making mistakes when you calculate probabilities. This region, R, is sketched below:



When working with joint density functions, **always sketch the region of non-zero density.**

To check that $f_{X,Y}(x,y)$ is a density, we note that

1. $f_{X,Y}(x,y) \geqslant 0 \quad \forall (x,y) \in \mathbb{R}^2$
2. The integral of $f_{X,Y}(x,y)$ over $\mathbb{R}^2$ is equal to 1:

$$\iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_{y=0}^{1}\left(\int_{x=0}^{1-y} 60xy^2\,\mathrm{d}x\right)\mathrm{d}y \qquad \text{(Note the limits of integration!)}$$

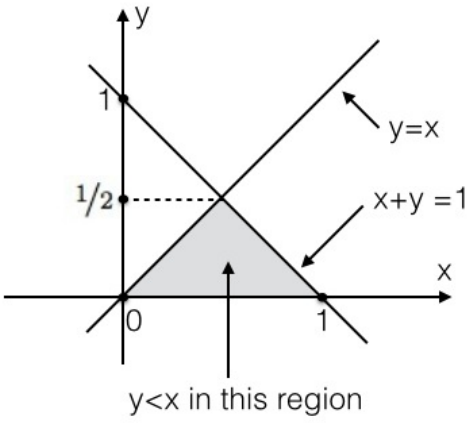
$$= \int_{y=0}^{1}\left[30x^2y^2\right]_{x=0}^{1-y}\,\mathrm{d}y = \int_{y=0}^{1} 30(1-y)^2y^2\,\mathrm{d}y$$

$$= 30\int_{y=0}^{1}(y^2-2y^3+y^4)\,\mathrm{d}y = 30\left[\frac{y^3}{3}-\frac{y^4}{2}+\frac{y^5}{5}\right]_{y=0}^{1} = 1$$

Determine $\mathbb{P}(Y < X)$

First step: To compute this probability, we have to integrate the joint density function $f_{X,Y}(x,y)$ over the region where $f_{X,Y}(x,y) > 0$ and $y < x$. So the first step is to sketch the region in the plane where these inequalities are satisfied in order to determine the region of integration. This sketch will help you to work out the correct limits of integration.

Here is a sketch of A, the region of integration:



Next step: The next step is to set up the double integral. To do this you will need to decide the order of integration. You should use the order of integration that makes the calculation as

uncomplicated as possible:

$$\mathbb{P}(Y < X) = \iint_A f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y = \int_{y=0}^{1/2} \left(\int_{x=y}^{1-y} 60xy^2\ \mathrm{d}x \right) \mathrm{d}y$$

Finally: Compute the double integral to obtain

$$\mathbb{P}(Y < X) = \frac{5}{16} \qquad \text{(Check this!)}$$

Remark: You can also compute $\mathbb{P}(Y < X)$ by integrating with respect to y first and then with respect to x. If you do the calculation this way, then you will need to set up *two* integrals:

$$\mathbb{P}(Y < X) = \int_{x=0}^{1/2} \left(\int_{y=0}^{x} 60xy^2\ \mathrm{d}y \right) \mathrm{d}x\ +\ \int_{x=1/2}^{1} \left(\int_{y=0}^{1-x} 60xy^2\ \mathrm{d}y \right) \mathrm{d}x$$

Example 6.12. Suppose that X and Y are jointly continuous random variables with joint density function $f_{X,Y}(x,y)$ given in example 6.11. Let $Z = X + Y$. Find the distribution function F_Z and density f_Z for the random variable Z.

Step 1: Determine ImZ
Since the joint density for X and Y is given by:

$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

we must have $0 < X + Y < 1$. So $ImZ = (0,1)$.
Step 2: Determine F_Z
Since $ImZ = (0,1)$, we know that $F_Z(z) = 0$ for $z < 0$ and $F_Z(z) = 1$ for $z \geqslant 1$.
For $0 \leqslant z < 1$, we use the definition of the distribution function and Z to obtain:

$$F_Z(z) = \mathbb{P}(Z \leqslant z) = \mathbb{P}(X + Y \leqslant z)$$

To determine $\mathbb{P}(X + Y \leqslant z)$ we have to integrate the joint density $f(x,y)$ for X and Y over the region in the plane that corresponds to $0 < x + y \leqslant z$, i.e.

$$\begin{aligned}
\mathbb{P}(Z \leqslant z) = \mathbb{P}(X + Y \leqslant z) &= \int_{x=0}^{z} \left(\int_{y=0}^{z-x} f(x,y)\mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{z} \left(\int_{y=0}^{z-x} 60xy^2 \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{z} \left[20xy^3 \right]_{y=0}^{z-x} \mathrm{d}x \\
&= \int_{x=0}^{z} 20x(z-x)^3 \mathrm{d}x \\
&= 20 \int_{x=0}^{z} xz^3 - 3x^2z^2 + 3x^3z - x^4\ \mathrm{d}x \\
&= 20 \left[\frac{x^2z^3}{2} - x^3z^2 + \frac{3x^4z}{4} - \frac{x^5}{5} \right]_{x=0}^{z} \\
&= 20 \left(\frac{z^5}{2} - z^5 + \frac{3z^5}{4} - \frac{z^5}{5} \right) \\
&= z^5
\end{aligned}$$

So the distribution function of Z is

$$F_Z(z) = \begin{cases} 0 & z < 0 \\ z^5 & 0 \leqslant z < 1 \\ 1 & z \geqslant 1 \end{cases}$$

Step 3: Determine f_Z.

We differentiate F_Z to obtain $f_Z(z) = 5z^4$ for $0 < z < 1$ and $f_Z(z) = 0$ otherwise.

Example 6.13. Let X and Y be two continuous random variables. Determine the distribution of $Z = X + Y$.

We need to determine

$$F_Z(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(X + Y \leq z).$$

If we write this in terms of the joint probability density function

$$F_Z(z) = \mathbb{P}(X + Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{X,Y}(x, y) \, \mathrm{d}x \, \mathrm{d}y.$$

To find the probability density function of Z we differentiate this expression with respect to z, by using the fundamental theorem of calculus we have

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} \frac{d}{dz} \left[\int_{-\infty}^{z-y} f_{X,Y}(x, y) \, \mathrm{d}x \, \mathrm{d}y \right] = \int_{-\infty}^{\infty} f_{X,Y}(z - y, y) \, \mathrm{d}y.$$

By substituting $x = z - y$ we can alternatively write this as

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z - y, y) \, \mathrm{d}y = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x) \, \mathrm{d}x.$$

We have used very little information about X and Y up to this point. If we now impose that X and Y are independent then we can write the density of Z as

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y) \, \mathrm{d}y = \int_{-\infty}^{\infty} f_X(x) f_Y(x - y) \, \mathrm{d}x.$$

This operation is known as the convolution of f_X and f_Y, denoted by $(f_X * f_Y)(z)$. The convolution is a symmetric operation in the sense that $(f_X * f_Y)(z) = (f_Y * f_X)(z)$.

While the convolution integral may not always yield a closed-form expression, there are several important cases where the result is well-known:

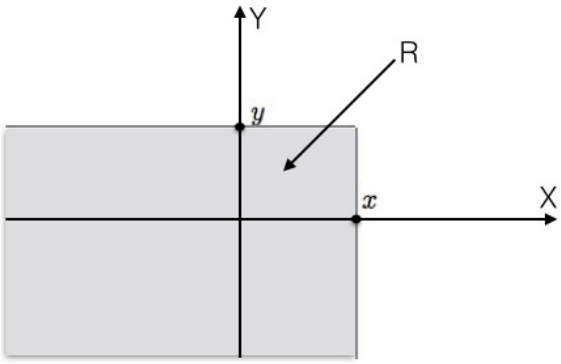
- If $X \sim \mathrm{N}\left(\mu_x, \sigma_x^2\right)$ and $Y \sim \mathrm{N}\left(\mu_y, \sigma_y^2\right)$ are independent then $X + Y \sim \mathrm{N}\left(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2\right)$.
- If $X \sim \mathrm{Exp}(\lambda)$ and $Y \sim \mathrm{Exp}(\lambda)$ are independent then $X + Y \sim \mathrm{Gamma}\,(2, \lambda)$.
- If $X \sim \mathrm{Gamma}\,(\alpha, \lambda)$ and $Y \sim \mathrm{Gamma}\,(\beta, \lambda)$ are independent then $X + Y \sim \mathrm{Gamma}\,(\alpha + \beta, \lambda)$.

6.4.1 Relationship between joint distribution function and joint density function

By definition of the joint distribution function $F_{X,Y}(x, y)$ and the joint density $f_{X,Y}(x, y)$, we have for $x, y \in \mathbb{R}$

$$F_{X,Y}(x, y) \stackrel{def}{=} \mathbb{P}(X \leqslant x, Y \leqslant y)$$

$$F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u, v) \, \mathrm{d}u \, \mathrm{d}v = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v) \, \mathrm{d}v \, \mathrm{d}u$$



where the region of integration corresponds to the (infinite) shaded region R below:
So, given the joint density $f_{X,Y}$, we can find the joint distribution function $F_{X,Y}$ by integrating the joint density $f_{X,Y}$ over an appropriate region.

Conversely, if $F_{X,Y}$ is *continuous* and is differentiable 'almost everywhere', then

$$f_{X,Y}(x,y) = \frac{\partial}{\partial x}\frac{\partial}{\partial y}F_{X,Y}(x,y) = \frac{\partial}{\partial y}\frac{\partial}{\partial x}F_{X,Y}(x,y)$$

wherever $F_{X,Y}$ is differentiable.

6.4.2 Marginal density functions

Suppose that X and Y are jointly continuous with joint density $f_{X,Y}$, then the **(marginal) density function** for X is given by

$$f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y \qquad \text{(integrate out } y\text{)}$$

and the **(marginal) density function** for Y is given by

$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}x \qquad \text{(integrate out } x\text{)}$$

Proof. With $f_X(x)$ given by the above formula, for any $a, b \in \mathbb{R}$ with $a \leqslant b$ we have

$$\int_a^b f_X(x)\ \mathrm{d}x \overset{def}{=} \int_{x=a}^{b}\left(\int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y\right)\ \mathrm{d}x = \mathbb{P}(a \leqslant X \leqslant b,\ -\infty < Y < \infty)$$
$$= \mathbb{P}(a \leqslant X \leqslant b) \qquad \text{(since there is no restriction on } Y\text{)}$$

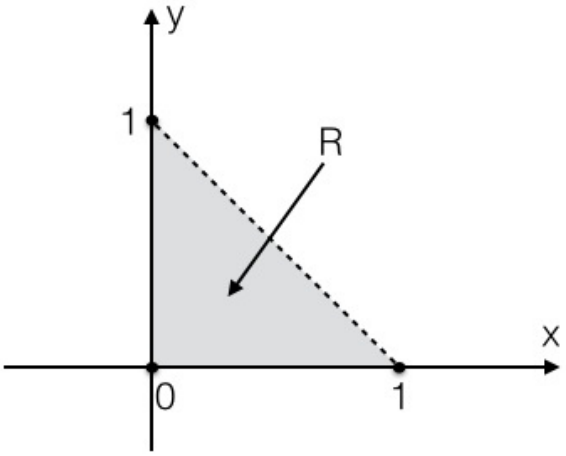
Furthermore, since the joint density satisfies $f_{X,Y}(x,y) \geqslant 0$ for all $x, y \in \mathbb{R}$, it follows that $f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y \geqslant 0$.
So the function $f_X(x)$ as defined above satisfies the properties of the density function for X.
A similar argument shows that the function $f_Y(y)$ as defined above is the density function for the random variable Y.

Example 6.11 continued. Suppose that X and Y are jointly continuous random variables with joint density function given by

$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Determine the marginal density f_X for X.

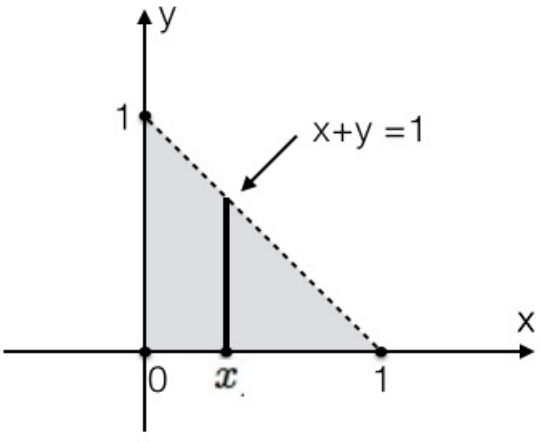
First step: Look at the region R where the joint density $f_{X,Y}(x,y)$ is non-zero.
We can determine from this region ImX, the set of possible values for X.



From this picture it is clear that Im$X = (0,1)$ since points where $f_{X,Y}(x,y) > 0$ occur for $0 < x < 1$. So we have $f_X(x) = 0$ if $x \leqslant 0$ or $x \geqslant 1$.

For $0 < x < 1$,

$$\begin{aligned} f_X(x) &= \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y \\ &= \int_{y=0}^{1-x} f_{X,Y}(x,y)\ \mathrm{d}y \qquad \text{(The limits of integration are based on the diagram below)} \\ &= \int_{y=0}^{1-x} 60xy^2\ \mathrm{d}y \\ &= \left[20xy^3\right]_{y=0}^{1-x} = 20x(1-x)^3 \\ &= 20x - 60x^2 + 60x^3 - 20x^4 \end{aligned}$$



Remark: You can check (*exercise*) that this is a valid density:

$$\int_{-\infty}^{\infty} f_X(x)\ \mathrm{d}x = \int_0^1 (20x - 60x^2 + 60x^3 - 20x^4)\ \mathrm{d}x = 1.$$

Exercise: Use f_X to show that $\mathbb{P}(\frac{1}{4} \leqslant X \leqslant \frac{1}{2}) = \frac{57}{128}$.

Expected value and variance of X

Once we have computed $f_X(x)$, the marginal density for X, we can use this to compute $\mathbb{E}(X)$ and $\mathbb{V}\mathrm{ar}(X)$:

$$\begin{aligned}
\mathbb{E}(X) = \int x f_X(x)\ \mathrm{d}x &= \int_0^1 x(20x - 60x^2 + 60x^3 - 20x^4)\ \mathrm{d}x \\
&= \int_0^1 20x^2 - 60x^3 + 60x^4 - 20x^5\ \mathrm{d}x \\
&= \left[\frac{20x^2}{3} - 15x^4 + 12x^5 - \frac{10x^6}{3}\right]_0^1 \\
&= \frac{20}{3} - 15 + 12 - \frac{10}{3} = \frac{1}{3}
\end{aligned}$$

Exercise: Show $\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - \big(\mathbb{E}(X)\big)^2 = \frac{2}{63}$

Exercise: Show that the marginal density f_Y for Y is given by

$$f_Y(y) = \begin{cases} 30y^2 - 60y^3 + 30y^4 & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Use $f_Y(y)$

1. to compute $\mathbb{P}(\frac{1}{4} \leqslant Y \leqslant \frac{1}{2})$, and

2. to show that $\mathbb{E}(Y) = \frac{1}{2}$ and $\mathbb{V}\mathrm{ar}(Y) = \frac{1}{28}$

Remark
The example above illustrates the general principle that given the joint density $f_{X,Y}(x, y)$ for the variables X and Y, we can always find the marginal density functions f_X and f_Y for the individual variables X and Y. Just as in the discrete case, the converse is *not* true. Given f_X and f_Y, it is not possible to determine the joint density $f_{X,Y}(x, y)$ for X and Y.

Independent jointly-continuous random variables

Recall that X and Y are independent if and only if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \qquad \forall (x, y) \in \mathbb{R}^2$$

So if X and Y are jointly continuous random variables with joint density function $f_{X,Y}(x, y)$, then, by differentiating the equation above, we see that X and Y are *independent* if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \qquad \forall (x, y) \in \mathbb{R}^2$$

i.e. joint density function = product of the marginal density functions **if independent**.

More generally, the random variables $X_1, X_2, \ldots, X_n$ with joint density function $f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n)$ are *independent* if and only if

$$f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \ldots f_{X_n}(x_n) \qquad \forall (x_1, \ldots x_n)$$

Remarks

1. If we know that continuous random variables X and Y are independent, then we can find their joint density function from their marginal probability density functions using the relationship $f_{X,Y}(x, y) = f_X(x) f_Y(y)$. Can only do this if we know that X, Y are independent.
2. When the variables $X_1, X_2, \ldots, X_n$ depend on *different experiments* which are *physically independent*, then it is reasonable to assume that $X_1, X_2, \ldots, X_n$ are (statistically) independent and we can compute their joint density function by multipying together their marginal density functions:

$$f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n) = \underbrace{f_{X_1}}_{\substack{\text{density} \\ \text{for } X_1}}(x_1) \cdot \underbrace{f_{X_2}}_{\substack{\text{density} \\ \text{for } X_2}}(x_2) \cdots \underbrace{f_{X_n}}_{\substack{\text{density} \\ \text{for } X_n}}(x_n)$$

3. In general, knowing the marginal density functions $f_{X_1}, f_{X_2}, \ldots, f_{X_n}$ is not sufficient to be able to find the joint probability density function $f_{X_1, X_2, \ldots, X_n}$
4. *Given* the joint density $f_{X,Y}$ for X and Y (say), we can compute f_X and f_Y and check whether the condition for independence is satisfied.
   - If $f_{X,Y}(x, y) = f_X(x) f_Y(y) \quad \forall (x, y) \in \mathbb{R}^2$ then X, Y are independent
   - If $f_{X,Y}(x, y) \neq f_X(x) f_Y(y)$ for some $(x, y) \in \mathbb{R}^2$ then X, Y are not independent

   Example 6.11 continued. In Example 6.11, we were given the joint density function
   $f_{X,Y}(x, y) = 60xy^2$ for $x > 0,\ y > 0,\ x + y < 1$
   and have seen that the marginal density functions corresponding to this are
   $f_X(x) = 20x - 60x^2 + 60x^3 - 20x^4$ for $0 < x < 1$
   $f_Y(y) = 30y^2 - 60y^3 + 30y^4$ for $0 < y < 1$
   Clearly $f_{X,Y}(x, y) \neq f_X(x) f_Y(y)$, so the random variables X and Y are not independent. (In this example, you can check this simply by looking at the regions of non-zero density; for any point (x, y) such that $0 < x < 1$, $0 < y < 1$ and $x + y > 1$ we have $f_{X,Y}(x, y) = 0$ but $f_X(x) f_Y(y) \neq 0$.)

6.4.3 Conditional probabilities and conditional densities

Example 6.11 continued. Suppose X and Y have joint density function

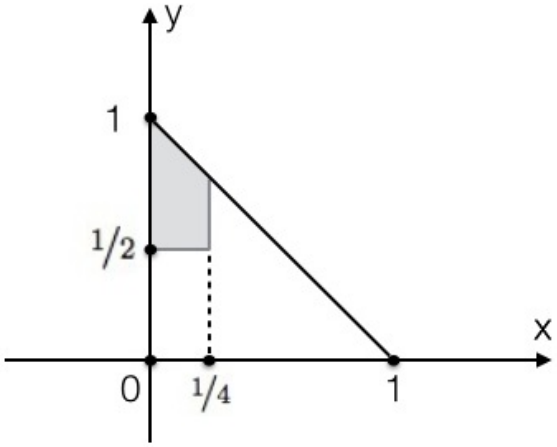
$$f_{X,Y}(x, y) = \begin{cases} 60xy^2 & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Then from the definition of conditional probability for *events* we have

$$\mathbb{P}(Y \geqslant 1/2 \mid X \leqslant 1/4) = \frac{\mathbb{P}(Y \geqslant 1/2,\ X \leqslant 1/4)}{\mathbb{P}(X \leqslant 1/4)}$$

To compute the right hand side of the equation above, we note that:

- The numerator is the probability of a joint event, so we must use the joint density $f_{X,Y}$ to compute it.
- The denominator is the probability of a marginal event, so we can use the marginal density f_X to compute it.



$$\begin{aligned}
\mathbb{P}(Y \geqslant 1/2,\ X \leqslant 1/4) &= \int_{x=0}^{1/4} \left(\int_{y=1/2}^{1-x} f_{X,Y}(x,y)\ \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{1/4} \left(\int_{y=1/2}^{1-x} 60xy^2\ \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{1/4} \left[20xy^3 \right]_{y=1/2}^{1-x} \mathrm{d}x \\
&= \int_{x=0}^{1/4} 20x(1-x)^3 - \frac{5x}{2}\ \mathrm{d}x \\
&= \int_{x=0}^{1/4} 20x(1-3x+3x^2-x^3) - \frac{5x}{2}\ \mathrm{d}x \\
&= \int_{x=0}^{1/4} \frac{35x}{2} - 60x^2 + 60x^3 - 20x^4\ \mathrm{d}x \\
&= \left[\frac{35x^2}{4} - 20x^3 + 15x^4 - 4x^5 \right]_{x=0}^{1/4} = \frac{35}{64} - \frac{5}{16} + \frac{15}{256} - \frac{1}{256} = \frac{37}{128}
\end{aligned}$$

$$\mathbb{P}(X \leqslant 1/4) = \int_0^{1/4} f_X(x)\ \mathrm{d}x = \int_0^{1/4} (20x - 60x^2 + 60x^3 - 20x^4)\ \mathrm{d}x = \frac{47}{128}\ (\textit{exercise: check this})$$

So:

$$\mathbb{P}(Y \geqslant 1/2 \mid X \leqslant 1/4) = \frac{\mathbb{P}(Y \geqslant 1/2,\ X \leqslant 1/4)}{\mathbb{P}(X \leqslant 1/4)} = \frac{37}{47} \approx 0.787$$

Question:
Suppose we have *observed* that $X = \frac{1}{4}$. Can we determine $\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4)$ *even though* $\mathbb{P}(X = 1/4) = 0$?

Answer:
We can compute $\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4)$ by using the conditional density for Y *given* $X = 1/4$.

We define the **conditional density** as follows:

Definition
Suppose X and Y are jointly continuous with joint density function $f_{X,Y}(x,y)$. Then the **conditional density of Y given $X = x$** is defined as

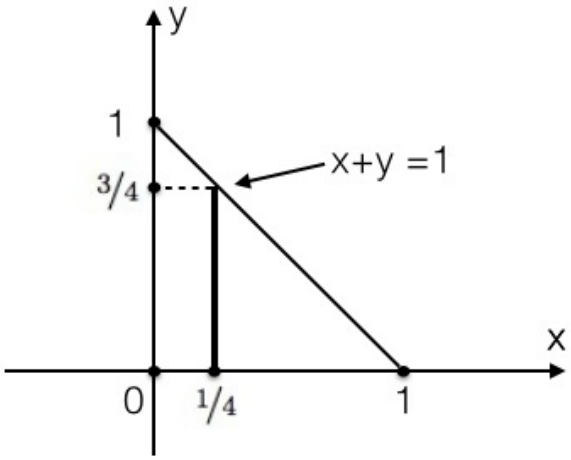
$$f_{Y|X}(y|x) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_X(x)} & \text{provided } f_X(x) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Likewise, the **conditional density of X given $Y = y$** is defined by

$$f_{X|Y}(x|y) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_Y(y)} & \text{provided } f_Y(y) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Example 6.11 continued. We compute $\mathbb{P}(Y \geqslant 1/2 |\ X = 1/4)$ using the conditional density as follows:

Step 1: The first step is to determine the conditional density function $f_{Y|X}(y \mid 1/4)$. When working out the conditional density, we have to be careful to think about where the conditional density $f_{Y|X}(y \mid 1/4)$ is non-zero. A good way to do this is to sketch the region where the joint density $f_{X,Y}(x,y)$ is non-zero and then draw the line corresponding to the *given information* that $X = 1/4$:



We can see from the diagram above that given $X = 1/4$ we must have $0 < Y < 3/4$, and that means that we must have $f_{Y|X}(y \mid 1/4) = 0$ for $y \leqslant 0$ or $y \geqslant 3/4$.

For $0 < y < 3/4$, the conditional density $f_{Y|X}(y \mid 1/4)$ is found as

$$f_{Y|X}(y \mid 1/4) = \frac{f_{X,Y}(1/4, y)}{f_X(1/4)} = \frac{60(1/4)y^2}{20(1/4) - 60(1/4)^2 + 60(1/4)^3 - 20(1/4)^4} = \frac{64}{9}y^2$$

That is,

$$f_{Y|X}(y \mid 1/4) \quad = \quad \begin{cases} 64y^2/9 & 0 < y < 3/4 \\ 0 & \text{otherwise} \end{cases}$$

Remark: It is easy to check that $f_{Y|X}(y \mid 1/4)$ is a valid density: it is clearly non-negative, and

$$\int_{y=0}^{3/4} f_{Y|X}(y \mid 1/4)\ \mathrm{d}y = \int_{y=0}^{3/4} \frac{64}{9}y^2\ \mathrm{d}y = \left[\frac{64y^3}{27}\right]_{y=0}^{3/4} = 1$$

Step 2: The next step is to use the conditional density $f_{Y|X}(y \mid 1/4)$ to compute the conditional probability $\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4)$. Just as we can integrate density functions to compute unconditional probabilities, we can integrate *conditional* density functions to compute *conditional* probabilities:

$$\begin{aligned}\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4) &= \int_{y=1/2}^{3/4} f_{Y|X}(y \mid 1/4)\ \mathrm{d}y \\ &= \int_{y=1/2}^{3/4} \frac{64y^2}{9}\ \mathrm{d}y = \left[\frac{64}{27}y^3\right]_{1/2}^{3/4} = 1 - \frac{8}{27} = \frac{19}{27} \approx 0.704\end{aligned}$$

Note: The upper limit in the integral above is $3/4$ because the conditional density $f_{Y|X}(y \mid 1/4)$ is only non-zero for $0 < y < 3/4$.

Remarks

1. In general, we can compute $\mathbb{P}(Y \in A \mid X = x)$ by integrating the conditional density $f_{Y|X}(y|x)$ over A, i.e.

$$\mathbb{P}(Y \in A \mid X = x) = \int_A f_{Y|X}(y|x)\ \mathrm{d}y$$

   Likewise,

$$\mathbb{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)\ \mathrm{d}x$$

2. Suppose that the random variables X and Y are jointly continuous and independent. Then

$$f_{X|Y}(x|y) \overset{def}{=} \frac{f_{X,Y}(x,y)}{f_Y(y)} \overset{indep}{=} \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x) \quad (\text{provided } f_Y(y) > 0).$$

   It follows that if X and Y are independent,

$$\mathbb{P}(X \in A \mid\ Y = y) = \int_A f_{X|Y}(x|y)\ \mathrm{d}x = \int_A f_X(x)\ \mathrm{d}x = \mathbb{P}(X \in A)$$

   That is, if X and Y are independent, then *conditional probabilities* for X given information about Y are the same as *unconditional probabilities* for X. Similarly, for X, Y independent,

   $f_{Y|X}(y|x) = f_Y(y) \qquad (\text{provided } f_X(x) > 0)$
   $\mathbb{P}(Y \in B \mid\ X = x) = \mathbb{P}(Y \in B).$

3. 'Chain Rule' for densities. It follows from the definition given above that

$$\boxed{f_{X,Y}(x,y) = f_X(x)f_{Y|X}(y|x)}$$

and

$$f_{X,Y}(x,y) = f_Y(y) f_{X|Y}(x|y)$$

This is a *density* version of the Chain Rule. It means that we can compute the joint density from a *marginal density* and the appropriate *conditional density*

Recall the **Law of Total Probability (Partition Theorem)**: Suppose events $E_1, E_2, \ldots$ are disjoint with $\Omega = \cup_i E_i$, and $\mathbb{P}(E_i) > 0 \ \forall i$. Then for *any* event F, we have

$$\mathbb{P}(F) = \sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)$$

For density functions, a corresponding result is as follows.

Law of Total Probability (Partition Theorem) for densities
Suppose that X is a random variable with density f_X and suppose that B is an event. Then

$$\mathbb{P}(B) = \int_{\mathbb{R}} \mathbb{P}(B \mid X = x) f_X(x) \ \mathrm{d}x$$

Special Case
Suppose X and Y are jointly continuous random variables. Then

$$\mathbb{P}(Y \in A) = \int_{\mathbb{R}} \mathbb{P}(Y \in A \mid X = x) f_X(x) \ \mathrm{d}x$$

Proof.

$$\begin{aligned}
&\int_{\mathbb{R}} \mathbb{P}(Y \in A |\ X = x) f_X(x) \ \mathrm{d}x \\
&= \int_{\mathbb{R}} \left(\int_A f_{Y|X}(y|x) \ \mathrm{d}y \right) f_X(x) \ \mathrm{d}x \\
&= \int_A \left(\int_{\mathbb{R}} f_{Y|X}(y|x) f_X(x) \ \mathrm{d}x \right) \mathrm{d}y \qquad \text{(change order of integration)} \\
&= \int_A \left(\int_{\mathbb{R}} f_{X,Y}(x,y) \ \mathrm{d}x \right) \mathrm{d}y \qquad \text{(since } f_{X,Y}(x,y) = f_X(x) f_{Y|X}(y|x)\text{)} \\
&= \int_A f_Y(y) \ \mathrm{d}y \qquad \left(\text{since } \int_{\mathbb{R}} f_{X,Y}(x,y) \ \mathrm{d}x = f_Y(y) \right) \\
&= \mathbb{P}(Y \in A)
\end{aligned}$$

Example 6.14. Application of the Law of Total Probability. Let T denote the random time that Anne leaves her flat in the morning, measured in hours past 7 am, and let X denote the random time, measured in hours past 7am, when Anne's alarm clock rings. Suppose that the density for X is given by

$$f_X(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and suppose that, given $X = x$, the *conditional density* for T is given by

$$f_{T|X}(t|x) = \begin{cases} 2(t-x) & x < t < x+1 \\ 0 & \text{otherwise} \end{cases}$$

Note: We can see from the definition of the density f_X that $ImX = (0, 1)$. That is, Anne's alarm clock rings at some time between 7am and 8am. Also, it follows from the definition of the conditional density $f_{T|X}(t|x)$ that *given* $X = x$, we must have $x < T < x + 1$. That is, Anne leaves her flat at some time between the time when her alarm clock rings, and one hour after her alarm clock rings.

Now suppose that in order to catch her bus to the university, Anne must leave her flat before 8.30am. Determine the probability that tomorrow Anne will leave her flat before 8.30am, i.e. determine $P(T < 3/2)$.

To do this calculation, we can use the Law of Total Probability:.

$$\mathbb{P}(T < 3/2) = \int_0^1 \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x$$

Observe: If $x < 1/2$, then $\mathbb{P}(T < 3/2 \mid X = x) = 1$ because from the definition of the conditional density $f_{T|X}(t|x)$, we know that we always have $T < x + 1 < 3/2$ for $x < 1/2$.

If $1/2 \leq x < 1$, then

$$\begin{aligned}
\mathbb{P}(T < 3/2 \mid X = x) &= \int_{t=x}^{3/2} f_{T|X}(t \mid x)\ dt \\
&= \int_{t=x}^{3/2} 2(t - x)\ dt \\
&= \left[(t - x)^2\right]_{t=x}^{3/2} \\
&= \left(\frac{3}{2} - x\right)^2
\end{aligned}$$

So applying the Law of Total Probability we obtain:

$$\begin{aligned}
\mathbb{P}(T < 3/2) &= \int_{x=0}^{1} \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x \\
&= \int_{x=0}^{1/2} \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x + \int_{x=1/2}^{1} \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x \\
&= \int_{x=0}^{1/2} 1 \cdot 2x\ \mathrm{d}x \quad + \quad \int_{x=1/2}^{1} \left(\frac{3}{2} - x\right)^2 \cdot 2x\ \mathrm{d}x \\
&= \left[x^2\right]_{x=0}^{1/2} + \int_{x=1/2}^{1} \left(\frac{9}{4} - 3x + x^2\right) \cdot 2x\ \mathrm{d}x \\
&= \frac{1}{4} + \int_{x=1/2}^{1} \frac{9x}{2} - 6x^2 + 2x^3\ \mathrm{d}x = \frac{1}{4} + \left[\frac{9x^2}{4} - 2x^3 + \frac{x^4}{2}\right]_{x=1/2}^{1} \\
&= \frac{1}{4} + \left(\frac{9}{4} - 2 + \frac{1}{2}\right) - \left(\frac{9}{16} - \frac{1}{4} + \frac{1}{32}\right) = \frac{1}{4} + \frac{3}{4} - \frac{11}{32} = \frac{21}{32} \approx 0.656
\end{aligned}$$

6.4.4 Conditional expectation

We can use the conditional density $f_{Y|X}(y|x)$ to compute the **conditional expectation of Y** given $X = x$ as follows:

$$\boxed{\mathbb{E}(Y \mid X = x) = \int_{\mathbb{R}} y\ f_{Y|X}(y|x)\ \mathrm{d}y}$$

Likewise, we can compute the **conditional expectation of X** given $Y = y$ by

$$\mathbb{E}(X \mid Y = y) = \int_{\mathbb{R}} x \; f_{X|Y}(x|y) \; \mathrm{d}x$$

Example 6.11 continued. For the joint density given in example 6.11, we showed previously that the conditional density $f_{Y|X}(y \mid 1/4)$ is given by $f_{Y|X}(y \mid 1/4) = 64y^2/9$ for $0 < y < 3/4$ (and zero elsewhere). Using this, we can compute

$$\begin{aligned}\mathbb{E}(Y|\; X = 1/4) = \int_{\mathbb{R}} y \; f_{Y|X}(y|1/4) \; \mathrm{d}y &= \int_0^{3/4} y \left(\frac{64y^2}{9}\right) \mathrm{d}y \\ &= \int_0^{3/4} \frac{64y^3}{9} \; \mathrm{d}y = \left[\frac{16}{9}y^4\right]_0^{3/4} = \frac{9}{16} = 0.5625\end{aligned}$$

Generalisation: More generally, if $g(Y)$ is a function of the random variable Y and $h(X)$ is a function of the random variable X, then we can use the conditional densities for X and Y to compute:

$$\mathbb{E}\big(g(Y)|\; X = x\big) = \int_{\mathbb{R}} g(y) \; f_{Y|X}(y|x) \; \mathrm{d}y$$
$$\mathbb{E}\big(h(X)|\; Y = y\big) = \int_{\mathbb{R}} h(x) \; f_{X|Y}(x|y) \; \mathrm{d}x$$

Exercise: Show that when X and Y are jointly continuous and independent, then $\mathbb{E}(Y|\; X = x) = \mathbb{E}(Y)$ and $\mathbb{E}(X|\; Y = y) = \mathbb{E}(X)$.

Law of Total Expectation (Partition Theorem for Expectation)
Suppose that X and Y are jointly continuous with joint density function $f_{X,Y}(x, y)$.
Then

$$\boxed{\mathbb{E}(X) = \int_{\mathbb{R}} \mathbb{E}(X \mid Y = y) f_Y(y) \, \mathrm{d}y}$$

Proof: Exercise.

Example 6.14 continued. Using the Law of Total Expectation, we can compute $\mathbb{E}(T)$, the expected time that Anne leaves her flat:

$$\mathbb{E}(T) \overset{\text{Law of Total Expectation}}{=} \int_{\mathbb{R}} \mathbb{E}(T \mid X = x) f_X(x) \, \mathrm{d}x = \int_0^1 \mathbb{E}(T \mid X = x) \; 2x \; \mathrm{d}x \qquad (*)$$

Using the conditional density $f_{T|X}(t|x) = 2(t - x)$ for $x < t < x + 1$, we compute

$$\begin{aligned}\mathbb{E}(T \mid X = x) = \int_{t=x}^{x+1} t f_{T|X}(t|x) \; dt &= \int_{t=x}^{x+1} 2t(t - x) \; dt \\ &= \int_{t=x}^{x+1} 2t^2 - 2tx \; dt \\ &= \left[\frac{2t^3}{3} - t^2 x\right]_{t=x}^{x+1} \\ &= \left(\frac{2(x+1)^3}{3} - (x+1)^2 x\right) - \left(\frac{2x^3}{3} - x^3\right) \\ &= x + \frac{2}{3}\end{aligned}$$

and substituting this into $(*)$ yields

$$\mathbb{E}(T) = \int_0^1 \left(x + \frac{2}{3}\right) 2x \ \mathrm{d}x = \int_0^1 2x^2 + \frac{4x}{3} \ \mathrm{d}x = \left[\frac{2x^3}{3} + \frac{2x^2}{3}\right]_0^1 = \frac{4}{3} \text{ hours}$$

Recall that T is measured in hours past 7am, so the expected time that Anne leaves her flat is 8:20am.

Chapter 7

Expectation and Covariance

7.1 Expected value of functions of random variables X and Y

Suppose the random variable $Z = g(X, Y)$ is a function of two random variables X and Y and we are interested in computing $\mathbb{E}[Z]$. As with functions of a single random variable, there are two ways to compute $\mathbb{E}(Z)$. One way is to first find the probability mass function (discrete case) or probability density function (continuous case) of Z, and then compute $\mathbb{E}(Z)$ from this, as follows.

Discrete case: If X and Y are discrete random variables, then so is the random variable $Z = g(X, Y)$ and you can use the joint probability mass function $p_{X,Y}(x, y)$ to compute the probability mass function $p_Z(z)$ for Z. Once you have determined $p_Z(z)$, you can use this to compute $\mathbb{E}(Z)$:

$$\mathbb{E}(Z) = \sum_{Z \in \mathrm{Im} Z} z p_Z(z)$$

Continuous case: If X and Y are jointly continuous random variables with a joint density $f_{X,Y}(x, y)$ and the function $g : \mathbb{R}^2 \to \mathbb{R}$ is continuous, then the random variable Z is a continuous random variable, and you can use the joint density $f_{X,Y}(x, y)$ to determine $F_Z(z)$, the distribution function for Z. Then, to obtain the density $f_Z(z)$, differentiate $F_Z(z)$. Once you have determined $f_Z(z)$, you can use this to compute

$$\mathbb{E}(Z) = \int_{\mathbb{R}} z f_Z(z) \, \mathrm{d}z$$

The second way to compute $\mathbb{E}(Z)$ is more direct, without needing to first find the probability mass function or density for Z. If you do not need to compute probabilities for the random variable Z, then you do not need to determine the probability mass function or density for Z, and instead you should compute $\mathbb{E}(Z)$ as follows.

Discrete case: If X and Y are discrete random variables with joint probability mass function $p_{X,Y}(x, y)$ then

$$\boxed{\mathbb{E}(Z) = \mathbb{E}\big(g(X, Y)\big) = \sum_{\substack{x \in \mathrm{Im} X \\ y \in \mathrm{Im} Y}} g(x, y) p_{X,Y}(x, y)}$$

Continuous case: If X and Y are continuous random variables with joint density $f_{X,Y}(x,y)$ then

$$\boxed{\mathbb{E}(Z) = \mathbb{E}\big(g(X,Y)\big) = \iint_{\mathbb{R}^2} g(x,y) f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y}$$

Example 6.11 continued. Suppose that X and Y have joint density function

$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x>0,\ y>0,\ x+y<1 \\ 0 & \text{otherwise} \end{cases}$$

Let $Z = XY$. Then

$$\begin{aligned}
\mathbb{E}(Z) = \mathbb{E}(XY) &= \iint xy f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \\
&= \int_{y=0}^{1} \int_{x=0}^{1-y} 60x^2y^3\ \mathrm{d}x\ \mathrm{d}y \\
&= \int_{y=0}^{1} \left[20x^3y^3\right]_{x=0}^{1-y}\ \mathrm{d}y \\
&= \int_{y=0}^{1} 20(1-y)^3y^3\ \mathrm{d}y \\
&= \int_{y=0}^{1} 20\left(y^3 - 3y^4 + 3y^5 - y^6\right)\mathrm{d}y \\
&= \left[5y^4 - 12y^5 + 10y^6 - \frac{20y^7}{7}\right]_{y=0}^{1} = \frac{1}{7} \approx 0.143
\end{aligned}$$

Note: We can also compute $\mathbb{E}(X)$ and $\mathbb{E}(Y)$ *without* first computing the marginal densities $f_X(x)$ and $f_Y(y)$ for X and Y as follows:

$$\mathbb{E}(X) = \iint_{\mathbb{R}^2} x f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \text{ and } \mathbb{E}(Y) = \iint_{\mathbb{R}^2} y f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

Two Important Results

1. Suppose X and Y are *any* two random variables, and $a, b \in \mathbb{R}$. Then

$$\boxed{\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)}$$

   More generally, if $X_1, X_2, \ldots, X_n$ are *any* random variables, and $a_1, a_2, \ldots, a_n \in \mathbb{R}$, then

$$\mathbb{E}(a_1X_1 + a_2X_2 + \cdots + a_nX_n) = a_1\mathbb{E}(X_1) + a_2\mathbb{E}(X_2) + \cdots + a_n\mathbb{E}(X_n)$$

2. Suppose X and Y are *independent* random variables. Then

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$$

   More generally, if $X_1, X_2, \ldots, X_n$ are *independent* random variables, then

$$\mathbb{E}(X_1X_2 \ldots X_n) = \mathbb{E}(X_1)\mathbb{E}(X_2) \ldots \mathbb{E}(X_n)$$

Proof. We will prove these two results in the case where X and Y are jointly continuous with density $f_{X,Y}(x,y)$, but the results are true regardless of whether X, Y are discrete or continuous or not.

1. With X, Y jointly continuous, we have

$$\begin{aligned}\mathbb{E}(aX+bY) &= \iint_{\mathbb{R}^2}(ax+by)f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \\ &= \iint_{\mathbb{R}^2} axf_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y + \iint_{\mathbb{R}^2} byf_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \\ &= \int_{\mathbb{R}} ax\left(\int_{\mathbb{R}} f_{X,Y}(x,y)\ dy\right)dx + \int_{\mathbb{R}} by\left(\int_{\mathbb{R}} f_{X,Y}(x,y)\ dx\right)dy \\ &= a\int_{\mathbb{R}} xf_X(x)\ \mathrm{d}x + b\int_{\mathbb{R}} yf_Y(y)\ \mathrm{d}y \\ &= a\mathbb{E}(X)+b\mathbb{E}(Y)\end{aligned}$$

2. With X, Y jointly continuous and *independent*, we have

$$\begin{aligned}\mathbb{E}(XY) &= \iint_{\mathbb{R}^2} xyf_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_{y=-\infty}^{\infty}\int_{x=-\infty}^{\infty} xyf_X(x)f_Y(y)\,\mathrm{d}x\,\mathrm{d}y \qquad \text{(since } X,\ Y \text{ are independent)} \\ &= \int_{y=-\infty}^{\infty} yf_Y(y)\left(\int_{x=-\infty}^{\infty} xf_X(x)\mathrm{d}x\right)\mathrm{d}y \\ &= \int_{y=-\infty}^{\infty} yf_Y(y)\cdot\mathbb{E}(X)\,\mathrm{d}y \\ &= \mathbb{E}(X)\left(\int_{y=-\infty}^{\infty} yf_Y(y)\,\mathrm{d}y\right) = \mathbb{E}(X)\mathbb{E}(Y)\end{aligned}$$

Remarks

1. The calculations which establish these result in the case where X and Y are *discrete* random variables are similar. You can do them as an exercise.

2. If $X_1, X_2, \ldots, X_n$ are any random variables, and $g_1, g_2, \ldots, g_n$ are functions, then

$$\mathbb{E}\big(g_1(X_1)+g_2(X_2)+\cdots+g_n(X_n)\big)=\mathbb{E}\big(g_1(X_1)\big)+\mathbb{E}\big(g_2(X_2)\big)+\cdots+\mathbb{E}\big(g_n(X_n)\big)$$

3. If $X_1, X_2, \ldots, X_n$ are (jointly) *independent* random variables, then so are the random variables $g_1(X_1), g_2(X_2), \ldots, g_n(X_n)$ where for $k = 1, 2, ..., n$, $g_k : \mathbb{R} \to \mathbb{R}$ is function.
So we also have

$$\mathbb{E}\big(g_1(X_1)g_2(X_2)\ldots g_n(X_n)\big)=\mathbb{E}\big(g_1(X_1)\big)\mathbb{E}\big(g_2(X_2)\big)\ldots\mathbb{E}\big(g_n(X_n)\big)$$

7.2 Covariance and correlation

Definition
Suppose X is a random variable with expected value $\mathbb{E}(X) = \mu_X$ and Y is a random variable with expected value $\mathbb{E}(Y) = \mu_Y$. Then $\mathbb{C}\mathrm{ov}(X, Y)$, the **covariance** of X and Y, is defined as

$$\boxed{\mathbb{C}\mathrm{ov}(X,Y)=\mathbb{E}\big((X-\mu_X)(Y-\mu_Y)\big)}$$

Remark One reason the covariance is useful is that its *sign* tells us something about how the variables X and Y are associated:

- If $\mathbb{C}\text{ov}(X,Y) > 0$ then we say that X and Y are *positively associated*. If the random variables X and Y are positively associated, then *larger* values of X tend to be associated with *larger* values of Y.
  Example: Suppose that X = an individual's height, Y = the same individual's weight, then we would expect $\mathbb{C}\text{ov}(X,Y) > 0$.

- If $\mathbb{C}\text{ov}(X,Y) < 0$ then we say that X and Y are *negatively associated*. If the random variables X and Y are negatively associated, then *larger* values of X tend to be associated with *smaller* values of Y (and vice versa).
  Example: Suppose that X = heating fuel consumption in a house and Y = outside temperature, then we would expect $\mathbb{C}\text{ov}(X,Y) < 0$.

If $\mathbb{C}\text{ov}(X,Y) = 0$, we say that X and Y are **uncorrelated**.

Important simplification:
Using the properties of expectation, we obtain

$$\begin{aligned}\mathbb{C}\text{ov}(X,Y) &= \mathbb{E}\big((X-\mu_X)(Y-\mu_Y)\big) = \mathbb{E}(XY - \mu_X Y - \mu_Y X + \mu_X\mu_Y) \qquad \text{(algebra)}\\ &= \mathbb{E}(XY) - \mu_X\mathbb{E}(Y) - \mu_Y\mathbb{E}(X) + \mu_X\mu_Y \qquad \text{(by properties of expectation)}\\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)\end{aligned}$$

Remark: We usually *compute* $\mathbb{C}\text{ov}(X,Y)$ using $\mathbb{C}\text{ov}(XY) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$.

Special case
If random variables X and Y are *independent*, then

$$\mathbb{C}\text{ov}(X,Y) \overset{\text{(from above)}}{=} \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) \overset{\text{(independence)}}{=} \mathbb{E}(X)\mathbb{E}(Y) - \mathbb{E}(X)\mathbb{E}(Y) = 0,$$

i.e. if the random variables X and Y are *independent*, then X and Y are *uncorrelated*.
Watch out: The *converse* of this is **not true**. That is, $\mathbb{C}\text{ov}(X,Y) = 0$ *does not imply* that the variables X and Y are independent.

Properties of $\mathbb{C}\text{ov}(X,Y)$: It is straightforward (*exercise*) to show that

1. $\mathbb{C}\text{ov}(X,Y) = \mathbb{C}\text{ov}(Y,X)$

2. $\mathbb{C}\text{ov}(X+Y,Z) = \mathbb{C}\text{ov}(X,Z) + \mathbb{C}\text{ov}(Y,Z)$

3. $\mathbb{C}\text{ov}(aX+b,cY+d) = ac\mathbb{C}\text{ov}(X,Y)$

Discussion
The *sign* of $\mathbb{C}\text{ov}(X,Y)$ gives an indication of the 'direction' of the association between X and Y (i.e. whether there is positive or negative association), but the numerical value of $\mathbb{C}\text{ov}(X,Y)$ is *not* a good measure of the *strength* of the association between the random variables X and Y. The problem is that $\mathbb{C}\text{ov}(X,Y)$ is not a *scale-free* measure of the association between X and Y. Here is an example that illustrates this problem.

Example 7.1. Suppose a student is selected at random and the student's height and foot length are measured. Based on these measurements, let

$X =$ student's height in metres
$Y =$ student's foot length in metres
$U =$ student's height in centimetres (so $U = 100X$)
$V =$ student's foot length in centimetres (so $V = 100Y$)

Then we have

$$\mathbb{C}\text{ov}(U, V) = \mathbb{C}\text{ov}(100X, 100Y) = 10000\mathbb{C}\text{ov}(X, Y)$$

So $\mathbb{C}\text{ov}(U, V)$ is ten thousand times greater than $\mathbb{C}\text{ov}(X, Y)$, but the 'strength' of the association between 'height' and 'foot length' should not depend on the *scale* that we use to measure these physical quantities.

Conclusion:
The example above illustrates why we need a measure of the association between variables that does not depend on the *scale* that we use for the variables. A *scale-free* measure of the strength of the association between two random variables X and Y is provided by the *correlation coefficient*.

Definition 7.1. The *correlation coefficient* $\rho(X, Y)$ for X and Y is defined by

$$\boxed{\rho(X, Y) = \frac{\mathbb{C}\text{ov}(X, Y)}{\sqrt{\mathbb{V}\text{ar}(X)\mathbb{V}\text{ar}(Y)}} = \frac{\mathbb{C}\text{ov}(X, Y)}{\sigma_X \sigma_Y}}$$

Here σ_X is the standard deviation of X and σ_Y is the standard deviation of Y.
Example 7.1 continued. For the height and foot length variables in example 7.1,

$$\begin{aligned}
\rho(U, V) &= \frac{\mathbb{C}\text{ov}(U, V)}{\sqrt{\mathbb{V}\text{ar}(U)\mathbb{V}\text{ar}(V)}} = \frac{\mathbb{C}\text{ov}(100X, 100Y)}{\sqrt{\mathbb{V}\text{ar}(100X)\mathbb{V}\text{ar}(100Y)}} \\
&= \frac{(100)^2\mathbb{C}\text{ov}(X, Y)}{\sqrt{(100)^2\mathbb{V}\text{ar}(X)(100)^2\mathbb{V}\text{ar}(Y)}} \qquad \text{(by the properties of variance and covariance)} \\
&= \frac{\mathbb{C}\text{ov}(X, Y)}{\sqrt{\mathbb{V}\text{ar}(X)\mathbb{V}\text{ar}(Y)}} = \rho(X, Y)
\end{aligned}$$

So: $\rho(U, V) = \rho(X, Y)$. In other words, the *correlation* between 'height' and 'foot length' does not depend on the *scale* used to measure these quantities.

Properties of the correlation coefficient $\rho(X, Y)$:

1. $\rho(aX + b, cY + d) = \rho(X, Y) \quad \forall a, b, c, d \in \mathbb{R}$ s.t. $ac > 0$.
   (i.e $\rho(X, Y)$ is scale and location invariant)

2. Whenever $\rho(X, Y)$ exists (i.e. $\mathbb{V}\text{ar}(X), \mathbb{V}\text{ar}(Y) \neq 0$) then

   $$-1 \leqslant \rho(X, Y) \leqslant 1.$$

   Also if $\rho(X, Y) = \pm 1$, then $Y = aX + b$ for some $a, b \in \mathbb{R}$.

Proof.

1.
$$\begin{aligned}\rho(aX+b,cY+d) &= \frac{\mathbb{C}\text{ov}(aX+b,cY+d)}{\sqrt{\mathbb{V}\text{ar}(aX+b)\mathbb{V}\text{ar}(cY+d)}}\\ &= \frac{ac\mathbb{C}\text{ov}(X,Y)}{\sqrt{a^2\mathbb{V}\text{ar}(X)c^2\mathbb{V}\text{ar}(Y)}}\\ &= \frac{\mathbb{C}\text{ov}(X,Y)}{\sqrt{\mathbb{V}\text{ar}(X)\mathbb{V}\text{ar}(Y)}}\\ &= \rho(X,Y)\end{aligned}$$

2. Let $\hat{X} = \frac{X-\mathbb{E}(X)}{\sqrt{\mathbb{V}\text{ar}(X)}}$, $\hat{Y} = \frac{Y-\mathbb{E}(Y)}{\sqrt{\mathbb{V}\text{ar}(Y)}}$. Then it follows from Property 1 above that
$$\rho(X,Y) = \rho(\hat{X},\hat{Y}) = \mathbb{E}(\hat{X}\hat{Y}) \qquad (\text{since } \mathbb{E}(\hat{X}) = \mathbb{E}(\hat{Y}) = 0)$$
Next we note that
$$0 \leqslant \mathbb{E}\Big(\big(\hat{X}-\hat{Y}\big)^2\Big) = \mathbb{E}(\hat{X}^2) - 2\mathbb{E}(\hat{X}\hat{Y}) + \mathbb{E}(\hat{Y}^2) = 2 - 2\mathbb{E}(\hat{X}\hat{Y}) \qquad (*)$$
since $\mathbb{E}(\hat{X}^2) = \mathbb{E}(\hat{Y}^2) = 1$. So, it follows from $(*)$ that $\rho(X,Y) = \mathbb{E}(\hat{X}\hat{Y}) \leqslant 1$.

Similarly, we have
$$0 \leqslant \mathbb{E}\Big(\big(\hat{X}+\hat{Y}\big)^2\Big) = \mathbb{E}(\hat{X}^2) + 2\mathbb{E}(\hat{X}\hat{Y}) + \mathbb{E}(\hat{Y}^2) = 2 + 2\mathbb{E}(\hat{X}\hat{Y})$$
So we must have $\rho(X,Y) = \mathbb{E}(\hat{X}\hat{Y}) \geqslant -1$. It follows that $-1 \leq \rho(X,Y) \leq 1$.

Next, suppose $\rho(X,Y) = \mathbb{E}(\hat{X}\hat{Y}) = 1$. Then it follows from $(*)$ that $\mathbb{E}\Big(\big(\hat{X}-\hat{Y}\big)^2\Big) = 0$. We note that since $\mathbb{E}\Big(\big(\hat{X}-\hat{Y}\big)^2\Big) = 0$, we must also have $\hat{X} = \hat{Y}$. In other words, by definition of $\hat{X}$ and $\hat{Y}$, we have
$$\frac{X-\mathbb{E}(X)}{\sqrt{\mathbb{V}\text{ar}(X)}} = \frac{Y-\mathbb{E}(Y)}{\sqrt{\mathbb{V}\text{ar}(Y)}} \qquad (**).$$
Solving the equation $(**)$ for Y, we obtain
$$Y = \underbrace{\frac{\sqrt{\mathbb{V}\text{ar}(Y)}}{\sqrt{\mathbb{V}\text{ar}(X)}}}_{a} X + \underbrace{\mathbb{E}(Y) - \mathbb{E}(X)\frac{\sqrt{\mathbb{V}\text{ar}(Y)}}{\sqrt{\mathbb{V}\text{ar}(X)}}}_{b}$$
as desired.
Exercise: Show that if $\rho(X,Y) = -1$, then $Y = aX + b$ for some $a, b \in \mathbb{R}$.

7.3 Variance of a linear sum of random variables X and Y

In this section we investigate the variance of a linear sum of random variables. We already know that for random variables X and Y and $a, b \in \mathbb{R}$
$$\mathbb{E}(aX+bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

Using the properties of variance and the definition of the covariance, we can now determine the corresponding result for $\mathbb{V}\text{ar}(aX + bY)$:

$$\begin{aligned}\mathbb{V}\text{ar}(aX + bY) &= \mathbb{E}\big((aX + bY)^2\big) - \big(\mathbb{E}(aX + bY)\big)^2 \\ &= a^2\mathbb{E}(X^2) + 2ab\mathbb{E}(XY) + b^2\mathbb{E}(Y^2) - a^2\big(\mathbb{E}(X)\big)^2 - 2ab\mathbb{E}(X)\mathbb{E}(Y) - b^2\big(\mathbb{E}(Y)\big)^2 \\ &= a^2\mathbb{V}\text{ar}(X) + 2ab\mathbb{C}\text{ov}(X, Y) + b^2\mathbb{V}\text{ar}(Y) \qquad (**)\end{aligned}$$

In particular,

$$\boxed{\mathbb{V}\text{ar}(X + Y) = \mathbb{V}\text{ar}(X) + 2\mathbb{C}\text{ov}(X, Y) + \mathbb{V}\text{ar}(Y)}$$

Remarks:

1. The above result shows that to compute $\mathbb{V}\text{ar}(X+Y)$ it is *not* enough to know the variances of each of the variables X and Y. We also need to know the covariance $\mathbb{C}\text{ov}(X, Y)$.

2. If X and Y are *independent* then $\mathbb{C}\text{ov}(X, Y) = 0$ and equation $(**)$ above becomes

   $$\mathbb{V}\text{ar}(aX + bY) = a^2\mathbb{V}\text{ar}(X) + b^2\mathbb{V}\text{ar}(Y).$$

   More generally, if $X_1, X_2, \ldots, X_n$ are *independent*, then

   $$\mathbb{V}\text{ar}(a_1X_1 + a_2X_2 + \cdots + a_nX_n) = a_1^2\mathbb{V}\text{ar}(X_1) + a_2^2\mathbb{V}\text{ar}(X_2) + \cdots + a_n^2\mathbb{V}\text{ar}(X_n)$$

   and, in particular,

   $$\mathbb{V}\text{ar}(X_1 + \cdots + X_n) = \mathbb{V}\text{ar}(X_1) + \cdots + \mathbb{V}\text{ar}(X_n) \text{ for } \textit{independent } X_1, X_2, \ldots, X_n$$

Example 7.2. Suppose X is the number of successes in n independent, identical trials with $\mathbb{P}(\text{'success'}) = p$ for each trial. That is, X is a binomial random variable, $X \sim \text{Bin}(n, p)$, with probability mass function $p_X(k) = \binom{n}{k}p^k(1-p)^{n-k}$ for $k = 0, 1, 2, ..., n$.
In chapter 3, we used the probability mass function to compute $\mathbb{E}(X)$; we can now compute $\mathbb{E}(X)$ much more easily using properties of expectation.

Recall that if we define the indicator random variables $I_1, I_2, \ldots, I_n$ to be

$$I_k = \begin{cases} 1 & \text{if } k^{\text{th}} \text{ trial is success (with probability } p) \\ 0 & \text{otherwise (with probability } 1-p) \end{cases}$$

then we can write X as the sum of these indicator random variables:

$$X = I_1 + I_2 + \cdots + I_n$$

and for $k = 1, 2, \ldots, n$, we have

$$\begin{aligned} \mathbb{E}(I_k) &= p \times 1 + (1-p) \times 0 = p \\ \mathbb{E}(I_k^2) &= p \times 1^2 + (1-p) \times 0^2 = p \\ \Rightarrow \quad \mathbb{V}\text{ar}(I_k) &= p - p^2 = p(1-p) \end{aligned}$$

So

$$\begin{aligned} \mathbb{E}(X) &= \mathbb{E}(I_1 + I_2 + \cdots + I_n) \\ &= \mathbb{E}(I_1) + \mathbb{E}(I_2) + \cdots + \mathbb{E}(I_n) \\ &= p + p + \cdots + p \\ &= np \end{aligned}$$

Note that we have not yet made use of independence. We will now assume that the indicator variables $I_1, I_2, \ldots, I_n$ are *independent* because the trials that the indicators are based on are physically independent trials. Then

$$\begin{aligned}
\mathbb{V}\text{ar}(X) &= \mathbb{V}\text{ar}(I_1 + I_2 + \cdots + I_n) \\
&= \mathbb{V}\text{ar}(I_1) + \mathbb{V}\text{ar}(I_2) + \cdots + \mathbb{V}\text{ar}(I_n) \qquad \text{(by independence of } I_1, I_2, \ldots, I_n\text{)} \\
&= p(1-p) + p(1-p) + \cdots + p(1-p) \\
&= np(1-p)
\end{aligned}$$

So $X \sim \text{Bin}(n,p) \Rightarrow \mathbb{E}(X) = np,\ \mathbb{V}\text{ar}(X) = np(1-p)$

Example 7.3. Statistical inference result. Suppose $X_1, X_2, \ldots X_n$ are i.i.d. (independent, identically distributed) random variables with

$$\mathbb{E}(X_k) = \mu \text{ and } \mathbb{V}\text{ar}(X_k) = \sigma^2 \quad \text{ for } k = 1, 2, ..., n.$$

In statistical inference we use the sample mean

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

to *estimate* the (population) mean μ.

The sample mean $\bar{X}$ is *also* a random variable, and it is important to know $\mathbb{E}(\bar{X})$ and $\mathbb{V}\text{ar}(\bar{X})$, the expected value and variance of the sample mean.

$$\begin{aligned}
\mathbb{E}(\bar{X}) &= \mathbb{E}\Big(\frac{X_1 + \cdots + X_n}{n}\Big) \\
&= \frac{\mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n)}{n} \qquad \text{(by linearity of expectation)} \\
&= \frac{\mu + \cdots + \mu}{n} = \frac{n\mu}{n} = \mu
\end{aligned}$$

Remark: This means that $\bar{X}$ is an *unbiased* estimator of μ.

$$\begin{aligned}
\mathbb{V}\text{ar}(\bar{X}) &= \mathbb{V}\text{ar}\Big(\frac{X_1 + \cdots + X_n}{n}\Big) \\
&= \frac{1}{n^2}\mathbb{V}\text{ar}(X_1 + \cdots + X_n) \qquad \text{(property of variance)} \\
&= \frac{1}{n^2}\big(\mathbb{V}\text{ar}(X_1) + \cdots + \mathbb{V}\text{ar}(X_n)\big) \qquad \text{(by independence of } X_1, X_2, \ldots X_n\text{)} \\
&= \frac{1}{n^2}(\sigma^2 + \cdots + \sigma^2) = \frac{\sigma^2}{n}
\end{aligned}$$

Remark: Notice that as $n \to \infty$, $\mathbb{V}\text{ar}(\bar{X}) \to 0$. In terms of statistical inference, this means that the larger the value of n (i.e. the more observations we have), the greater the precision of the estimator $\bar{X}$ for μ.

Chapter 8

Inequalities and Limit Theorems

8.1 Two important inequalities

Sometimes a random variable, X, can have a distribution which is difficult to determine explicitly, but it is still possible to determine moments of X (e.g. by using linearity of expected value). In such situations, we can still get some information about *probabilities* associated with X by using probability inequalities based on moments of X.

8.1.1 Markov's inequality

Suppose X is a *non-negative* random variable and suppose $\varepsilon > 0$ is a constant. Then

$$\mathbb{P}(X \geqslant \varepsilon) \leqslant \frac{\mathbb{E}(X)}{\varepsilon}$$

Proof
We start by defining the indicator variable

$$I_\varepsilon = \begin{cases} 1 & \text{if } X \geqslant \varepsilon \\ 0 & \text{otherwise} \end{cases}$$

and note that $\varepsilon I_\varepsilon \leqslant X$, since $I_\varepsilon = 1$ only when $X \geq \varepsilon$.
So it follows from the properties of expected value that $\mathbb{E}(\varepsilon I_\varepsilon) \leqslant \mathbb{E}(X)$. $(*)$
Next, we compute:

$$\mathbb{E}(\varepsilon I_\varepsilon) = \varepsilon \mathbb{E}(I_\varepsilon) = \varepsilon \mathbb{P}(I_\varepsilon = 1) = \varepsilon \mathbb{P}(X \geqslant \varepsilon)$$

So, substituting $\varepsilon \mathbb{P}(X \geqslant \varepsilon)$ for $\mathbb{E}(\varepsilon I_\varepsilon)$ in $(*)$, we obtain

$$\varepsilon \mathbb{P}(X \geqslant \varepsilon) \leqslant \mathbb{E}(X)$$

and the result follows.

Remark It is important to note that Markov's inequality can only be used when we know that the random variable X takes *non-negative* values (can you see where we use the fact that X is non-negative in the proof above?). So be careful when you use Markov's inequality and always check that X is a non-negative random variable first.

Example 8.1. Suppose $X_1, X_2, \ldots, X_5$ are independent, identically distributed random variables, each being exponentially distributed with mean 2. Use Markov's inequality to bound the probability that the sum of these random variables is greater than 13.

Solution: Let $S = X_1 + X_2 + \cdots + X_5$. Then S is a non-negative random variable, with $\mathbb{E}[S] = \sum_{i=1}^{5} \mathbb{E}[X_i] = 10$. Applying Markov's inequality,

$$\begin{aligned} \mathbb{P}(S > 13) &= \mathbb{P}(S \geqslant 13) && \text{(since } S \text{ is a continuous r.v.)} \\ &\leqslant \frac{\mathbb{E}[S]}{13} && \text{(Markov's inequality)} \\ &= \frac{10}{13} \end{aligned}$$

Note that we did not need to know the distribution of the random variable S to derive the above bound.

Although Markov's inequality can only be used when X is a non-negative random variable, we can apply it to obtain an inequality that can be used for more general random variables.

8.1.2 Chebyshev's inequality

Suppose X is a random variable with $\mathbb{E}(X) = \mu$ and $\mathbb{V}\mathrm{ar}(X) < \infty$, and suppose $\varepsilon > 0$ is a constant. Then

$$\mathbb{P}(|X - \mu| \geqslant \varepsilon) \leqslant \frac{\mathbb{V}\mathrm{ar}(X)}{\varepsilon^2}$$

Proof. The inequality follows from an application of Markov's inequality to the *non-negative* random variable $(X - \mu)^2$ as follows:

$$\mathbb{P}(|X - \mu| \geqslant \varepsilon) = \mathbb{P}((X - \mu)^2 \geqslant \varepsilon^2) \leqslant \frac{\mathbb{E}\big((X - \mu)^2\big)}{\varepsilon^2} = \frac{\mathbb{V}\mathrm{ar}(X)}{\varepsilon^2}$$

remembering that $\mathbb{E}((X - \mu)^2) = \mathbb{V}\mathrm{ar}(X)$, by definition of $\mathbb{V}\mathrm{ar}(X)$.

Remark. In statistical applications we are often interested in determining the probability that a random variable X is 'far away' (i.e. more than ε) from its mean μ. Chebyshev's inequality gives us a way to bound this probability.

Example 8.2. Suppose that $X \sim \Gamma(200, 2)$. Use Chebyshev's inequality to find a lower bound on the probability $\mathbb{P}(90 < X < 110)$.

Solution: For $X \sim \Gamma(200, 2)$, we have $\mathbb{E}[X] = \frac{200}{2} = 100$, and $\mathbb{V}\mathrm{ar}(X) = \frac{200}{2^2} = 50 < \infty$. Applying Chebyshev's inequality with $\varepsilon = 10$, we obtain

$$\mathbb{P}(|X - 100| \geqslant 10) \leqslant \frac{50}{10^2} = \frac{1}{2}$$

and so

$$\mathbb{P}(90 < X < 110) = \mathbb{P}(|X - 100| < 10) = 1 - \mathbb{P}(|X - 100| \geqslant 10) \geqslant \frac{1}{2}$$

There is probability at least 0.5 that the random variable X is within 10 of its mean, by Chebyshev's inequality.

Application of Chebyshev's Inequality: Weak Law of Large Numbers
Suppose $X_1, X_2, \ldots$ are independent, identically distributed random variables with $\mathbb{E}(X) = \mu$ and $\mathbb{V}\mathrm{ar}(X) = \sigma^2 < \infty$ and let $S_n = X_1 + \cdots + X_n$. Then for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(\left| \underbrace{\frac{S_n}{n}}_{\text{sample mean}} - \mu \right| \geqslant \varepsilon \right) = 0.$$

Proof
First, we note that

$$\begin{aligned}
\mathbb{E}\left(\frac{S_n}{n}\right) &= \frac{\mathbb{E}(X_1 + \cdots + X_n)}{n} \\
&= \frac{\mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n)}{n} && \text{(property of expectation)} \\
&= \frac{\mu + \cdots + \mu}{n} = \mu && \text{(since the variables have the same mean)}
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{V}\mathrm{ar}\Big(\frac{S_n}{n}\Big) &= \frac{1}{n^2}\mathbb{V}\mathrm{ar}(X_1 + \cdots + X_n) && \text{(property of variance)} \\
&= \frac{1}{n^2}\big(\mathbb{V}\mathrm{ar}(X_1) + \cdots + \mathbb{V}\mathrm{ar}(X_n)\big) && \text{(since } X_1, \ldots, X_n \text{ are independent)} \\
&= \frac{1}{n^2}(\sigma^2 + \cdots + \sigma^2) = \frac{\sigma^2}{n} && \text{(since the variables have the same variance)}
\end{aligned}$$

So, by Chebyshev:

$$0 \leqslant \mathbb{P}\left(\left| \frac{S_n}{n} - \mu \right| \geqslant \varepsilon \right) \leqslant \frac{\mathbb{V}\mathrm{ar}(\frac{S_n}{n})}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \qquad (*)$$

It follows from $(*)$ that $\lim_{n\to\infty} \mathbb{P}\Big(\big|\frac{S_n}{n} - \mu\big| \geqslant \varepsilon\Big) = 0$.
Remark The Weak Law of Large Numbers is used in various statistical applications.

8.2 Central Limit Theorem

One of the most important results in probability theory and statistics is the Central Limit Theorem.

Central Limit Theorem (CLT)
Suppose that $X_1, X_2, \ldots$ are independent and identically distributed random variables with $\mathbb{E}(X_i) = \mu$, $\mathbb{V}\mathrm{ar}(X_i) = \sigma^2 < \infty$, and let $S_n = X_1 + \ldots X_n$ for $n = 1, 2, \ldots$
Then for *any* $a < b$,

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leqslant b \right) = \mathbb{P}\left(a \leqslant Z \leqslant b\right) \quad (*) \qquad \text{where } Z \sim \mathrm{N}\left(0, 1\right)$$

Proof: Beyond the scope of this course (requires use of *generating functions*).

Remark. Another way to describe the Central Limit Theorem is to say that

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \quad \textit{converges in distribution to} \quad Z \sim \mathrm{N}\left(0, 1\right)$$

as $n \to \infty$. We say this because equality $(*)$ above tells us that probabilities for $\frac{S_n - n\mu}{\sqrt{n\sigma^2}}$ converge to probabilities for $Z \sim \mathrm{N}(0,1)$.

Remark. Typically we use the Central Limit Theorem to justify the statement:
For n *large* and for independent and identically distributed random variables $X_1, X_2, \ldots, X_n$ with $\mathbb{E}(X_i) = \mu$, $\mathbb{V}\mathrm{ar}(X_i) = \sigma^2 < \infty$, and $S_n = X_1 + \ldots X_n$, we have, by the Central Limit Theorem,

$$\mathbb{P}\left(a \leqslant \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leqslant b\right) \approx \mathbb{P}(a \leqslant Z \leqslant b)$$

where $Z \sim \mathrm{N}(0,1)$.

Central Limit Theorem applied to Binomial distribution
Suppose that $0 < p < 1$ and that X_n is a Binomial $\mathrm{Bin}(n,p)$ random variables for $n = 1, 2, \ldots$. Then for any $a < b$ we have

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{X_n - np}{\sqrt{np(1-p)}} \leqslant b\right) = \mathbb{P}(a \leqslant Z \leqslant b) \qquad \text{where } Z \sim \mathrm{N}(0,1)\,.$$

Proof. Recall that the Binomial variable X_n counts the number of successes in n independent trials with $\mathbb{P}(\text{success}) = p$. So we can write $X_n = I_1 + I_2 + \cdots + I_n$ where

$$I_k = \begin{cases} 1 & k^{\text{th}} \text{ trial is success} \\ 0 & \text{otherwise} \end{cases}$$

and the indicator variables $I_1, I_2, \ldots$ are independent. We also know that $\mathbb{E}(I_k) = p$ and $\mathbb{V}\mathrm{ar}(I_k) = p(1-p)$. So, by the Central Limit Theorem,

$$\lim_{n\to\infty} \mathbb{P}\left(a \leqslant \frac{X_n - np}{\sqrt{np(1-p)}} \leqslant b\right) = \lim_{n\to\infty} \mathbb{P}\left(a \leq \frac{I_1 + \cdots + I_n - np}{\sqrt{np(1-p)}} \leqslant b\right) = \mathbb{P}(a \leqslant Z \leqslant b)$$

Remark. Typically we use the above result to justify the statement:
For $X_n \sim \mathrm{Bin}(n,p)$ with n *large*, we have, by the Central Limit Theorem,

$$\mathbb{P}\left(a \leqslant \frac{X_n - np}{\sqrt{np(1-p)}} \leqslant b\right) \approx \mathbb{P}(a \leqslant Z \leqslant b)$$

where $Z \sim \mathrm{N}(0,1)$.
This is referred to as the **Normal Approximation to the Binomial**.

Example 8.3. 100 new light bulbs have independent lifetimes that are exponentially distributed with mean 5 weeks. The light bulbs are used one at a time and when a bulb fails, it is immediately replaced. Suppose the first bulb is installed at time $t = 0$. What is the (approximate) probability that all of the 100 new bulbs will have been used and have failed before the end of 550 weeks?

Solution: To answer this question, we begin by defining some notation. For $i = 1, 2, ..., 100$, let X_i equal the lifetime of ith light bulb. From the information given in the problem, we know that the variables $X_1, X_2, \ldots, X_{100}$ are *independent*. For $k = 1, 2, \ldots, 100$, we know that X_i is exponentially distributed with mean 5 weeks, so $X_i \sim \mathrm{Exp}(1/5)$, and $\mathbb{V}\mathrm{ar}(X_i) = 25$.

The time until all of the light bulbs have been used and have failed is $X_1 + X_2 + \cdots + X_{100}$ and we want to compute

$$\mathbb{P}(X_1 + X_2 + \cdots + X_{100} \leqslant 550)$$

Since $X_1, X_2, \ldots, X_n$ are i.i.d., we can use the Central Limit Theorem (CLT) to obtain

$$\begin{aligned}
\mathbb{P}(X_1 + X_2 + \cdots + X_{100} \leqslant 550) &= \mathbb{P}\left(\frac{X_1 + X_2 + \cdots + X_{100} - 100\mathbb{E}(X_i)}{\sqrt{100\mathbb{V}\mathrm{ar}(X_i)}} \leqslant \frac{550 - 100\mathbb{E}(X_i)}{\sqrt{100\mathbb{V}\mathrm{ar}(X_i)}}\right) \\
&\approx \mathbb{P}\left(Z \leqslant \frac{50}{\sqrt{2500}}\right) \qquad \text{by the CLT, with } Z \sim \mathrm{N}(0,1) \\
&= \mathbb{P}(Z \leqslant 1) = 0.8413 \qquad \text{from the Statistical Tables}
\end{aligned}$$

8.3 Continuity correction; more examples of CLT

The Central Limit Theorem allows us to use Normal distribution probabilities to approximate both continuous distributions and discrete distributions. When approximating a *discrete* distribution using the (*continuous*) Normal distribution, there is an important refinement, called a *continuity correction*.

Example 8.4. Suppose that $X \sim B(200, 0.5)$, and we want to approximate $\mathbb{P}(X = 105)$ using the CLT.

Note that **because X takes only integer values,** $\mathbb{P}(X = 105) = \mathbb{P}(104.5 \leq X \leq 105.5)$.

Now we have $\mathbb{E}(X) = 100$, $\mathbb{V}\mathrm{ar}(X) = 50$, and so by the CLT,

$$\begin{aligned}
\mathbb{P}(X = 105) &= \mathbb{P}(104.5 \leqslant X \leqslant 105.5) \qquad \text{(continuity correction)} \\
&= \mathbb{P}\left(\frac{104.5 - 100}{\sqrt{50}} \leqslant \frac{X - 100}{\sqrt{50}} \leqslant \frac{105.5 - 100}{\sqrt{50}}\right) \\
&\approx \mathbb{P}(0.64 \leqslant Z \leqslant 0.78) \qquad \text{where } Z \sim N(0,1), \text{ by CLT} \\
&= \Phi(0.78) - \Phi(0.64) \\
&= 0.7823 - 0.7389 \qquad \text{(from the Statistical Tables)} \\
&= 0.0434
\end{aligned}$$

More generally, if X is an integer-valued random variable, then for any integer values a, b, we have

$$\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a - 0.5 \leqslant X \leqslant b + 0.5)$$

Note that the interval $\{a \leqslant X \leqslant b\}$ *includes* both end-points, so we *extend* the interval at both ends. Compare with

$$\mathbb{P}(a < X < b) = \mathbb{P}(a + 0.5 \leqslant X \leqslant b - 0.5)$$

In this case, the interval $\{a < X < b\}$ *excludes* both end-points, so we *contract* the interval at both ends.

Example 8.5. An electronic instrument functions if at least 40 of its 70 components are functioning. Components function independently and the lifetime of each component is exponentially distributed with mean 100 months (so $\lambda = 1/100$). What is the (approximate) probability that the instrument is still functioning after 80 months?

Define Y to be the number of components that are still functioning after 80 months. Since the components operate independently and all have the same lifetime distribution, the random variable Y is Binomial, $Y \sim \mathrm{Bin}(70, p)$, where $p = \mathbb{P}(\text{a component is still functioning after 80 months})$. Since the lifetime, X, of a component is exponentially distributed with $\lambda = 1/100$, we have

$$p = \mathbb{P}(X > 80) = e^{-80/100} = e^{-0.8}$$

Then we have

$\mathbb{P}$(instrument functioning after 80 months) $=\mathbb{P}(Y \geqslant 40) = \mathbb{P}(Y \geqslant 39.5)$

since Y can take only integer values, and the interval $\{Y \geqslant 40\}$ includes the value 40, so we have used a *continuity correction* by extending the interval to $\{Y \geqslant 39.5\}$.
Now

$$\mathbb{P}(Y \geqslant 39.5) = \mathbb{P}\left(\frac{Y - 70p}{\sqrt{70p(1-p)}} \geqslant \frac{39.5 - 70p}{\sqrt{70p(1-p)}}\right)$$

$$\approx \mathbb{P}\left(Z \geqslant \frac{39.5 - 70e^{-0.8}}{\sqrt{70e^{-0.8}(1-e^{-0.8})}}\right) \qquad \text{(CLT for Binomial, with } Z \sim \mathrm{N}(0,1)\text{)}$$

$$= \mathbb{P}(Z \geqslant 1.93) = 1 - \Phi(1.93) = 0.0268 \qquad \text{(from the Statistical Tables)}$$

Example 8.6. Suppose that we wish to estimate the proportion, p, of voters who intend to vote for Party A in an upcoming election. To estimate p, we select n people at random (without replacement) from the population of registered voters. Let

$$X_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ person selected intends to vote for Party A} \\ 0 & \text{otherwise} \end{cases}$$

We note that $\mathbb{E}(X_i) = p$ and $\mathbb{V}\mathrm{ar}(X_i) = p(1-p)$ and since the population is *very large*, we will *assume* that the variables $X_1, X_2, \ldots$ are independent.

To estimate the population proportion p, we will use the sample mean $\bar{X}(n) = \frac{X_1 + \cdots + X_n}{n}$, where n is the size of the selected sample.

Question: How large should n be to ensure that

$$\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big) \geqslant 0.95?$$

Answer: We begin by using the Central Limit Theorem to estimate $\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big)$:

$$\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big)$$

$$= \mathbb{P}\big(-0.05 \leqslant \bar{X}(n) - p \leqslant 0.05\big) \qquad \text{(algebra)}$$

$$= \mathbb{P}\big(-0.05n \leqslant X_1 + \cdots + X_n - np \leqslant 0.05n\big) \qquad \text{(since } n\bar{X}(n) = X_1 + \cdots + X_n\text{)}$$

$$= \mathbb{P}\left(\frac{-0.05n}{\sqrt{np(1-p)}} \leqslant \frac{X_1 + \cdots + X_n - np}{\sqrt{np(1-p)}} \leqslant \frac{0.05n}{\sqrt{np(1-p)}}\right)$$

$$\geqslant \mathbb{P}\left(\frac{-0.05\sqrt{n}}{\sqrt{0.25}} \leqslant \frac{X_1 + \cdots + X_n - np}{\sqrt{np(1-p)}} \leqslant \frac{0.05\sqrt{n}}{\sqrt{0.25}}\right) \qquad \text{(since } p(1-p) \leqslant 0.25\text{)}$$

$$\approx \mathbb{P}(-0.1\sqrt{n} \leqslant Z \leqslant 0.1\sqrt{n}) \qquad \text{(by the Central Limit Theorem, with } Z \sim N(0,1)\text{)}$$

Now, we want $\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big) \geqslant 0.95$ and, from the calculation above, we can see that this will be achieved if $\mathbb{P}(-0.1\sqrt{n} \leqslant Z \leqslant 0.1\sqrt{n}) = 0.95$. Using Table 5 from the Cambridge Statistical Tables, we see that for the standard normal variable Z,

$$\mathbb{P}(-1.96 \leqslant Z \leqslant 1.96) = 0.95.$$

So, $\mathbb{P}(-0.1\sqrt{n} \leqslant Z \leqslant 0.1\sqrt{n}) = 0.95$ if and only if $0.1\sqrt{n} = 1.96$. Solving $0.1\sqrt{n} = 1.96$, we obtain $n = 385$, and so a sample size $n = 400$ would be large enough to achieve $\mathbb{P}\big(|\bar{X}(n) - p| \leqslant 0.05\big) \geqslant 0.95$.

8.4 Poisson approximation to the binomial distribution

A different distributional approximation (not based on the Central Limit Theorem) is the following.
Suppose $\lambda > 0$ and for $n = 1, 2, \ldots$ let $Y_n \sim \text{Bin}(n, \frac{\lambda}{n})$.
Then the variables $Y_1, Y_2, \ldots$ converge in *distribution* to a Poisson variable $X \sim \text{Poi}(\lambda)$.

Remark: Convergence of $Y_1, Y_2, ...$ to X in distribution means that

$$\lim_{n\to\infty} \mathbb{P}(Y_n = k) = \mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Proof. Since $Y_n \sim B\left(n, \frac{\lambda}{n}\right)$, then for $k \leq n$ we have

$$\begin{aligned}
\mathbb{P}(Y_n = k) &= \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} \\
&= \frac{n!}{(n-k)!\,k!}\left(\frac{\lambda^k}{n^k}\right)\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k} \\
&= \frac{\lambda^k}{k!}\left(\frac{n(n-1)(n-2)\cdots(n-k+1)}{n^k}\right)\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k} \\
&= \frac{\lambda^k}{k!}\,1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right)\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k} \qquad (*)
\end{aligned}$$

Now as $n \to \infty$,

$$\begin{aligned}
1\cdot\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) &\to 1 \\
\left(1-\frac{\lambda}{n}\right)^k &\to 1 \\
\left(1-\frac{\lambda}{n}\right)^n &\to e^{-\lambda}
\end{aligned}$$

Substituting into $(*)$, we obtain

$$\lim_{n\to\infty} \mathbb{P}(Y_n = k) = \frac{\lambda^k}{k!}e^{-\lambda}$$

as required.

Example 8.7. National Lottery. To win the UK National Lottery, the 6 numbers on your ticket have to match the 6 balls drawn from 59 balls, numbered $1, 2, \ldots, 59$. There are $\binom{59}{6}$

different possible subsets of size 6 from the 59 balls and all of these possible subsets are assumed to be equally likely to be selected, so

$$\mathbb{P}(\text{win}) = \frac{1}{\binom{59}{6}} = \frac{1}{45\,057\,474} \approx \frac{1}{N}$$

where $N = 45$ million.

Now suppose that $2N$ lottery tickets are sold and let Y equal the number of jackpot winners. What is the probability that there are 3 jackpot winners?

Assuming the numbers on the $2N$ lottery tickets are independent, we have

$$Y \sim \text{Bin}(2N, 1/N) = \text{Bin}(2N, 2/2N)$$

By the Poisson approximation to the Binomial, we have that probabilities for the Binomial variable Y can be approximated by probabilities for X, where X is a Poisson random variable with $\lambda = (2N) \cdot \left(\frac{2}{2N}\right) = 2$. So

$$\mathbb{P}(Y = 3) \approx \mathbb{P}(X = 3) = \frac{e^{-2} \cdot 2^3}{3!} = 0.18$$

Note: here we are approximating a *discrete* distribution (Binomial) by another *discrete* distribution (Poisson), so a continuity correction would **not** be appropriate.